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CELLULAR MAXIMAL-DENSITY FACTORING: JOINT SHADINGS, STABLE REFINEMENTS, AND A CONDITIONAL KAKEYA APPLICATION

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Abstract. Maximal-density factoring must often make one parent object carry several forms of information at once: child mass, two multiplicity levels, a density estimate, and the ability to survive later geometric refinements. These properties are not stable under arbitrary deletion. We introduce a cellular factoring method that places positive child mass on a weighted bipartite graph of labelled parents and half-open spatial cells. After regularizing edge weights and cell degrees, one edge set defines both the child shading and the parent shading. Compatibility and pointwise multiplicity bounds are then exact, while any later selection of whole parent–cell incidences lifts quantitatively back to the children. The combinatorial core does not require Euclidean geometry: it extends to finite measurable atomic partitions with comparable atom measures and admits an explicit finite-iteration loss.

We combine this joint construction with a weighted planar incidence estimate, a resolution-limited cellwise angular selection, a labelled slab estimate, and a global synchronization of tubelet and fine-tube mass. The resulting rectangular parent estimate has shading exponent $2-\sigma$, which is sharp uniformly over the class considered here. With the stated GWZ factoring data and corresponding $K_F(\beta)$ or $K_{KT}(\beta)$, we obtain the two small-middle interfaces used in the reduction. For the very-non-sticky branch, assuming both $K_{KT}(\beta)$ and $K_F(\beta)$, the stated GWZ Lemma 9.1 hypotheses together with the source-uniformity and refinement interface (I1)–(I4) yield the terminal gain for the represented fine-tube input; this conditional application supplies the corresponding input to the surrounding reduction, while the sharp-convex-parent problem lies outside its scope.

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Notation and conventions. For a finite labelled family $\mathcal{F}$ of measurable carriers of positive finite volume, $|\mathcal{F}|$ denotes its cardinality; two labels may have the same geometric carrier. A *shading* is an assignment of a measurable set $Y(F) \subset F$ to each label $F \in \mathcal{F}$. We use

$$\begin{aligned} U(\mathcal{F},Y) &:= \bigcup_{F\in\mathcal{F}} Y(F), \qquad M(Y) := \sum_{F\in\mathcal{F}} |Y(F)|, \\ D(\mathcal{F}) &:= \sum_{F\in\mathcal{F}} |F|, \qquad \lambda(\mathcal{F},Y) := \frac{M(Y)}{D(\mathcal{F})}, \qquad \mu(\mathcal{F},Y) := \frac{M(Y)}{|U(\mathcal{F},Y)|}. \end{aligned} \tag{1.1}$$

The last two ratios are set to zero when their denominators vanish. Thus M, D, and μ count coincident carriers with multiplicity, while U is their physical union. For a measurable set E, let $N_a(E) = \{x : \operatorname{dist}(x,E) < a\}$. Let $\mathfrak{K}_n$ denote the class of measurable convex sets $K \subset \mathbb{R}^n$ with $0 < |K| < \infty$. For $K \in \mathfrak{K}_n$, define

$$\Delta(\mathcal{F},K) := \frac{\sum_{F\in\mathcal{F}:\,F\subset K} |F|}{|K|}, \qquad \Delta_{\max}(\mathcal{F}) := \sup_{K\in\mathfrak{K}_n} \Delta(\mathcal{F},K). \tag{1.2}$$

If every member of $\mathcal{F}$ lies in $K_0 \in \mathfrak{K}_n$, define

$$C_F(\mathcal{F},K_0) := \sup_{\substack{K\in\mathfrak{K}_n \\ K\subset K_0}} \frac{\Delta(\mathcal{F},K)}{\Delta(\mathcal{F},K_0)}. \tag{1.3}$$

We set $\Delta_{\max}(\varnothing) = 0$, and set $C_F(\mathcal{F}, K_0) = 0$ when $\Delta(\mathcal{F}, K_0) = 0$ (in particular for the empty family). Thus when one ambient body K_0 has been fixed by the surrounding statement, we abbreviate $C_F(\mathcal{F}) := C_F(\mathcal{F}, K_0)$; no ambient body is suppressed when more than one is in play. Thus neither a null or lower-dimensional convex set nor an infinite-volume set is ever an admissible test body, and no ratio in (1.2)–(1.3) is undefined. All sums in (1.1)–(1.3) are over labels. We call the closed $c_{\mathrm{cap}}w$-neighbourhood of a carrier segment of length exactly one a *source-standard tube of width* w; both endpoint caps are included. The unit-scale tubes and their cross-scale ancestors in the direct source interfaces of Sections 9–10 are source-standard. A local $w \times w \times r$ tubelet, on the other hand, may have carrier length in $[c_{\mathrm{len}}r, C_{\mathrm{len}}r]$; this is the only place where "comparable length" is used. The four constants $c_{\mathrm{cap}}, C_{\mathrm{cap}}, c_{\mathrm{len}}, C_{\mathrm{len}}$ are fixed once and for all. Every containment statement includes the caps. A *fixed enlargement* means a dilation by a constant depending only on dimension and these four structural constants. It may change the cap constant, but it does not change or merge the labelled carrier. Thus every use of $O(1)$, "comparable", or a fixed enlargement in a tube, plank, or slab interface has the same declared structural dependence. We write $A \lesssim_{\log} B$ when $A \le C[\log(2/\delta)]^C B$ for a structural exponent, and write $A = \delta^{o(1)}B$ only when a displayed uniform function $\omega(\delta) \to 0$ gives $\delta^{\omega(\delta)}B \le A \le \delta^{-\omega(\delta)}B$. For an explicitly named secondary scale $s \in \{\alpha, \rho, r_1\}$, the notation $s^{o(1)}$ has the identical meaning with one displayed uniform function $\omega_s(s) \to 0$; the polynomial scale windows in the surrounding statement are used whenever it is converted to a power of δ. For results that absorb source/refinement losses into terminal powers—in particular propositions 9.6 and 9.7 and the terminal results in Section 10—a *loss profile* $\mathfrak{p}$ consists of the fixed polynomial-complexity, bounded-fibre, and structural constants together with fixed moduli controlling every source/refinement occurrence of $o(1)$, including ω_0, ω_{br}, and the derived envelopes ω_{ref} and ω_{src}. The profile is fixed before any cutoff δ_0 is chosen. Thus $\delta_0 = \delta_0(\beta, \zeta, \mathfrak{p})$ may depend on these fixed convergence rates and constants, while the positive gain exponents below remain functions only of the displayed geometric parameters. Every fixed δ-indexed input class with common subpower bounds determines such a profile by taking the finite maximum of its named moduli. No cutoff uniform over all possible rates $\omega(\delta) \to 0$ is asserted. A *child point* always means a point of the currently displayed child union $U(\mathcal{T}, Y)$; an unrestricted ambient point is called simply a point. We denote a structural dilation of a carrier by $C_{\mathrm{car}}F$ and a structural enlargement of an ambient ball by $C_{\mathrm{amb}}B$. Thus source carrier separation is never conflated with overlap of tagged ambient balls. When all tubes in a displayed family are congruent, $|T|$ denotes their common geometric volume; $|\mathbb{T}|$ always denotes labelled cardinality. Whenever an ambient ball is attached to a label, the construction is performed first in the tagged disjoint union (3.4b); physical recombination occurs only through lemma 3.5.

1. Introduction

The central question of this paper is one of simultaneous control. Suppose short shaded tubes are organized through larger convex parents. Can the parent output retain the child mass, regularize both the inner and outer multiplicities, satisfy the required density estimate, and remain stable under the geometric selections that follow? In the streamlined three-dimensional Kakeya reduction of Guth–Wang–Zahl [GWZ26], built on the convex-set framework of Wang–Zahl [WZ25b, WZ26, WZ25a], all four demands occur in the same factorization. Treating them separately loses the compatibility needed at the next scale.

The difficulty is not merely that a refinement may lose mass. Parent mass and child mass live on different objects, and the usual conclusions are not monotone under arbitrary deletion. Nearby child support does not determine the multiplicity of a clipped parent neighbourhood at a different point; after a further child selection, constant multiplicity and exact induced

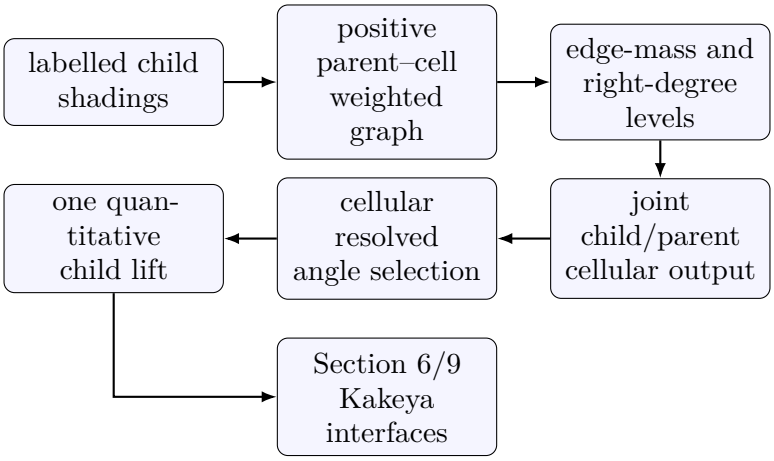


FIGURE 1. The cellular mechanism. After regularization, every admissible selection acts on complete positive parent–cell incidences.

compatibility may both fail. The finite configurations in section 2 isolate these mechanisms. Their role is to identify the structure that a stable factorization must retain.

Our organizing principle is to make the incidence, rather than either shading alone, the indivisible object. Fix a half-open grid of side $h = c_0 a$, where a is the shortest parent scale. A labelled parent B and a cell Q are joined precisely when the children assigned to B carry positive shaded mass in Q. We regularize first the positive edge weights and then the cell degrees. The surviving edges define the child shading by restriction and the parent shading by whole cells. Thus mass retention, inner multiplicity, parent multiplicity, compatibility, and the pointwise product bound all come from one finite weighted graph.

The same representation makes subsequent refinement quantitative. An admissible parent refinement is a subcollection of complete parent–cell edges. Equal cell volumes convert retention of labelled parent mass into retention of edge count, while comparable positive edge weights convert that edge count back into child mass. In particular, the typical-angle descent from [GWZ26] may be run once per cell, with deterministic tie-breaking, so that its output is again an edge subset and requires only one lift. At the angular resolution a/b, parallel labels are distinguished through the resolution-floored angular separation

$$\operatorname{angle}_{a/b}(B, B') = \max\{a/b, \operatorname{angle}(TB, TB')\}.$$

The lower cutoff records the geometric resolution of the parent family.

Two further ideas connect the cellular theorem to the multiscale application. A tubelet-level refinement need not preserve mass at the level of the represented fine tubes, so we choose one global fine-multiplicity level after the coarse tubelet levels are fixed. This makes cellular coarse mass quantitatively equivalent to fine incidence mass. The angular selection may also lower parent degree by A_{ang}^{-K}; to meet the next lemma's output-degree hypothesis, we split at the buffered threshold

$$H_{\mathrm{ang}} := 4A_{\mathrm{ang}}^{K}, \qquad \alpha := a/r_1.$$

When $d_0 \geq H_{\mathrm{ang}}\alpha^{-\eta_{\mathrm{pl}}}$, the high branch enters the plank-reduction lemma; the complementary range $d_0 < H_{\mathrm{ang}}\alpha^{-\eta_{\mathrm{pl}}}$ falls within a direct low-degree estimate. The buffer is what prevents an uncovered middle band.

The density exponent comes from a separate analytic layer. The planar incidence theorem of [DOV22], transferred through a weighted layer-cake argument, accommodates repeated projected parameters and measurable shadings. It yields the rectangular shading power $\lambda^{2-\sigma}$;

the ordinary Frostman endpoint is supplied by the quadratic Córdoba method [Cór77]. Our contribution is the joint cellular interface in which these estimates remain compatible with mass lifting, together with the two small-middle applications and the conditional very-non-sticky closure under (I1)–(I4).

Main results and architecture. The method has an abstract core and a Euclidean realization. The atomic theorem, theorem 5.6, applies to an arbitrary finite measurable partition and uses neither a metric nor convexity. Its density-equipped Euclidean companion, theorem 8.2, starts from labelled children in comparable rectangular parents of polynomially bounded complexity. Both theorems produce compatible child and parent shadings from a single positive edge set. Schematically,

$$\begin{aligned} &M(Y_1) \geq \mathfrak{L}^{-1} M(Y), \\ &\mu_{\mathrm{child}} \sim \mu_{\mathrm{in}}, \qquad \mu_{\mathrm{parent}} \sim d_0, \\ &\mu_{\mathrm{total}} \lesssim \mu_{\mathrm{in}} d_0. \end{aligned}$$

Every permitted later parent refinement then lifts quantitatively to the children. Under the biased complete-child hypothesis, the Euclidean output also satisfies

$$\lambda(\widetilde{\mathbb{B}}', Z) \gtrsim C_0^{-(2-\sigma)} (a/b)^{\varepsilon} \kappa_D^{-(2-\sigma)} \mathfrak{L}^{-(2-\sigma)} \lambda(\mathbb{T}, Y)^{2-\sigma}.$$

The Kakeya application is Corollary 10.21. It assumes $K_{KT}(\beta)$, $K_F(\beta)$, the uniform bounds in (10.33), and the conditional source interface (I1)–(I4) of proposition 10.19. Once tubelet and outer-ball parameters have been synchronized locally and across whole tags, these hypotheses close the very-non-sticky branch on the represented original fine-tube input. The cellular theorems themselves are unconditional; only this multiscale application depends on (I1)–(I4).

Sections 2–8 develop the reusable theorem. Section 2 derives the necessary stability requirements from finite obstructions. Sections 3–5 build the positive parent–cell graph, regularize its two multiplicity scales, prove whole-incidence lifting, and isolate the atomic core. Sections 6–7 add the rectangular density estimate and the cellwise angular protocol; Section 8 assembles them. Sections 9–10 then apply this machinery to the two small-middle factorizations and to the very-non-sticky reduction. Section 11 proves uniform sharpness of the density exponent, and Section 12 identifies the sharp-convex boundary of the method. Appendices A–D contain the weighted planar transfer, the labelled slab estimate, the multiset filling lemma, and the quantitative source map.

Relation to previous work and scope. Grid cellularization, dyadic graph regularization, the DOV exponent, and the ordinary typical-angle descent are established ingredients. The point here is their organization around a positive parent–cell edge set whose child and parent outputs survive the same downstream selections. Cohen's notes [Coh26] develop a related but different submodular/global decomposition, while Guth's survey [Gut26] places the surrounding reduction in a broader context. Both serve as comparison sources for the present construction.

Zeraoulia Rafik's note [Zer26] contains a broader set of source-level bookkeeping observations and an independent Córdoba-type proof of the shaded slab estimate. The genuine overlap is the slab L^2 calculation in section B. The positive parent–cell construction, whole-incidence lifting, and tagged synchronization are developed independently here.

The local note [Zer26b] identifies that an unrestricted C_F condition is stronger than the all-convex definition (1.3) supports. Accordingly, the application uses the restricted source slab-test condition required by GWZ Lemma 6.1; Proposition 9.7 and its conclusion are unchanged.

The rectangular scope is structural. A comparable outer box preserves the labelled counting conditions needed in the WZ application, but its union lower bound need not restrict uniformly to an arbitrary sharp convex parent. We therefore keep rectangular parents and verify the fixed-dilation downstream lemmas in that category.

2. Obstructions to non-cellular refinement

Before constructing a stable factorization, we isolate two tests that any candidate must pass. The first asks whether nearby child support determines the multiplicity of a clipped parent shading. The second asks whether compatibility and multiplicity survive a refinement made after the parent conclusions have been established. These are independent failures of monotonicity, and the cellular construction will answer them by different features: exact cell ownership for the first, and indivisible incidences for the second. For the first test, let

$$\mathbb{V}' = \bigsqcup_{W \in \mathbb{W}'} \mathbb{V}'_W$$

and let Y be a child shading. The induced parent shading in [GWZ26] has the form

$$Z(W) = W \cap N_a\big(U(\mathbb{V}'_W, Y')\big)\,. \tag{2.1}$$

Suppose one controls, on child points x, the nearby-support quantity

$$\widetilde{\mu}(x) = \#\{W : B(x,a) \cap U(\mathbb{V}'_W, Y') \neq \varnothing\}. \tag{2.2}$$

The point of the example is that (2.2) and the multiplicity of (2.1) are different functions: the former sees proximity to child support, while the latter also sees membership in the individual parent.

Proposition 2.1 (Nearby support does not regularize the induced parent)**.** *For every $M \geq 4$, there are M congruent labelled rectangular parents, one shaded child in each parent, and a common level on which $\widetilde{\mu} = M$, while the multiplicity of the induced parent shading assumes both values 1 and M.*

Proof. Fix $a > 0$, put $d = a/(4M)$, and let $c_j = jd$ for $0 \leq j < M$. Choose congruent labelled parents

$$W_j = [c_j - a/2, c_j + a/2] \times [-b/2, b/2] \times [-r/2, r/2], \qquad a \leq b \leq r.$$

Inside W_j, take the congruent convex child

$$V_j = [c_j - a/4, c_j + a/4] \times [-b/4, b/4] \times [-r/4, r/4]$$

and shade it only on

$$S_j = B((c_j, 0, 0), d/10).$$

The S_j are pairwise disjoint. At every child point, the inner and outer multiplicities equal one, while all M support balls lie within distance a. Hence $\widetilde{\mu} = M$ on the complete child union.

The singleton family $\{V_j\}$ is 8-Frostman in W_j: a convex test body either contains no child or has volume at least $|V_j| = |W_j|/8$. The exact induced shading is $Z_j = W_j \cap N_a(S_j)$. The point

$$y_{\text{core}} = ((c_0 + c_{M-1})/2, 0, 0)$$

belongs to all Z_j. In contrast,

$$y_{\text{edge}} = (c_0 - a/2 + d/4, 0, 0)$$

belongs to Z_0 and lies strictly to the left of every W_j, $j \geq 1$. Thus the induced multiplicity is M at the core and 1 at the edge. □

Remark 2.2. The mismatch comes from clipping by W. Nearby child support gives $x \in N_a(U(\mathbb{V}'_W, Y'))$, whereas parent membership at y also requires $y \in W$. An average multiplicity estimate cannot replace the pointwise conclusion needed later.

Proposition 2.3 (A final child refinement is not automatically harmless)**.** *Each of the following monotonicity assertions fails for a finite labelled configuration:*

(N1) *constant inner multiplicity survives an arbitrary child deletion;*

(N2) *the induced parent shading is unchanged after such a deletion;*
(N3) *constant parent multiplicity survives when the induced shadings are recomputed;*
(N4) *the pointwise inner–outer multiplicity product survives; and*
(N5) *retaining a fixed fraction of labelled parent volume retains a fixed fraction of child shading mass.*

In particular, none of these properties can be transferred through a final "mass-preserving refinement" without an additional structural hypothesis.

Proof. All examples use only two measurable pieces.

For inner multiplicity, take two identical child shadings on a set A. Delete the second child on a positive-measure subset $A_0 \subset A$. The new inner multiplicity is one on A_0 and two on $A \setminus A_0$, which proves (N1). If the two children have one common parent whose shading is all of A, the outer multiplicity stays one and the inner–outer product takes the values one and two. This also proves (N4).

For exact parent shading, let one child shading have separated pieces $A_1 \cup A_2$. Keeping only A_1 changes

$$W \cap N_a(A_1 \cup A_2) \quad \text{to} \quad W \cap N_a(A_1).$$

Retaining the old parent shading instead breaks the exact induced identity, which proves (N2). Here is a literal two-parent incidence table that also proves (N3). Take disjoint a-neighbourhoods of A_1, A_2, and take two labelled parents W_1, W_2 containing both neighbourhoods. Initially give each parent one child shading $A_1 \cup A_2$. Refine only the W_1-child to A_1, leave the W_2-child unchanged, and recompute the induced parent shadings. On the two positive-measure atoms the complete before/after table is

	$Y^{\mathrm{in}}_{W_1}$	$Y^{\mathrm{in}}_{W_2}$	$Z^{\mathrm{in}}(W_1)$	$Z^{\mathrm{in}}(W_2)$	$Y^{\mathrm{out}}_{W_1}$	$Y^{\mathrm{out}}_{W_2}$	$Z^{\mathrm{out}}(W_1)$	$Z^{\mathrm{out}}(W_2)$
A_1	1	1	1	1	1	1	1	1
A_2	1	1	1	1	0	1	0	1

(2.3a)

where a Z-entry records membership in the exact recomputed set $W_j \cap N_a(Y_{W_j})$. Thus the induced shading of W_1 genuinely changes, and labelled parent multiplicity changes from two on both atoms to two on A_1 and one on A_2. No informal transfer between different pieces is being used.

Finally, give two parent pieces equal volume but child masses ε and 1. A parent selection retaining the first piece keeps half the labelled parent mass and only $\varepsilon/(1+\varepsilon)$ of child mass. This tends to zero. This proves (N5). A joint construction is therefore necessary. □

Proposition 2.4 (Failure of parentwise restoration after angular refinement)**.** *Even the following finite matched-output interface does not imply compatibility after the whole old child shading of each retained parent label is restored: exact child–parent incidence inclusion, constant inner multiplicity in each labelled parent, constant total parent multiplicity, common labelled parent-mass and local-union-volume levels, and the pointwise angle-class rule followed by a global labelled-parent-mass pigeonhole used in* [GWZ26, Proposition 5.1, p. 12 and Lemma 6.11]*. These are precisely the displayed properties used at the tangential lift. Thus the implication between source equations (95)–(96) and (100)–(101) is not derivable from that displayed interface alone.*

Proof. Fix $0 < \vartheta < 2^{-8}$, put $x_A = (0, -1/4, 0)$, $x_B = (0, 1/4, 0)$, and choose $0 < r_0 \ll \delta$. Let $A = B(x_A, r_0)$ and $B_* = B(x_B, r_0)$. Take three labelled rectangular parents whose labelled tangent planes are

$$P_1 = \mathrm{span}(e_2, e_3), \quad P_2 = \mathrm{span}(\cos(16\vartheta)e_2 + \sin(16\vartheta)e_1, e_3),$$
$$P_3 = \mathrm{span}(\cos(\vartheta)e_2 + \sin(\vartheta)e_1, e_3). \tag{2.4a}$$

The first parent is long enough in its e_2-direction to meet both balls; the second and third are translated to meet A and B_*, respectively. Thus the separation of the two centres is parallel to the long e_2-axis of the first parent. This coordinate correction changes neither the three planes in (2.4a), nor the incidence table below, nor any of the pairwise angle values. Choose all three shortest widths much larger than r_0. Vertical δ-tubes through the two centres then contain the indicated balls and are contained in the corresponding parents. Use one labelled child of W_1 shaded by A, a second shaded by B_*, one child of W_2 shaded by A, and one child of W_3 shaded by B_*. Empty portions of the geometric tubes play no role.

On the two positive-measure atoms the complete incidence table is

	Y_{W_1}	Y_{W_2}	Y_{W_3}	$Z(W_1)$	$Z(W_2)$	$Z(W_3)$
A	1	1	0	1	1	0
B_*	1	0	1	1	0	1

(2.4b)

where each entry means that the corresponding shading equals the entire atom. Thus the inner multiplicity in every labelled parent is one, the parent multiplicity is exactly two on both atoms, the labelled parent masses of the two atoms agree, and parent-scale local union volumes agree. These are the matched-output properties used at the later lift. If a literal neighbourhood model is desired, first replace A, B_* by still smaller concentric balls and take the displayed sets as measurable subshadings of the induced clipped neighbourhoods; the table is unchanged.

At A the only parent pair has angle 16ϑ, whereas at B_* the only pair has angle ϑ. These lie in disjoint dyadic classes. The legal global angle-class pigeonhole that keeps the 16ϑ-class therefore retains precisely the two A-incidences, half of the labelled parent mass, and constant parent multiplicity two. It deletes the B_*-part of $Z(W_1)$. Restoring the whole old child shading for the retained label W_1 restores its B_*-child, although $B_* \cap Z_{\mathrm{ang}}(W_1) = \varnothing$. Hence the asserted incidence inclusion fails on all of B_*.

The construction has a precise logical scope: the displayed implication is not a consequence of the listed interface alone. It makes no assertion about Proposition 5.1, the typical-angle lemma, or Main Lemma 2 of [GWZ26]. What it identifies is the missing invariant. In theorem 5.1, parent and child shadings are restricted on the same cells, so restoration is replaced by a quantitative lift along surviving edges. □

The three propositions point to one common design rule. A parent shading should not be reconstructed after a child selection, and a child shading should not be restored after a parent selection. Both must instead be read from a shared incidence object. Sections 3–5 build that object and prove the exact stability statements suggested by these examples.

3. Cellular parent shadings and tagged geometry

The shared object has four layers. A child belongs to a sharp parent; the parent is assigned a permanent fixed enlargement; space is partitioned into half-open cells; and positive child mass turns a parent–cell pair into a weighted edge. The parent and child shadings will both be recovered from those edges. Half-open cells make boundary ownership unique, while tagged copies prevent local constructions in different ambient balls from being identified before their multiplicities have been compared.

Fix $0 < c_0 \le 10^{-3}$, set $h = c_0 a$, and let

$$\mathcal{Q}_h = \{h(k + [0,1)^3) : k \in \mathbb{Z}^3\}$$

be the global half-open grid. Every point belongs to exactly one cell. For $E \subset \mathbb{R}^3$, define

$$\mathcal{C}_h(E) = \bigcup_{\substack{Q \in \mathcal{Q}_h \\ Q \cap E \neq \varnothing}} Q. \tag{3.1}$$

Lemma 3.1 (Cellular neighbourhood comparison)**.** *For every $E \subset \mathbb{R}^3$,*

$$E \subset \mathcal{C}_h(E) \subset N_{\sqrt{3}h}(E). \tag{3.2}$$

If $h \asymp a$, then

$$|N_a(E)| \lesssim |\mathcal{C}_h(E)| \lesssim |N_{Ca}(E)|, \tag{3.3}$$

with constants depending only on the comparison between h and a.

Proof. The unique cell containing $x \in E$ appears in (3.1). If y lies in an active cell, it is within that cell's diameter $\sqrt{3}h$ of a point of E, proving (3.2).

For the nontrivial inequality in (3.3), let $\mathcal{A} = \{Q : Q \cap E \neq \varnothing\}$. Then

$$N_a(E) \subset \bigcup_{Q \in \mathcal{A}} N_{a+\sqrt{3}h}(Q).$$

Each last neighbourhood meets only $O(1)$ grid cells when $h \asymp a$. Summing first over an increasing finite exhaustion $\mathcal{A}_1 \subset \mathcal{A}_2 \subset \cdots$ gives the same bound with a constant independent of the exhaustion. The corresponding unions increase to the full union; continuity from below (equivalently, monotone convergence applied to their indicators) therefore gives the result. In every later application $\mathcal{A}$ is finite, but this last sentence makes the stated arbitrary-set version literal. □

Lemma 3.2 (Parent boundary)**.** *Let B be an arbitrarily oriented $a \times b \times r$ rectangular box, $a \le b \le r$, and let $E \subset B$. There is a fixed enlargement $\widetilde{B}$ with side lengths comparable to a, b, r such that*

$$\mathcal{C}_h(E) \subset \widetilde{B}, \qquad |\widetilde{B}| \asymp |B|. \tag{3.4}$$

Proof. If e_1, e_2, e_3 are the side directions and x_B is the centre, take

$$\widetilde{B} = \left\{ x_B + \sum_{j=1}^{3} s_j e_j : |s_j| \le \tfrac{1}{2}(a,b,r)_j + \sqrt{3}h \right\}.$$

Every active cell point is within $\sqrt{3}h$ of B, and $h = c_0 a \le c_0 b \le c_0 r$. This proves both assertions. □

Lemma 3.3 (Convex parallel bodies)**.** *Let $K \subset \mathbb{R}^n$ be convex and suppose $B(z,r) \subset K$, where $r > 0$. For every $\lambda \ge 0$,*

$$K + B(0, \lambda r) \subset z + (1+\lambda)(K - z). \tag{3.4p}$$

Consequently

$$|K + B(0,\lambda r)| \le (1+\lambda)^n |K|. \tag{3.4q}$$

Both conclusions remain valid with closed or open balls and are insensitive to boundary points.

Proof. The case $\lambda = 0$ is immediate. If $x \in K$ and $|y| \le \lambda r$, then $z + y/\lambda \in B(z,r) \subset K$. Convexity gives

$$\frac{x - z}{1+\lambda} + \frac{\lambda}{1+\lambda}\frac{y}{\lambda} \in K - z.$$

Multiplication by $1+\lambda$ proves (3.4p), and the determinant of the homothety gives (3.4q). The open-ball case follows by approximation. This is the elementary parallel-body inclusion used below; for standard convex body conventions and Minkowski addition see [Sch14, Chs. 1 and 3]. □

To make later angular statements unambiguous, every labelled rectangular parent is henceforth supplied with an ordered orthonormal side frame $(e_a(B), e_b(B), e_r(B))$, corresponding to side lengths $a \le b \le r$. The frame is part of the label when side lengths coincide. We write

$$TB := \operatorname{span}\{e_b(B), e_r(B)\}, \tag{3.4a}$$

and use the following unoriented plane metric. If P, P' are two-planes with arbitrarily chosen unit normals $n_P, n_{P'}$, set

$$d_{\mathrm{pl}}(P, P') := \arccos |\langle n_P, n_{P'}\rangle| \in [0, \pi/2]. \tag{3.4a$'$}$$

This is independent of the signs of the normals and is the standard metric on the projective plane $\mathbb{RP}^2$. In particular it satisfies the triangle inequality. We use its capped version

$$\operatorname{angle}(P, P') := \min\{1, d_{\mathrm{pl}}(P, P')\}. \tag{3.4a$''$}$$

The map $t \mapsto \min\{1, t\}$ is increasing and subadditive, so angle is again a metric; it is comparable to d_{pl} and takes values in $[0, 1]$. All plane diameters and plane caps below, including those in Appendix C, use this metric. Thus TB in (7.1) is defined even for repeated geometric boxes or equal side lengths.

Remark 3.4 (Labels and ambient balls). Coincident geometric boxes with different labels remain different parents and different summands. A fixed enlargement may leave an ambient ball that the original parent touches; every application below correspondingly works in a fixed dilation of that ball. We never restrict the enlarged conclusion back to a sharp parent boundary.

The next convention prevents the carrier from changing type during a refinement. A parent label used below is the triple

$$\mathbf{B} = (B^\sharp, \widetilde{B}, \mathfrak{e}_B), \tag{3.4d}$$

where $B^\sharp$ is the sharp box to which the children were assigned, $\widetilde{B}$ is the fixed enlargement in lemma 3.2, and $\mathfrak{e}_B$ is the ordered frame in (3.4a). We continue to write B for the label, but a parent shading $Z(B)$ is, by definition, a measurable subset of the permanent carrier $\widetilde{B}$, never a shading clipped to $B^\sharp$. Conditions concerning child containment or complete-child density refer to $B^\sharp$; parent volumes in the cellular output refer to $\widetilde{B}$. Since their volumes are comparable, this convention costs only the fixed constant in (3.4), while preventing a later change of type.

Lemma 3.5 (Tagged-cell recombination). *Let $\mathcal{A}$ be a finite set of tags. For each $A \in \mathcal{A}$, fix a bounded measurable support region $A^+ \subset \mathbb{R}^3$, and assume that the family $(A^+)_{A\in\mathcal{A}}$ has overlap at most κ_{amb}. Work first in the disjoint union*

$$X^{\mathrm{tag}} = \bigsqcup_{A\in\mathcal{A}} \{A\} \times \mathbb{R}^3. \tag{3.4b}$$

Give each tagged component its own full copy (A, Q) of every half-open grid cell Q; only the finitely many cells meeting A^+ can be active. Then every tagged point belongs to exactly one tagged cell, so all degree and product identities of theorem 4.2 are literal in X^{tag}. If $E_A \subset A^+$ are measurable and $\pi(A, x) = x$, then

$$|\pi(\bigsqcup_A \{A\} \times E_A)| \le \sum_A |E_A| \le \kappa_{\mathrm{amb}} |\pi(\bigsqcup_A \{A\} \times E_A)|. \tag{3.4c}$$

Moreover, if a tagged pointwise multiplicity is at most M in every component, its recombined physical multiplicity is at most $\kappa_{\mathrm{amb}} M$. Thus projection back to physical space costs only the displayed overlap constant; it never turns a tagged identity into an unrecorded identity in physical space.

Proof. Uniqueness of (A, Q) follows from uniqueness of Q inside each tagged copy. The first inequality in (3.4c) is subadditivity. For the second, write $N(x) = \#\{A : x \in E_A\}$. Then

$$\sum_A |E_A| = \int_{\cup_A E_A} N(x)\,dx \le \kappa_{\mathrm{amb}} |\cup_A E_A|.$$

The multiplicity assertion is the same calculation pointwise: at most κ_{amb} tagged components contribute at one physical point. □

3.1. **The positive parent–cell graph.** We now pass from geometry to a finite weighted graph. Let $\mathbb{B}$ be a finite nonempty labelled family of permanent rectangular parents as in (3.4d), and let

$$\mathbb{T} = \bigsqcup_{B\in\mathbb{B}} \mathbb{T}_B$$

be a finite indexed family. Every child $T \in \mathbb{T}_B$ has the unique labelled parent B, is contained in $B^\sharp$, and has a measurable shading $Y(T) \subset T$. Write

$$M(Y) = \sum_B \sum_{T\in\mathbb{T}_B} |Y(T)|, \qquad f_B^Y(x) = \sum_{T\in\mathbb{T}_B} \mathbf{1}_{Y(T)}(x). \tag{3.5}$$

Assume throughout Sections 3–5 that $0 < M(Y) < \infty$. In particular $N_{\max} \ge 1$, the selected inner mass is positive, and the active positive-edge graph below is nonempty. We also fix the global denominator and shading density notation

$$D(\mathbb{T}) = \sum_B \sum_{T\in\mathbb{T}_B} |T|, \qquad \lambda(\mathbb{T}, Y) = \frac{M(Y)}{D(\mathbb{T})}. \tag{3.5a}$$

After the common inner-level selection of section 4, define

$$\omega_{BQ} = \int_Q f_B^{Y_0}(x)\,dx = \sum_{T\in\mathbb{T}_B} |Y_0(T) \cap Q|. \tag{3.6}$$

The active graph is

$$E^+ = \{(B, Q) : \omega_{BQ} > 0\}. \tag{3.7}$$

Only positive weights define edges. This exclusion is structural: a positive-measure shading in one cell together with an isolated point in another would otherwise create a geometric touching edge carrying no child mass, and such an edge could survive into the parent output without anything to lift.

4. Positive-incidence regularization

The geometric input is now a finite weighted bipartite graph. Its left vertices are labelled parents, its right vertices are cells, and its edge weights are child masses. We select a single subgraph in three steps: one common inner-multiplicity level, one positive edge-weight level, and one saturated right-degree level. Saturation means that every edge at a retained right vertex is kept, so the degree selected before the output is still the degree of the output. All families remain indexed; coincident labels are never merged. Let

$$N_{\max} = \max_B |\mathbb{T}_B|.$$

4.1. **The common inner level.** For $0 \le k \le \lceil \log_2 N_{\max} \rceil$, set

$$X_{B,k} = \{x : 2^k \le f_B^Y(x) < 2^{k+1}\}.$$

Since

$$M(Y) = \sum_k \sum_B \int_{X_{B,k}} f_B^Y,$$

one value k_0 carries at least the reciprocal of

$$L_{\text{in}} = 1 + \lceil \log_2 N_{\max} \rceil \tag{4.1}$$

of the mass. Put $\mu_{\text{in}} = 2^{k_0}$ and

$$Y_0(T) = Y(T) \cap X_{B,k_0}, \qquad T \in \mathbb{T}_B. \tag{4.2}$$

At a fixed pair (B, x), (4.2) keeps all or none of the B-child incidences. Hence

$$M(Y_0) \ge L_{\text{in}}^{-1} M(Y), \qquad \mu_{\text{in}} \le f_B^{Y_0}(x) < 2\mu_{\text{in}} \tag{4.3}$$

on every nonempty retained inner union.

4.2. **Edge weights and right degrees.**

Lemma 4.1 (Positive edge-weight regularization)**.** *Let E be a finite nonempty edge set with weights $\omega_e > 0$, and write $\Omega = \sum_E \omega_e$, $m = |E|$. There are $E_1 \subset E$ and $\omega_0 > 0$ such that*

$$\omega_0 \le \omega_e < 2\omega_0 \quad (e \in E_1) \tag{4.4}$$

and

$$\sum_{e \in E_1} \omega_e \ge \frac{\Omega}{2(1 + \lceil \log_2(2m) \rceil)}. \tag{4.5}$$

Proof. Delete weights below $\Omega/(2m)$; their sum is below $\Omega/2$. The remaining ratio between the largest and smallest weights is at most $2m$. The heaviest dyadic class proves both assertions. □

Apply lemma 4.1 to the active graph E^+ in (3.7). For a right cell incident to E_1, write

$$d_1(Q) = \#\{B : (B, Q) \in E_1\}.$$

Select a dyadic value d_0 whose cell class carries the greatest E_1-weight, and retain *all* E_1-edges at those cells:

$$E_2 = \{(B, Q) \in E_1 : d_0 \le d_1(Q) < 2d_0\}. \tag{4.6}$$

Because every E_1-weight is comparable, the loss in (4.6) is at most

$$L_{\text{deg}} = 2(1 + \lceil \log_2 |E_1| \rceil). \tag{4.7}$$

The saturation in (4.6) is important: the degree inside E_2 is still $d_1(Q)$.

Define

$$L_{J1} := L_{\text{in}}, \qquad L_{R1} := 2(1 + \lceil \log_2(2|E^+|) \rceil), \qquad L_{R2} := L_{\text{deg}}, \tag{4.8a}$$

and

$$L_{\text{reg}} := L_{J1} L_{R1} L_{R2}, \qquad \mathfrak{L} := L_{\text{reg}}. \tag{4.8}$$

Thus (J1) means the whole-incidence inner multiplicity level, (R1) the positive edge-weight level of lemma 4.1, and (R2) the saturated right-degree level (4.6). Their inputs, retained factors, and outputs are exactly (4.1)–(4.7); the symbols are not placeholders for further refinements. Then

$$\sum_{(B,Q) \in E_2} \omega_{BQ} \ge \mathfrak{L}^{-1} M(Y). \tag{4.9}$$

4.3. **Reading both shadings from one graph.** Put

$$\mathbb{B}' = \{B : \text{there exists } Q \text{ with } (B, Q) \in E_2\}. \tag{4.9a}$$

For $B \in \mathbb{B}'$ put

$$\mathcal{Q}_B = \{Q : (B, Q) \in E_2\},$$

and define

$$Y_1(T) = Y_0(T) \cap \bigcup_{Q \in \mathcal{Q}_B} Q, \quad T \in \mathbb{T}_B, \qquad Z(B) = \bigcup_{Q \in \mathcal{Q}_B} Q. \tag{4.10}$$

For $B \notin \mathbb{B}'$, set $\mathcal{Q}_B = \varnothing$, $Z(B) = \varnothing$, and $Y_1(T) = \varnothing$ for $T \in \mathbb{T}_B$. Thus all output shadings are total functions on the original labelled families. Every selected edge has positive weight, so

$$\boxed{Z(B) = \mathcal{C}_h(U(\mathbb{T}_B, Y_1)).} \tag{4.11}$$

This identity is the first payoff of the construction: the parent shading is exactly the cellularization of its retained child union, with no separate metric reconstruction.

Theorem 4.2 (Cellular Joint Factoring)**.** *The output in (4.10) has the following properties.*

(P1) $M(Y_1) \geq \mathfrak{L}^{-1} M(Y)$.
(P2) *If* $x \in Y_1(T)$ *and* $T \in \mathbb{T}_B$*, then* $x \in Z(B)$.
(P3) *On* $U(\mathbb{T}_B, Y_1)$*,*

$$\mu_{\text{in}} \leq \sum_{T \in \mathbb{T}_B} \mathbf{1}_{Y_1(T)}(x) < 2\mu_{\text{in}}.$$

(P4) *On* $U(\mathbb{B}, Z)$*,*

$$d_0 \leq \sum_{B} \mathbf{1}_{Z(B)}(x) < 2d_0.$$

(P5) *At every child point,*

$$\sum_{T \in \mathbb{T}} \mathbf{1}_{Y_1(T)}(x) < 4\mu_{\text{in}} d_0.$$

Moreover $Z(B)$ *is contained in a labelled fixed enlargement* $\widetilde{B}$ *with the same side-length scales.*

Proof. Disjointness of half-open cells and (3.6) give

$$M(Y_1) = \sum_{(B,Q) \in E_2} \omega_{BQ},$$

so (P1) is (4.9). The unique cell of a point in $Y_1(T)$ is selected for the parent B, proving (P2).

For (P3), an edge (B, Q) retains every B-child incidence inside Q. Thus $f_B^{Y_1}(x) = f_B^{Y_0}(x)$ on the final inner union, and (4.3) applies. For (P4), the unique cell $Q(x)$ of a parent point satisfies

$$\sum_{B} \mathbf{1}_{Z(B)}(x) = \#\{B : (B, Q(x)) \in E_2\},$$

which lies in $[d_0, 2d_0)$ by (4.6). Finally,

$$\begin{aligned} \sum_{T \in \mathbb{T}} \mathbf{1}_{Y_1(T)}(x) &= \sum_{B : x \in U(\mathbb{T}_B, Y_1)} \sum_{T \in \mathbb{T}_B} \mathbf{1}_{Y_1(T)}(x) \\ &< 2\mu_{\text{in}} \#\{B : x \in Z(B)\} < 4\mu_{\text{in}} d_0. \end{aligned}$$

This proves (P5). The enlargement statement is lemma 3.2. □

Corollary 4.3 (Average multiplicity product)**.** *Let $\mathbb{B}'$ be the retained parent family and put*

$$\mu_{\text{child}} := \max_{B\in\mathbb{B}'} \mu(\mathbb{T}_B, Y_1).$$

Then

$$\mu(\mathbb{T}, Y) \lesssim \mathfrak{L}\,\mu(\mathbb{B}', Z)\,\mu_{\text{child}}, \tag{4.12}$$

where $\mu(\mathcal{F}, Y) = M(Y)/|U(\mathcal{F}, Y)|$.

Proof. Every retained parent $B \in \mathbb{B}'$ is incident to at least one edge of E_2. That edge has positive weight, so $U(\mathbb{T}_B, Y_1) \neq \varnothing$; in particular the local average multiplicity used below has a nonzero denominator. Integrating (P3)–(P5) gives, uniformly for $B \in \mathbb{B}'$, $\mu(\mathbb{T}_B, Y_1) \sim \mu_{\text{in}}$, while $\mu(\mathbb{B}', Z) \sim d_0$, and $\mu(\mathbb{T}, Y_1) \lesssim \mu_{\text{in}} d_0$. Since $M(Y_1) \geq \mathfrak{L}^{-1} M(Y)$ and $U(\mathbb{T}, Y_1) \subset U(\mathbb{T}, Y)$, $\mu(\mathbb{T}, Y) \leq \mathfrak{L}\mu(\mathbb{T}, Y_1)$. □

5. Refinement by whole parent–cell incidences

Joint factoring would be of limited use if it were destroyed by the next geometric selection. The relevant stability statement is therefore a lifting theorem. A later step may discard parent incidences, but it must do so edge by edge. Equal cell volumes turn retained labelled parent mass into retained edge count, and the regularized positive weights turn retained edge count into retained child mass. No geometric hypothesis beyond this atomic structure is needed.

Theorem 5.1 (Whole-incidence lifting)**.** *Let $0 < c \leq 1$, let $E^\star \subset E_2$, and define*

$$Z^\star(B) = \bigcup_{Q:(B,Q)\in E^\star} Q. \tag{5.1}$$

Suppose the labelled parent mass satisfies

$$\sum_B |Z^\star(B)| \geq c \sum_B |Z(B)|. \tag{5.2}$$

For $T \in \mathbb{T}_B$, set

$$Y^\star(T) = Y_1(T) \cap \bigcup_{Q:(B,Q)\in E^\star} Q. \tag{5.3}$$

Then

$$M(Y^\star) \geq \frac{c}{2} M(Y_1). \tag{5.4}$$

The common inner multiplicity, exact cellular compatibility, and the original pointwise product upper bound survive.

Proof. Each labelled parent–cell edge contributes exactly h^3 to the parent sum, so (5.2) implies $|E^\star| \geq c|E_2|$. Also

$$M(Y^\star) = \sum_{e\in E^\star} \omega_e \geq \omega_0 |E^\star|, \qquad M(Y_1) < 2\omega_0 |E_2|,$$

which proves (5.4). A selected edge keeps every child incidence of its parent in that cell. Therefore

$$Z^\star(B) = \mathcal{C}_h(U(\mathbb{T}_B, Y^\star))$$

and the inner multiplicity is unchanged on the new union. If $d^\star(Q)$ is the new parent degree, then

$$\sum_T \mathbf{1}_{Y^\star(T)}(x) \leq 2\mu_{\text{in}} d^\star(Q(x)) \leq 4\mu_{\text{in}} d_0.$$

□

Corollary 5.2 (Restoring a constant refined parent degree)**.** *Choose a dyadic right-degree class of $E^\star$ having maximal edge count, keep every edge at each cell in that class, let $Y^{\star\star}$ be the resulting child shading, and put*

$$L_\star := 1 + \lceil \log_2 |E^\star| \rceil.$$

Then the output again has constant parent degree and the complete comparison chain is

$$M(Y^{\star\star}) \geq \frac{1}{2L_\star} M(Y^\star) \geq \frac{c}{4L_\star} M(Y_1) \geq \frac{c}{4L_\star \mathfrak{L}} M(Y). \tag{5.5}$$

Here Y is the input to theorem 4.2, Y_1 is its output, $Y^\star$ is the immediate output of theorem 5.1, and c is exactly the parent-mass fraction in (5.2), measured relative to Z. Thus the lift supplies $c/2$, the subsequent degree regularization supplies $(2L_\star)^{-1}$, and the earlier joint construction supplies $\mathfrak{L}^{-1}$; none of these baselines is implicit.

Proof. Pigeonhole the positive values of $d^\star(Q) = \#\{B : (B, Q) \in E^\star\}$. The selected degree class retains the reciprocal of the $L_\star$ classes in edge count. Edge weights remain in $[\omega_0, 2\omega_0)$, so this converts to $M(Y^{\star\star}) \geq (2L_\star)^{-1} M(Y^\star)$. Combine theorem 5.1 with (P1) of theorem 4.2 to obtain the remaining two inequalities in (5.5). □

Remark 5.3 (Three refinement types). Deleting whole parent labels, deleting whole right cells, and deleting individual parent–cell incidences are all instances of $E^\star \subset E_2$. In the third case one usually applies corollary 5.2. No operation may delete only selected child labels inside a retained (B, Q) edge.

Proposition 5.4 (The cellular restriction is sharp)**.** *A positive-volume measurable parent refinement that cuts through cells need not retain any child mass.*

Proof. In one cell Q, support all child mass on a measurable half Q_+. Retain the disjoint half Q_- as the parent refinement. It keeps half the parent volume and zero child mass. □

Remark 5.5 (Iteration budget). For m successive whole-incidence lifts with parent fractions c_j,

$$M(Y^{(m)}) \geq 2^{-m} \prod_{j=1}^{m} c_j \, M(Y^{(0)}), \tag{5.5a}$$

before any degree re-regularizations. Thus a fixed number of controlled lifts is safe. Arbitrarily many harmless lifts are not: retaining half the edges at each stage leaves 2^{-m} of the mass.

5.1. **The geometry-free core.** The proof above only used Euclidean cubes to obtain equal-volume atoms and to reconstruct a geometric carrier. The regularization and lifting themselves see only a finite measurable partition, positive edge weights, and comparable atom measures. The following theorem records this geometry-free core and makes the finite-iteration loss explicit.

Theorem 5.6 (Atomic cellular factoring and lifting)**.** *Let $(X, \mathfrak{m})$ be a measure space. Let $X_0 \subset X$ be measurable, let $\mathcal{Q}$ be a finite measurable partition of X_0, and let $\mathcal{B}$ be a finite nonempty labelled parent set. For each $B \in \mathcal{B}$, let $\mathcal{T}_B$ be a finite labelled child set with measurable shadings $Y(T) \subset X_0$, and assume*

$$0 < M(Y) := \sum_B \sum_{T \in \mathcal{T}_B} \mathfrak{m}(Y(T)) < \infty, \qquad N_{\max} := \max_B |\mathcal{T}_B| < \infty. \tag{5.6}$$

Repeated geometric realizations, if any, are counted by labels. Put

$$f_B(x) = \sum_{T \in \mathcal{T}_B} \mathbf{1}_{Y(T)}(x).$$

Then there are $k_0 \in \{0, \dots, \lceil \log_2 N_{\max} \rceil\}$*, the common inner level* $\mu_{\text{in}} = 2^{k_0}$*, and the restricted shading*

$$X_{B,k_0} := \{x : 2^{k_0} \le f_B(x) < 2^{k_0+1}\}, \qquad Y_0(T) := Y(T) \cap X_{B,k_0} \quad (T \in \mathcal{T}_B). \tag{5.6a}$$

There are also a positive edge set $E_2 \subset \mathcal{B} \times \mathcal{Q}$*, and numbers* $\omega_0, d_0 > 0$ *with the following properties. If*

$$\begin{aligned} Y_1(T) &= Y_0(T) \cap \bigcup_{Q:(B,Q)\in E_2} Q \quad (T \in \mathcal{T}_B), \\ Z(B) &= \bigcup_{Q:(B,Q)\in E_2} Q, \end{aligned} \tag{5.7}$$

then

$$M(Y_1) \ge L_{\text{at}}^{-1} M(Y), \qquad L_{\text{at}} \le 4(1 + \lceil \log_2 N_{\max} \rceil)(1 + \lceil \log_2(2e_+) \rceil)(1 + \lceil \log_2 e_+ \rceil), \tag{5.8}$$

where e_+ *is the number of positive parent–atom incidences after the inner-level selection. Moreover:*

(A1) $Y_1(T) \subset Z(B)$ *for* $T \in \mathcal{T}_B$*, and*

$$Z(B) = \bigcup \{Q : \mathfrak{m}(Q \cap U(\mathcal{T}_B, Y_1)) > 0\}; \tag{5.9}$$

(A2) *on* $U(\mathcal{T}_B, Y_1)$*, the* B*-child multiplicity lies in* $[\mu_{\text{in}}, 2\mu_{\text{in}})$*;*
(A3) *on* $\bigcup_B Z(B)$*, the labelled parent multiplicity lies in* $[d_0, 2d_0)$*;*
(A4) *everywhere, the total child multiplicity is below* $4\mu_{\text{in}} d_0$*.*

Assume additionally that every atom occurring in E_2 *has measure in* $[q_0, \kappa_Q q_0]$*, with* $q_0 > 0$*. Let*

$$E^{(m)} \subset \cdots \subset E^{(1)} \subset E^{(0)} = E_2$$

be whole-incidence refinements, and define

$$Y^{(j)}(T) := Y_0(T) \cap \bigcup_{Q:(B,Q)\in E^{(j)}} Q, \qquad Z^{(j)}(B) := \bigcup_{Q:(B,Q)\in E^{(j)}} Q. \tag{5.9a}$$

Suppose

$$\sum_B \mathfrak{m}(Z^{(j)}(B)) \ge c_j \sum_B \mathfrak{m}(Z^{(j-1)}(B)), \qquad 0 < c_j \le 1. \tag{5.10}$$

Then

$$M(Y^{(m)}) \ge L_{\text{at}}^{-1} \prod_{j=1}^{m} \frac{c_j}{2\kappa_Q} M(Y). \tag{5.11}$$

At every stage of (5.9a), the common inner multiplicity, and the original total multiplicity upper bound remain valid. A constant refined parent degree can be restored by one saturated dyadic right-degree selection, exactly as in corollary 5.2. No topology, metric, convexity, or equal-volume assumption is used beyond the stated comparability of atom measures in the lifting part.

Proof. Because $\mathcal{Q}$ partitions X_0 and every input shading is contained in X_0, the atoms cover all of the mass in (5.6). In particular, after the inner-level selection, $\sum_{B,Q} \omega_{BQ}$ is exactly the retained labelled mass rather than the mass of only an unspecified part of X. Select the level k_0 of f_B simultaneously over all parents, keeping at least $(1 + \lceil \log_2 N_{\max} \rceil)^{-1}$ of $M(Y)$. For the restricted shading Y_0 in (5.6a), define

$$\omega_{BQ} := \sum_{T \in \mathcal{T}_B} \mathfrak{m}(Y_0(T) \cap Q), \qquad E^+ := \{(B, Q) : \omega_{BQ} > 0\}.$$

Apply lemma 4.1 to these positive weights and then retain a saturated dyadic right-degree class. The calculation (4.4)–(4.9) is purely a weighted bipartite-graph calculation and gives (5.8), with $\omega_e \in [\omega_0, 2\omega_0)$ on E_2. Disjointness of the atoms replaces disjointness of half-open cubes in (4.10).

Keeping every child incidence of B in a retained atom proves (5.9) and preserves the selected inner level. The right-degree class proves (A3), and summing at most $2\mu_{\mathrm{in}}$ children over fewer than $2d_0$ retained parents proves (A4).

For the lifting assertion, (5.10) and atom comparability imply

$$|E^{(j)}| \geq (c_j/\kappa_Q)|E^{(j-1)}|.$$

Since every surviving edge still has its original weight in $[\omega_0, 2\omega_0)$,

$$M(Y^{(j)}) \geq \omega_0|E^{(j)}| \geq \frac{c_j}{2\kappa_Q} M(Y^{(j-1)}).$$

Iteration and (5.8) give (5.11). All remaining assertions follow because an edge is always kept or deleted together with every child incidence it represents. □

6. Rectangular density estimates from planar incidences

The graph mechanism controls survival and compatibility, not the size of the parent union. The latter is an analytic question. We obtain the required rectangular density exponent by projecting the children to a planar line parameter space, applying a weighted form of the incidence estimate in [DOV22], and reconstructing volume inside the parent. Appendix A proves the weighted formulation needed for repeated parameters and arbitrary measurable shadings.

6.1. **Weighted planar input.** Let $1 < t \leq 2$, set $q = 3 - t$, and let $0 < h \leq 1$ and $L \geq 1$. In a bounded slope–intercept chart, suppose

$$\nu(B(p,\rho)) \leq L\rho^t, \qquad h \leq \rho \leq 1, \tag{6.1}$$

for an atomic probability measure $\nu = \sum_i \alpha_i \delta_{p_i}$. Repetitions among the p_i are allowed. If Y_i is an arbitrary measurable shading of the corresponding h-tube and

$$\Lambda = h^{-1} \sum_i \alpha_i |Y_i|,$$

then the weighted layer-cake corollary of DOV is

$$\left| \bigcup_i Y_i \right| \gtrsim_{\varepsilon,t} L^{-q} h^{\varepsilon} \Lambda^q. \tag{6.2}$$

The proof of (6.2) in section A uses two layer-cake reductions: the first removes atomic weights and the second removes spatial occupancy weights. The logarithmic costs are absorbed into h^{ε}. In particular, neither a minimum atom mass nor a bound on projection-fibre size is part of the input.

6.2. **From planar parameters back to a rectangular parent.** Let B be a labelled $a \times b \times r$ parent, $\delta \leq a \leq b \leq r$, containing $N \geq 1$ labelled $\delta \times \delta \times r$ children. Suppose that for every convex $K \subset B$,

$$\frac{\#\{T : T \subset K\}}{N} \leq C_0 \left(\frac{|K|}{|B|} \right)^{1+\sigma}, \qquad 0 < \sigma \leq 1. \tag{6.3}$$

Write $s = a/b$, $q = 2 - \sigma$, and let

$$\lambda_B = \frac{\sum_{T \in \mathbb{T}_B} |Y(T)|}{\sum_{T \in \mathbb{T}_B} |T|}.$$

Theorem 6.1 (Rectangular biased shading bound)**.** *For every $\varepsilon > 0$,*

$$|B \cap N_a(U(\mathbb{T}_B, Y))| \gtrsim_{\varepsilon,\sigma} C_0^{-q} s^{\varepsilon} \lambda_B^q |B|. \tag{6.4}$$

For every $\eta > 0$, before epsilon absorption the scale factor may be written (with an implicit constant also depending on η)

$$C_0^{-q} [\log(2/s)]^{-q} s^{q\eta} \lambda_B^q. \tag{6.5}$$

The proof has five geometric bridges. We choose finitely many normalized line charts, convert parameter balls to convex slabs, transfer the complete-child hypothesis to a Frostman estimate on parameters, thicken the projected shading without losing boundary mass, and finally reconstruct three-dimensional volume inside the parent. The intervening capped-carrier lemmas ensure that the same charts remain valid across the coarse ancestry used later. Once the structural tube and parent conventions are fixed, all constants are uniform.

Lemma 6.2 (Finite line-parameter charts)**.** *After a rigid motion, write the side coordinates of B as (x_1, x_2, x_3), with side lengths $a \le b \le r$. Let $0 < w \le a$, and let the children be labelled $w \times w \times r$ tubes contained in B. They are the disjoint union of at most three labelled subfamilies. On each nonempty subfamily there are distinct indices j, k, ℓ such that the carrier is a bounded graph over x_j. The capped carrier length is in $[c_{\mathrm{len}} r, C_{\mathrm{len}} r]$, with fixed structural constants, and the graph contains a relevant x_j-interval of length at least cr and at most Cr. If a cap meets a parent face, this interval is chosen one-sided from that cap. More precisely, the relevant interval is the x_j-projection of a carrier subsegment fixed inside one structural enlargement of B; the enlargement and the constants c, C are the same for all three charts and all labels. Every resulting affine graph line is extended to one common fixed bounded normalized u-interval; the physical carrier and its shading use only their relevant subinterval. Moreover,*

$$s_j \asymp r, \qquad s_k \asymp b, \qquad s_\ell \asymp a.$$

Under

$$S_{jk}(x_k, x_j) = (x_k/s_k, x_j/s_j) =: (v, u),$$

the projected carrier has equation $v = \beta_T + m_T u$, with $p(T) = (\beta_T, m_T)$ in one fixed bounded slope–intercept chart. Its normalized projected solid lies in a fixed-constant h_0-tube about that line, where

$$h_0 = w/s_k, \qquad h = a/s_k \asymp a/b, \qquad h_0 \le h \le 1.$$

Proof. Let $d(T) = \sum_i d_i e_i$ be the carrier direction and assign T to the least index j for which $|d_j| = \max_i |d_i|$. Then $|d_j| \ge 3^{-1/2}$. By the capped-tube convention, the carrier has length comparable to r and lies in the tube; hence a segment of length at least cr lies in a fixed enlargement of B. If a cap meets a face of B, take this segment one-sided from that endcap. Projecting its two endpoints to the i-th side gives

$$cr|d_i| \le Cs_i + Cw \le C' s_i, \tag{6.3c}$$

because $w \le a \le s_i$. Hence $s_j \ge c'r$, while $s_j \le r$, so $s_j \asymp r$. If $j = 3$, choose $k = 2, \ell = 1$. If $j = 2$, then $b = s_j \asymp r$, and choose $k = 3, \ell = 1$. If $j = 1$, then $a = s_j \asymp r$, hence all three side lengths are comparable and either remaining order has the asserted sizes. This proves uniformly that $s_j \asymp r, s_k \asymp b, s_\ell \asymp a$; it is the coordinate permutation omitted by the informal dominant-direction argument.

Since $|d_j| \ge 3^{-1/2}$, the carrier is a graph over the full relevant x_j-interval. The normalized slope

$$m_T = \frac{s_j d_k}{s_k d_j}$$

is bounded by (6.3c), and the normalized intercept is bounded because the carrier meets B. Normalized endpoint ordinates and (β_T, m_T) differ by a fixed invertible linear map. The

orthogonal solid radius contributes $O(w/s_k)$ after projection. A cap is contained in an $O(w)$-ball about its carrier endpoint and contributes the same amount, so neither changes the chart nor its width. Labels are never merged. The original fine-child application is $w = \delta$; the coarse application below is $w = \rho$. □

Lemma 6.3 (A parameter ball lifts to one convex slab)**.** *Fix a width w and a chart from lemma 6.2. For $h \le u \le 1$ and every parameter ball $D(p_0, u)$, there is a convex set $K_u \subset B$ which contains every complete child in the chart whose parameter lies in that ball, and*

$$\frac{|K_u|}{|B|} \lesssim u.$$

Proof. Let $v = L_0(u)$ represent p_0. In physical coordinates take

$$K_u = B \cap \{x : |x_k - s_k L_0(x_j/s_j)| \le C(us_k + w)\}.$$

This is the intersection of a box with two affine half-spaces. Bounded slope–intercept distance controls the extended graph on the common normalized interval fixed in lemma 6.2, hence also on every relevant carrier subinterval; Cw includes the solid radius and caps. Thus

$$|K_u| \lesssim s_\ell s_j (us_k + w).$$

Because $w/s_k \le h \le u$ and $|B| \asymp s_\ell s_j s_k$, the result follows. The restriction is one slab, so the relative volume is $O(u)$, not $O(u^2)$. □

Lemma 6.4 (Weighted marginal Frostman estimate)**.** *Let a nonempty chart contain $N_c = \vartheta N$ labelled children, counted with repetitions, and put*

$$\nu_c = \frac{1}{N_c} \sum_{T \text{ in the chart}} \delta_{p(T)}.$$

Then, for $h \le \rho \le 1$,

$$\nu_c(D(p, \rho)) \lesssim C_0 \vartheta^{-1} \rho^{1+\sigma}.$$

If instead the parent family obeys the ordinary complete-child condition

$$\frac{\#\{T : T \subset K\}}{N} \le C_F \frac{|K|}{|B|} \quad (K \subset B \text{ convex}), \tag{6.3a}$$

then the same argument gives

$$\nu_c(D(p, \rho)) \lesssim C_F \vartheta^{-1} \rho. \tag{6.3b}$$

No bound on the number of labels with the same parameter is required.

Proof. Every label counted on the left is a complete child of K_ρ. Hence (6.3) and lemma 6.3 give

$$\nu_c(D(p, \rho)) \le \frac{N}{N_c} C_0 \left(\frac{|K_\rho|}{|B|}\right)^{1+\sigma} \lesssim C_0 \vartheta^{-1} \rho^{1+\sigma}.$$

This estimates total atomic weight and is unchanged by repeated parameters or unbounded projection fibres. Under (6.3a), replace the power $1 + \sigma$ in the same calculation by one to obtain (6.3b). □

Lemma 6.5 (Clipped one-dimensional thickening)**.** *Let I be an interval with $c_I \le |I| \le C_I$, $0 < h_0 \le h \le 1$, and let $E \subset I$ be measurable with* $\operatorname{diam}(E) \le C_0 h_0$. *Fix $0 < c < c_I/4$. Then*

$$|(E + [-ch, ch]) \cap I| \ge c_1(c, C_0) \frac{h}{h_0} |E|. \tag{6.3d}$$

The assertion remains true for almost every section of a Borel measurable planar representative.

Proof. The statement is trivial when $|E| = 0$. Otherwise choose a density point $x \in E$. Since $ch \le c < c_I/4 \le |I|/4$, at least one of the two one-sided intervals of length ch from x lies in I. Thus the left side is at least ch. On the other hand $|E| \le \operatorname{diam}(E) \le C_0 h_0$, so $ch \ge (c/C_0)(h/h_0)|E|$. Choose a Borel representative of a planar measurable set. Fubini gives measurable sections for almost every transverse parameter; applying the one-dimensional estimate to those sections and then Cavalieri proves the final assertion. Completion of the measure handles a general Lebesgue-measurable representative. □

Lemma 6.6 (Projection of an arbitrary measurable shading)**.** *For a child T in a fixed chart, let*

$$A_T = S_{jk}(\pi_{jk} Y(T)),$$

where π_{jk} forgets x_ℓ. There is a measurable shading Z_T of a fixed-constant h-tube about $p(T)$, contained in the normalized parent projection $R_{jk} := S_{jk}(\pi_{jk} B)$, such that

$$\frac{|Z_T|}{h} \gtrsim \frac{|Y(T)|}{|T|}.$$

Consequently, if a chart carries at least J^{-1} of the total shaded mass and $N_c = \vartheta N$, then

$$h^{-1} N_c^{-1} \sum_T |Z_T| \gtrsim \frac{\lambda_B}{J\vartheta}.$$

Proof. The normalized parent projection R_{jk} is a fixed-comparable rectangle, so its transverse sections have one structural length $c_I \le |I| \le C_I$. Fix once and for all a structural $c \in (0, c_I/4)$, sufficiently small for the fibre reconstruction in lemma 6.10; every Z_T below uses this same c. If $|Y(T)| = 0$, take $Z_T = \varnothing$. Otherwise, after replacing the shading by a Borel representative modulo a null set, inner regularity gives a compact $K_T \subset Y(T)$ with $|K_T| \ge \frac{1}{2}|Y(T)|$. We perform the construction on K_T. Its projection is compact, its transverse Minkowski sum is compact, and its intersection with the parent rectangle is Borel; hence the resulting Z_T is measurable. The factor $1/2$ is absorbed in the structural implicit constant. Since $|d_j| \gtrsim 1$, every fibre of π_{jk} meets T in an x_ℓ-interval of length $O(\delta)$. Fubini and the planar Jacobian give

$$|A_T| \gtrsim \frac{|Y(T)|}{\delta s_j s_k}.$$

Every u-slice of A_T has diameter $O(h_0)$. Define the *parent-clipped* transverse thickening

$$Z_T = (A_T + [-ch, ch]e_v) \cap R_{jk}.$$

By lemma 6.5, including slices meeting either transverse face, and $h/h_0 = a/\delta$,

$$|Z_T| \gtrsim (h/h_0)|A_T|, \qquad \frac{|Z_T|}{h} \gtrsim \frac{|Y(T)|}{\delta^2 s_j} \asymp \frac{|Y(T)|}{|T|}.$$

Summing proves the second assertion. The clipping is essential for the in-parent fibre reconstruction below. □

Lemma 6.7 (Two-point comparison for capped carriers)**.** *Let $T = N_{c_T w}(I_T)$ and $S = N_{c_S \rho}(I_S)$ be capped solid tubes whose carrier lengths lie in fixed intervals comparable to r. Assume $T \subset S$ and $0 < w \le \rho \le c_{\mathrm{sep}} r$, where $c_{\mathrm{sep}} > 0$ is sufficiently small in terms of the fixed tube constants. Then there are $x_1, x_2 \in I_T$ and $y_1, y_2 \in I_S$ such that*

$$|x_1 - x_2| \ge cr, \qquad |x_i - y_i| \le C\rho, \qquad |y_1 - y_2| \ge c'r. \tag{6.6C}$$

Consequently the unoriented carrier directions differ by at most $C\rho/r$, and a point of I_T has distance $O(\rho)$ from I_S. The assertion includes containment through either endcap.

Proof. Choose two carrier points of I_T, one-sided if necessary, separated by a fixed fraction of its length. Since every carrier point of T belongs to the solid capped tube S, for each x_i there is a nearest $y_i \in I_S$ with $|x_i - y_i| \le C\rho$. The triangle inequality gives

$$|y_1 - y_2| \ge |x_1 - x_2| - |x_1 - y_1| - |x_2 - y_2| \ge cr - 2C\rho \ge c'r.$$

Thus $y_2 - y_1$ is a nonzero multiple of the coarse carrier direction, whereas $x_2 - x_1$ is a nonzero multiple of the fine direction. Comparing the two normalized secants gives the $O(\rho/r)$ angular error. If a nearest point is a carrier endpoint, the same calculation is one-sided and unchanged; no continuation beyond a cap is used. □

Lemma 6.8 (Hausdorff control for standardized capped ancestors)**.** *Let*

$$T = N_{c_T w}(I), \qquad S = N_{c_S \rho}(J), \qquad 0 < w \le \rho \le L,$$

where the two carrier segments have the same length $|I| = |J| = L$, and suppose $T \subset S$. Then

$$d_{\mathrm{H}}(I, J) \le C\rho, \qquad S \subset N_{C\rho}(T). \tag{6.6D}$$

The constant depends only on the fixed cap constants. In particular the conclusion applies to the source-standard fine/coarse ancestry in proposition 9.7; it is not asserted for arbitrary comparable-length local tubelets.

Proof. Let p_-, p_+ be the endpoints of I. Containment of the carrier in the solid S gives points $q_-, q_+ \in J$ with $|p_\pm - q_\pm| \le C_0\rho$. If $\rho \le L/(4C_0)$, then

$$|q_+ - q_-| \ge L - 2C_0\rho.$$

Writing $J = \{q_0 + tu : 0 \le t \le L\}$, the two parameters of q_- and q_+ therefore differ by at least $L - 2C_0\rho$. After possibly reversing the order on J, each endpoint of J is within $2C_0\rho$ of the corresponding $q_\pm$, and hence within $3C_0\rho$ of the corresponding endpoint of I. Linear interpolation between the paired endpoints gives both inclusions $I \subset N_{C\rho}(J)$ and $J \subset N_{C\rho}(I)$. If $\rho > L/(4C_0)$, choose any point of I and a nearest point of J; the diameter bound $L \lesssim \rho$ gives the same two inclusions with a larger structural constant. Finally every point of S is within $c_S\rho$ of J, while $I \subset T$, which proves the second assertion. Endpoint caps are already included in the two neighbourhoods, so no extension of either carrier is used. □

Lemma 6.9 (Proxy-chart inheritance)**.** *There is an absolute constant $0 < c_{\mathrm{dom}} \le c_{\mathrm{sep}}$, determined only by the finite charts in lemma 6.2 and the fixed cap/length constants in the carrier convention, with the following property. Let $\delta \le \rho \le a \le b \le r$, assume*

$$\rho/r \le c_{\mathrm{dom}}, \tag{6.6c‴}$$

and let a fine $\delta \times \delta \times r$ tube T be contained in a coarse $\rho \times \rho \times r$ tube S, and S belongs to one of the charts in lemma 6.2, then, after a fixed enlargement of that chart, T is a graph in the same dominant coordinate and

$$|p(T) - p(S)| \lesssim \rho/b \le a/b. \tag{6.6c″}$$

Consequently every projected measurable shading of T constructed in lemma 6.6 is contained in a fixed enlargement of the a/b-tube belonging to the proxy atom $p(S)$.

Proof. Lemma 6.7 gives directional separation $O(\rho/r)$ and transverse offset $O(\rho)$, including both endcaps. In the normalized coordinates of lemma 6.2, the slope error is $O((r/b)(\rho/r)) = O(\rho/b)$, and the intercept error is also $O(\rho/b)$. The coarse dominant component is bounded below by a fixed chart constant. Choose c_{dom} so that the directional error $C\rho/r$ is at most half that lower bound. The fine dominant component is then bounded below by half the same constant, so the same graph coordinate applies. Finally $\rho/b \le a/b$, and the projected fine shading already lies in an $O(a/b)$-thickening of its fine carrier. The triangle inequality proves the last assertion. □

Lemma 6.10 (Fibre reconstruction inside the parent)**.** *For the projected shadings in lemma 6.6,*

$$|B \cap N_a(\bigcup_T Y(T))| \gtrsim |B| \left|\bigcup_T Z_T\right|,$$

where planar measure is in normalized coordinates. The constant is uniform over all charts and over shadings meeting a face of B.

Proof. If $(v', u) \in Z_T$, then for some $(x_\ell, x_k, x_j) \in Y(T)$, the corresponding physical observed coordinate satisfies $|x'_k - x_k| \le ca$. The clipping keeps x'_k in the parent projection. Inside the x_ℓ-side of B, the interval

$$\{x'_\ell : |x'_\ell - x_\ell| \le ca\}$$

has length at least ca, even at a face. Choosing c small puts every (x'_ℓ, x'_k, x_j) in $B \cap N_a(Y(T))$. Therefore Fubini and $|\det S_{jk}^{-1}| = s_j s_k$ yield

$$|B \cap N_a(\bigcup_T Y(T))| \gtrsim a s_j s_k \left|\bigcup_T Z_T\right| \asymp |B| \left|\bigcup_T Z_T\right|.$$

□

Proof of theorem 6.1. The calculation now follows the projection–incidence–reconstruction chain. If $\lambda_B = 0$, the right side of (6.4) vanishes and the conclusion is immediate. Hence assume $\lambda_B > 0$. Choose a chart carrying at least J^{-1} of the total shaded mass, where $J \le 3$, and write $N_c = \vartheta N$. Lemmas 6.4 and 6.6 give Frostman constant $L \lesssim C_0/\vartheta$ and planar density at least $c\lambda_B/(J\vartheta)$. Apply (6.2) with $t = 1 + \sigma$, $q = 2 - \sigma$, and $h \asymp a/b$. The chart fraction cancels exactly:

$$\left|\bigcup_T Z_T\right| \gtrsim_{\varepsilon,\sigma} (C_0/\vartheta)^{-q} h^\varepsilon (\lambda_B/(J\vartheta))^q \gtrsim C_0^{-q} h^\varepsilon \lambda_B^q.$$

Lemma 6.10 now gives (6.4). In the unabsorbed planar estimate the right side is

$$C_0^{-q} [\log(2/h)]^{-q} h^{q\eta} \lambda_B^q.$$

This is (6.5), since $h \asymp s = a/b$. Taking $\eta = \varepsilon/(2q)$ and using

$$[\log(2/s)]^{-q} s^{\varepsilon/2} \gtrsim_{\varepsilon,q} s^\varepsilon$$

recovers (6.4). The compact approximation is removed by letting its mass loss tend to zero. All counts remain labelled, so repeated children and repeated projected parameters are included. □

Corollary 6.11 (Ordinary rectangular endpoint)**.** *Assume (6.3a). Then*

$$|B \cap N_a(U(\mathbb{T}_B, Y))| \gtrsim C_F^{-1} [\log(2/s)]^{-1} \lambda_B^2 |B|. \tag{6.6}$$

This statement allows labelled repetitions, coincident projected parameters, and arbitrary measurable shadings.

Proof. Repeat the chart selection, projection, parent clipping, and fibre reconstruction in the proof of theorem 6.1. The chart marginal now satisfies (6.3b). Apply the weighted labelled Córdoba estimate theorem A.5 with $L \lesssim C_F \vartheta^{-1}$ and planar density $\Lambda \gtrsim \lambda_B/(J\vartheta)$. Its lower bound is

$$(C_F \vartheta^{-1})^{-1} [\log(2/h)]^{-1} (\lambda_B/(J\vartheta))^2 = C_F^{-1} [\log(2/h)]^{-1} \lambda_B^2 (J^2\vartheta)^{-1}.$$

Since $0 < \vartheta \le 1$ and $J \le 3$, $(J^2\vartheta)^{-1} \ge J^{-2} \gtrsim 1$. Notice that the chart was selected by shaded mass, so no lower bound for its fraction of labels is being used. The parent-clipped fibre reconstruction contributes only the fixed chart Jacobian, and $h \asymp s$. This proves (6.6). In particular, the endpoint is not obtained by formally setting $t = 1$ in DOV. □

Lemma 6.12 (Coarse/thickened-child Córdoba endpoint)**.** *Let $\delta \le \rho \le a \le b \le r$, and let B be an $a \times b \times r$ labelled rectangular parent. Suppose a nonempty labelled family $\mathbb{S}_B$ of $\rho \times \rho \times r$ coarse tubes is contained in B, and a nonempty fine family $\mathbb{T}_B$ is given, with each fine child $T \in \mathbb{T}_B$ assigned to one $S(T) \in \mathbb{S}_B$ with $T \subset S(T)$. The fine and coarse carriers use one common capped length normalization at scale r: their carrier lengths lie in the same fixed interval $[c_{\mathrm{len}} r, C_{\mathrm{len}} r]$, with the same endpoint convention. Thus all ancestry comparisons below use one declared length window rather than unrelated "comparable length" constants. Suppose there are $m \ge 1$ and $R_{\mathrm{br}} \ge 1$ such that every coarse label S has n_S assigned fine labels with*

$$m \le n_S \le R_{\mathrm{br}} m. \tag{6.6c$'$}$$

Assume

$$\frac{\#\{S \in \mathbb{S}_B : S \subset K\}}{|\mathbb{S}_B|} \le C_{F,\rho} \frac{|K|}{|B|} \qquad (K \subset B \text{ convex}). \tag{6.6c}$$

Then arbitrary measurable fine shadings, repeated fine labels, and coincident fine carriers satisfy

$$|B \cap N_a(U(\mathbb{T}_B, Y))| \gtrsim C_{F,\rho}^{-1} R_{\mathrm{br}}^{-3} [\log(2b/a)]^{-1} \lambda_B^2 |B|. \tag{6.6d}$$

The conclusion asserts no fine-scale all-convex Frostman condition.

Proof. Put $h = a/b$, and fix a sufficiently small absolute $h_* > 0$. If $h \ge h_*$, partition the fine labels themselves by lemma 6.2 with $w = \delta$, and choose a chart carrying at least one third of the fine shaded mass. For the normalized equal-weight measure on that chart, $\nu(D(p,u)) \le 1 \le h_*^{-1} u$ for $h \le u \le 1$. The weighted labelled Córdoba endpoint theorem A.5 and fine fibre reconstruction give $|B \cap N_a(U)| \gtrsim \lambda_B^2 |B|$. Since $C_{F,\rho}, R_{\mathrm{br}} \ge 1$ and $\log(2b/a) = O_{h_*}(1)$, this is stronger than (6.6d).

Assume henceforth that $h < h_*$, where at the outset we also choose $h_* \le c_{\mathrm{dom}}$. Since $\rho \le a$ and $b \le r$,

$$\rho/r \le a/b = h < h_* \le c_{\mathrm{dom}}.$$

Thus lemma 6.9 is applicable. Partition the coarse labels into the at most three graph charts of lemma 6.2 with $w = \rho$, assigning all labelled descendants of a coarse label to its chart. Choose a chart carrying at least one third of the total fine shaded mass. If it contains N_c coarse labels and $N_f = \sum_S n_S$ fine labels, put equal mass $1/N_f$ on each descendant label, but place that atom at the parameter $p(S(T))$ of its coarse proxy.

For $h \le u \le 1$, the parameter-slab construction lemma 6.3 with transverse padding $C(ub + \rho)$ contains every coarse proxy whose parameter lies in a u-ball. Its relative volume is $O(u)$, because $\rho \le a = hb \le ub$. Applying (6.6c), and then dividing by the coarse chart fraction, proves the one-dimensional Frostman estimate for the repeated proxy measure with constant

$$L \lesssim C_{F,\rho} R_{\mathrm{br}} \vartheta^{-1}, \tag{6.6e}$$

where $\vartheta = N_c/|\mathbb{S}_B|$. Indeed a coarse label has proxy mass $n_S/N_f \le R_{\mathrm{br}}/N_c$. This is the only use of branching comparability.

By lemma 6.9, the projected shading Z_T constructed in lemma 6.6 is a shading of a fixed enlargement of the h-tube belonging to the proxy atom $p(S(T))$. This verifies the chart and carrier inheritance for every assigned fine label; no fine shading mass is merged or replaced. Since $N_f \le R_{\mathrm{br}} m N_c$, whereas the full family has at least $m|\mathbb{S}_B|$ descendants, selecting the chart by shaded mass gives planar density

$$\Lambda \gtrsim \lambda_B/(R_{\mathrm{br}} \vartheta). \tag{6.6f}$$

Apply the weighted labelled Córdoba endpoint theorem A.5 to these repeated proxy atoms. Combining (6.6e)–(6.6f) gives

$$L^{-1} \Lambda^2 \gtrsim C_{F,\rho}^{-1} R_{\mathrm{br}}^{-3} \vartheta^{-1} \lambda_B^2 \ge C_{F,\rho}^{-1} R_{\mathrm{br}}^{-3} \lambda_B^2.$$

Finally, lemma 6.10 applies to the original fine projected shadings, so parent clipping and boundary faces cost only a fixed constant. This gives (6.6d). □

6.3. **Density after joint regularization.** The rectangular estimate is parentwise, whereas the joint graph retains mass globally across parents. The next averaging lemma is the bridge: comparable child denominators convert the global retained-mass statement into a lower bound for the mean surviving density, after which Jensen supplies the parent union estimate.

Lemma 6.13 (Denominator-comparability averaging)**.** *Let $\mathbb{B}$ be a finite nonempty parent family, let $D_B > 0$, let $m_B \ge 0$ with $\sum_B m_B > 0$, and suppose*

$$\frac{\max_{B\in\mathbb{B}} D_B}{\min_{B\in\mathbb{B}} D_B} \le \kappa_D. \tag{6.6a}$$

Put $\lambda = (\sum_B m_B)/(\sum_B D_B)$. If $\mathbb{B}' \subset \mathbb{B}$ and $0 \le m'_B \le m_B$ satisfy

$$\sum_{B\in\mathbb{B}'} m'_B \ge \mathfrak{L}^{-1} \sum_{B\in\mathbb{B}} m_B,$$

then

$$\frac{1}{|\mathbb{B}'|} \sum_{B\in\mathbb{B}'} \frac{m'_B}{D_B} \ge \kappa_D^{-1}\mathfrak{L}^{-1}\lambda. \tag{6.6b}$$

Proof. Write $D_- = \min_B D_B$ and $D_+ = \max_B D_B$. Since $|\mathbb{B}'| \le |\mathbb{B}|$,

$$\frac{1}{|\mathbb{B}'|} \sum_{B\in\mathbb{B}'} \frac{m'_B}{D_B} \ge \frac{\mathfrak{L}^{-1}\sum_B m_B}{|\mathbb{B}|D_+} = \mathfrak{L}^{-1}\lambda\frac{\sum_B D_B}{|\mathbb{B}|D_+} \ge \mathfrak{L}^{-1}\lambda\frac{D_-}{D_+}.$$

This is (6.6b). Notice that the symmetric normalization $\kappa_D^{-1}D \le D_B \le \kappa_D D$ would give only $D_+/D_- \le \kappa_D^2$; it is not used here. □

Apply theorem 6.1 to the original labelled child family, assigning empty shadings to discarded labels. In the ordinary alternative use corollary 6.11; in the coarse/thickened-child alternative use lemma 6.12. By lemma 3.1 and (4.11), in the biased case

$$|Z(B)| \gtrsim C_0^{-q} s^{\varepsilon} \lambda_B^q |B|. \tag{6.7}$$

Suppose the original child denominators

$$D_B = \sum_{T\in\mathbb{T}_B} |T|$$

satisfy the ratio condition (6.6a). Apply lemma 6.13 with the P1 masses; this gives

$$\frac{1}{|\mathbb{B}'|} \sum_{B\in\mathbb{B}'} \lambda_B \ge \kappa_D^{-1}\mathfrak{L}^{-1}\lambda(\mathbb{T}, Y). \tag{6.8}$$

Jensen now proves the global estimate.

Corollary 6.14 (Dimension-sensitive parent density)**.** *For $0 < \sigma \le 1$,*

$$\lambda(\widetilde{\mathbb{B}}', Z) \gtrsim C_0^{-(2-\sigma)} s^{\varepsilon} \kappa_D^{-(2-\sigma)} \mathfrak{L}^{-(2-\sigma)} \lambda(\mathbb{T}, Y)^{2-\sigma}. \tag{6.9}$$

In the ordinary Frostman endpoint, the corresponding bound is

$$\lambda(\widetilde{\mathbb{B}}', Z) \gtrsim C_F^{-1} [\log(2/s)]^{-1} \kappa_D^{-2} \mathfrak{L}^{-2} \lambda(\mathbb{T}, Y)^2. \tag{6.10}$$

In the coarse/thickened-child alternative, replace C_F in (6.10) by $C_{F,\rho} R_{\mathrm{br}}^3$.

Proof. Parent volumes are comparable and $|\widetilde{B}| \asymp |B|$. Apply Jensen to (6.7) using (6.8). For (6.10), apply the quadratic local input (6.6), or (6.6d) in the coarse/thickened-child alternative; every mass loss is then squared, while branching comparability contributes the explicit R_{br}^{-3} from that lemma. □

7. Cellwise angular refinement

Angular refinement must preserve the object type created in Sections 3–5. The ordinary typical-angle descent is pointwise, but the lifting theorem acts on whole parent–cell incidences. We therefore freeze the incident parent family on each half-open cell, run the descent once on that finite family, and use deterministic tie-breaking. The result is a genuine edge subset, not a collection of unrelated pointwise choices.

Set

$$\alpha = a/r, \qquad q_{\mathrm{res}} = a/b.$$

For parent labels B, B', define

$$\mathrm{angle}_{q_{\mathrm{res}}}(B, B') = \max\{q_{\mathrm{res}}, \mathrm{angle}(TB, TB')\} \tag{7.1}$$

and for a nonempty labelled family $\mathcal{F}$,

$$M_q(\mathcal{F}) = \max_{B,B' \in \mathcal{F}} \mathrm{angle}_{q_{\mathrm{res}}}(B, B'). \tag{7.2}$$

Thus a singleton has resolved diameter q_{res}.

Put

$$L_0 = \log \alpha^{-1}, \qquad L_+ = \max\{1, L_0\},$$

and take

$$A_{\mathrm{ang}} = e^{L_+^{1/2}}, \qquad B_{\mathrm{ang}} = e^{L_+^{3/4}}. \tag{7.3}$$

Thus $A_{\mathrm{ang}}, B_{\mathrm{ang}} > 1$ on the full range $0 < \alpha \le 1$, including the endpoint $\alpha = 1$ covered by the definition of L_+. At a retained right cell Q, start from $\mathcal{F}_0(Q) = \{B : (B, Q) \in E_2\}$. Recursively, while the indicated collection is nonempty, define

$$\mathcal{F}_{j+1}(Q) = \mathrm{first}\left\{\mathcal{H} \subset \mathcal{F}_j(Q) : |\mathcal{H}| \ge A_{\mathrm{ang}}^{-1}|\mathcal{F}_j(Q)|,\ M_q(\mathcal{H}) \le B_{\mathrm{ang}}^{-1} M_q(\mathcal{F}_j(Q))\right\}. \tag{7.3a}$$

Here "first" uses a fixed total order on the finite power set. At $j = 0$, the threshold in (7.3a) is the relative quantity $B_{\mathrm{ang}}^{-1} M_q(\mathcal{F}_0(Q))$. Stop at the first J for which the displayed set is empty and put $\mathcal{F}(Q) = \mathcal{F}_J(Q)$ and $\theta(Q) = M_q(\mathcal{F}_J(Q))$.

Lemma 7.1 (Cellwise descent)**.** *The procedure stops after at most*

$$K = 1 + \lceil \log_{B_{\mathrm{ang}}}(q_{\mathrm{res}}^{-1}) \rceil \le 2 + L_+^{1/4} \tag{7.4}$$

successful selections, and

$$|\mathcal{F}(Q)| \ge A_{\mathrm{ang}}^{-K} |\mathcal{F}_0(Q)|. \tag{7.5}$$

If $\mathcal{H} \subset \mathcal{F}(Q)$ and $|\mathcal{H}| \ge A_{\mathrm{ang}}^{-1}|\mathcal{F}(Q)|$, then

$$\frac{\theta(Q)}{B_{\mathrm{ang}}} < M_q(\mathcal{H}) \le \theta(Q). \tag{7.6}$$

Proof. Every resolved diameter is at least q_{res}, whereas every successful step divides the current diameter by B_{ang}. This proves (7.4). Each selected family has at least an A_{ang}^{-1} fraction of its predecessor, proving (7.5). At the stopping stage, (7.6) is the negation of the next successful selection; its upper bound follows because diameter cannot increase on passing to a subfamily. □

The local descent must still be synchronized globally. Deterministic tie-breaking first makes $\mathcal{F}(Q)$ a well-defined edge subset. We then pigeonhole $\theta(Q)$ into at most

$$J_\theta = 1 + \lceil \log_2(q_{\mathrm{res}}^{-1}) \rceil$$

classes by labelled edge count. The surviving cell degrees lie between $A_{\mathrm{ang}}^{-K} d_0$ and $2d_0$; a second dyadic pigeonhole uses at most

$$J_d = 1 + \lceil \log_2(2A_{\mathrm{ang}}^{K}) \rceil$$

classes. Retain every edge at a selected cell.

Theorem 7.2 (Resolved ANGLE-CELL)**.** *There exist an edge subset $E_{\mathrm{ang}} \subset E_2$, a cellular parent shading Z_{ang}, a dyadic degree d_{ang}, and $\theta \in [q_{\mathrm{res}}, 1]$ such that:*

(A1) *every retained cell has parent degree in $[d_{\mathrm{ang}}, 2d_{\mathrm{ang}})$;*
(A2) *its resolved angular diameter lies in $[\theta, 2\theta)$;*
(A3) *for every retained cell Q, every $\mathcal{H} \subset \{B : (B, Q) \in E_{\mathrm{ang}}\}$ with*

$$|\mathcal{H}| \geq A_{\mathrm{ang}}^{-1} \#\{B : (B, Q) \in E_{\mathrm{ang}}\}$$

has resolved diameter larger than $\theta / B_{\mathrm{ang}}$;

(A4)

$$\sum_B |Z_{\mathrm{ang}}(B)| \geq c_{\mathrm{ang}} \sum_B |Z(B)|, \qquad c_{\mathrm{ang}} \geq \frac{A_{\mathrm{ang}}^{-K}}{J_\theta J_d}. \tag{7.7}$$

More explicitly, with an absolute constant C,

$$c_{\mathrm{ang}}^{-1} \leq C(1 + L_+)^2 \exp(L_+^{3/4} + 2L_+^{1/2}). \tag{7.7a}$$

Consequently $c_{\mathrm{ang}}^{-1} = O(1)$ on $L_0 \leq 1$, while $c_{\mathrm{ang}} = \alpha^{o(1)}$ as $\alpha \downarrow 0$.

The corresponding child lift retains at least $c_{\mathrm{ang}}/2$ of the pre-angle child mass.

Proof. lemma 7.1 gives the cellwise angular conclusions. Both global pigeonholes retain all edges at each selected cell, so the output degree is constant and the cellwise angular family is unchanged. Equal cell volumes identify labelled parent mass with edge count, proving (7.7). Finally apply theorem 5.1 once to the resulting edge set. The estimate

$$A_{\mathrm{ang}}^{K} \leq \exp(L_+^{3/4} + 2L_+^{1/2})$$

and the elementary bounds $J_\theta, J_d \leq C(1 + L_+)$ prove (7.7a). If $L_0 \to \infty$, then $L_+ = L_0$, and hence $\log c_{\mathrm{ang}}^{-1} = o(L_0)$. If $L_0 \leq 1$, every factor in (7.7a) is bounded absolutely. □

Remark 7.3. Here and below "raw" means the capped geometric angle angle, before the resolution floor q_{res} is imposed. The exact threshold at which (7.6) yields genuine raw separation, including for repeated parallel labels, is

$$\boxed{\theta > B_{\mathrm{ang}} q_{\mathrm{res}}}. \tag{7.8}$$

Indeed, then $\theta / B_{\mathrm{ang}} > q_{\mathrm{res}}$, so $M_q(\mathcal{H}) > \theta / B_{\mathrm{ang}}$ lies above the resolution floor, so the raw angular diameter is larger than $\theta / B_{\mathrm{ang}}$. At and below this threshold, including $\theta \asymp q_{\mathrm{res}}$, the conclusion is resolution-limited.

8. A stable cellular maximal-density factoring theorem

We now assemble the reusable Euclidean theorem. Its combinatorial engine is the atomic factoring-and-lifting principle of theorem 5.6; rectangular geometry supplies the carrier, and Section 6 supplies three alternative density inputs. The theorem records not only the output estimates but also the category in which they are stable: polynomially many labelled objects, positive parent–cell edges, and a fixed number of whole-incidence refinements. The angular theorem is one admissible protocol on this output, not an extra shading created by the factoring statement.

Lemma 8.1 (Labelled biased maximal-density factoring)**.** *Let $U \subset \mathbb{R}^n$ be a bounded convex body with nonempty interior, let $\mathcal{V}$ be a finite nonempty labelled family of congruent, full-dimensional compact convex bodies of positive n-dimensional volume contained in U, and fix $\gamma > 0$. Every convex hull used as a candidate parent is required to be full-dimensional (equivalently, only nonempty subfamilies with positive-volume hulls enter the maximization), so every candidate lies in the test-body class $\mathfrak{K}_n$ fixed before (1.2). Repeated geometric carriers are allowed and are counted with their labels. Fix finite sets of integers $\mathcal{K}_\Phi, \mathcal{K}_1, \dots, \mathcal{K}_n$. Assume that, for every nonempty labelled subfamily $\mathcal{A} \subset \mathcal{V}$, a maximizer $K_\mathcal{A}$ of the score below can be chosen so that*

$$\Phi_\mathcal{A}(K) := |K|^{-\gamma}\Delta(\mathcal{A}, K) = \frac{\sum_{V\in\mathcal{A}:\, V\subset K} |V|}{|K|^{1+\gamma}} \tag{8.0A}$$

satisfies $\Phi_\mathcal{A}(K_\mathcal{A}) \in [2^k, 2^{k+1})$ for some $k \in \mathcal{K}_\Phi$, and its ordered Löwner half-axes $\ell_j(K_\mathcal{A})$ belong to $[2^{k_j}, 2^{k_j+1})$ with $k_j \in \mathcal{K}_j$. Define

$$L_{\mathrm{bias}} := |\mathcal{K}_\Phi| \prod_{j=1}^{n} |\mathcal{K}_j|. \tag{8.0A$'$}$$

Then there are a labelled subfamily $\mathcal{V}' \subset \mathcal{V}$, a family $\mathcal{W}$ of convex parents with comparable Löwner dimensions, and a partition

$$\mathcal{V}' = \bigsqcup_{W\in\mathcal{W}} \mathcal{V}_W, \qquad V \subset W \quad (V \in \mathcal{V}_W), \tag{8.0B}$$

such that

$$\sum_{V\in\mathcal{V}'} |V| \ge L_{\mathrm{bias}}^{-1} \sum_{V\in\mathcal{V}} |V|. \tag{8.0C}$$

Writing w for the common parent volume up to a dimensional factor, the following hold.

(B1) *For $K \subset W$ convex,*

$$\Delta(\mathcal{V}_W, K) \le C \left(\frac{|K|}{|W|}\right)^{\gamma} \Delta(\mathcal{V}_W, W). \tag{8.0D}$$

Equivalently, because the children are congruent,

$$\frac{\#\mathcal{V}_W[K]}{\#\mathcal{V}_W} \le C \left(\frac{|K|}{|W|}\right)^{1+\gamma}. \tag{8.0E}$$

(B2) *The density achieved in every selected parent satisfies*

$$\Delta(\mathcal{V}_W, W) \ge c \left(\frac{|W|}{|U|}\right)^{\gamma} \Delta_{\max}(\mathcal{V}'). \tag{8.0F}$$

(B3) *The labelled parent family obeys*

$$\Delta_{\max}(\mathcal{W}) \le C \left(\frac{w}{|U|}\right)^{-\gamma}. \tag{8.0G}$$

The constants C, c depend only on n, γ and the factor-two class widths. More explicitly, if $|\mathcal{V}| \le \delta^{-C_{\mathrm{comp}}}$ and every positive score and every chosen half-axis above lies in $[\delta^{C_{\mathrm{win}}}, \delta^{-C_{\mathrm{win}}}]$, then each displayed index set may be taken to have at most $C(C_{\mathrm{comp}}, C_{\mathrm{win}}, n, \gamma)\log(2/\delta)$ elements. Hence $L_{\mathrm{bias}} \le C\log^{n+1}(2/\delta) = \delta^{-o(1)}$, with all dependencies explicit.

Proof. For completeness, we give the greedy construction. For a nonempty labelled subfamily $\mathcal{A} \subset \mathcal{V}$, maximize $\Phi_{\mathcal{A}}(K)$ over $K \in \mathfrak{K}_n$. A maximizer exists: replacing K by the convex hull of the members of $\mathcal{A}[K]$ does not change the numerator and can only decrease the denominator, and only finitely many labelled subfamilies occur. Fix an order on those finitely many hulls and on all dyadic classes, and use it to break every maximizing or class-selection tie. Thus the construction is a deterministic function of labelled carrier and containment data.

Set $\mathcal{V}^{(0)} = \mathcal{V}$. Having defined $\mathcal{V}^{(j)} \neq \varnothing$, choose a maximizer W_j, put

$$\mathcal{V}_{W_j} := \mathcal{V}^{(j)}[W_j], \qquad \phi_j := \Phi_{\mathcal{V}^{(j)}}(W_j), \tag{8.0H}$$

and remove all labels in $\mathcal{V}_{W_j}$. The process terminates and partitions $\mathcal{V}$. Pigeonhole the values ϕ_j and the Löwner half-axes of W_j, choosing the combined class that maximizes retained labelled child volume. This proves (8.0B)–(8.0C), makes ϕ_j comparable to one number ϕ, and makes $|W_j|$ comparable to w. It also proves the asserted polylogarithmic bound for L_{bias} in the stated finite windows.

Fix a selected W_j and a convex $K \subset W_j$. Since the children assigned to W_j are exactly the members of $\mathcal{V}^{(j)}$ contained in W_j,

$$\begin{aligned} \Delta(\mathcal{V}_{W_j}, K) &\le \Delta(\mathcal{V}^{(j)}, K) \\ &\le \phi_j |K|^\gamma = \left(\frac{|K|}{|W_j|}\right)^\gamma \Delta(\mathcal{V}_{W_j}, W_j). \end{aligned} \tag{8.0I}$$

This gives (8.0D), up to the fixed class-comparability constants. Multiplying by $|K|/|W_j|$ and using equal child volume gives (8.0E).

Let j_0 be the first selected greedy index. All labels of $\mathcal{V}'$ are still present in $\mathcal{V}^{(j_0)}$. Any test body in $\mathfrak{K}_n$ may be intersected with U, without losing a child of $\mathcal{V}'$; a positive-volume intersection remains in $\mathfrak{K}_n$, while a null intersection has zero numerator. Hence

$$\Delta_{\max}(\mathcal{V}') \le C\phi|U|^\gamma. \tag{8.0J}$$

On the other hand

$$\Delta(\mathcal{V}_{W_j}, W_j) = \phi_j |W_j|^\gamma \ge c\phi|W_j|^\gamma.$$

Combining these inequalities proves (8.0F). This is the quantitative near-tie rule: among almost equal density scores, a smaller parent receives the larger value of the volume-biased score.

Finally let $K \in \mathfrak{K}_n$. If $W \subset K$, then every child assigned to W is contained in K. By (8.0F), or directly by the common ϕ-class,

$$\begin{aligned} \Delta_{\max}(\mathcal{V}')|K| &\ge \sum_{W \in \mathcal{W}:\, W \subset K} \sum_{V \in \mathcal{V}_W} |V| \\ &\ge c\phi w^\gamma \sum_{W \in \mathcal{W}:\, W \subset K} |W|. \end{aligned}$$

Use (8.0J), divide by $|K|$, and take the supremum over K. This gives (8.0G). Every selection and sum was made on labels, so the proof is unchanged for repeated carriers. □

Theorem 8.2 (Quantitative cellular maximal-density factoring)**.** *Fix $C_{\mathrm{comp}} \ge 1$, $m_0 \in \mathbb{N}$, and $\varepsilon > 0$. Let $0 < \delta \le a \le b \le r \le 1$, put $h = c_0 a$, and let $\mathbb{T} = \bigsqcup_{B \in \mathbb{B}} \mathbb{T}_B$ be a finite labelled family with $M(Y) > 0$. Assume:*

(F1) *every B is a labelled rectangular box, with the ordered side frame fixed in (3.4a), of dimensions comparable to $a \times b \times r$, and every $T \in \mathbb{T}_B$ is a labelled $\delta \times \delta \times r$ child contained in B, with measurable shading $Y(T) \subset T$;*

(F2) *for some $D > 0$ and $\kappa_D \geq 1$,*

$$D \leq D_B := \sum_{T \in \mathbb{T}_B} |T| \leq \kappa_D D$$

for every parent;

(F3) *the finite active-cell set*

$$\mathcal{Q}_h(Y) = \left\{ Q \in \mathcal{Q}_h : \sum_{B \in \mathbb{B}} \sum_{T \in \mathbb{T}_B} |Y(T) \cap Q| > 0 \right\}$$

satisfies

$$|\mathbb{T}| + |\mathbb{B}| + |\mathcal{Q}_h(Y)| \leq \delta^{-C_{\mathrm{comp}}};$$

(F4) *one of the following three global alternatives is designated for the whole family. In the ordinary fine-child alternative every $\mathbb{T}_B$ satisfies, with one common constant C_F,*

$$\frac{\#\{T \in \mathbb{T}_B : T \subset K\}}{|\mathbb{T}_B|} \leq C_F \frac{|K|}{|B|} \qquad \text{for every convex } K \subset B, \tag{8.0}$$

In the biased alternative every $\mathbb{T}_B$ satisfies, for one fixed $0 < \sigma \leq 1$, the complete-child condition (6.3) with one common constant C_0. In the coarse/thickened-child alternative there is one scale $\delta \leq \rho \leq a$ and, for every B, a nonempty labelled family $\mathbb{S}_B$ of $\rho \times \rho \times r$ capped tubes contained in $B^\sharp$, satisfying (6.6c) with the common constant C_F. The fine and coarse carriers use the same declared length window $[c_{\mathrm{len}} r, C_{\mathrm{len}} r]$ and the same endcap convention. Each fine child T is assigned to one coarse label $S(T)$ with $T \subset S(T)$, and the assigned branching numbers satisfy (6.6c′) with one common factor R_{br}.

After the common inner-level selection, let E^+ be the positive graph (3.7) constructed from Y_0, put $N_{\max} = \max_B |\mathbb{T}_B|$, $e_+ = |E^+|$, and define

$$\mathfrak{L}_0 = 4(1 + \lceil \log_2 N_{\max} \rceil)(1 + \lceil \log_2(2e_+) \rceil)(1 + \lceil \log_2 e_+ \rceil). \tag{8.1}$$

Then there exist a retained positive edge set $E_2 \subset E^+$, its labelled rectangular parent family $\widetilde{\mathbb{B}}'$, a cellular parent shading Z, and a child refinement $(\mathbb{T}', Y_1)$ satisfying P1–P5 of theorem 4.2, with constants depending only on ε, σ where applicable, the fixed comparison constants, and C_{comp}, and with

$$M(Y_1) \geq \mathfrak{L}_0^{-1} M(Y) \tag{8.2}$$

Here the output label sets are literal:

$$\mathbb{B}' = \{B : \exists Q \ (B, Q) \in E_2\}, \qquad \mathbb{T}' = \bigsqcup_{B \in \mathbb{B}'} \mathbb{T}_B, \qquad \mathbb{T}'_{\mathrm{act}} = \{T \in \mathbb{T}' : Y_1(T) \neq \varnothing\} \tag{8.2′}$$

The shading Y_1 is a total function on the structural family $\mathbb{T}'$, and labels in $\mathbb{T}' \setminus \mathbb{T}'_{\mathrm{act}}$ have empty shading. Complete-child Frostman hypotheses are always read on the structural family $\mathbb{T}_B$, while mass, union, and pointwise multiplicity are unchanged if one restricts to $\mathbb{T}'_{\mathrm{act}}$. Thus an empty shading is never confused with deletion of a complete carrier from a denominator. Explicitly, $Y_1(T) \subset Z(B)$ for $T \in \mathbb{T}_B$, and there are $\mu_{\mathrm{in}}, d_0 \geq 1$ such that on the corresponding unions

$$\begin{aligned} \mu_{\mathrm{in}} &\leq \sum_{T \in \mathbb{T}_B} \mathbf{1}_{Y_1(T)} < 2\mu_{\mathrm{in}}, \\ d_0 &\leq \sum_B \mathbf{1}_{Z(B)} < 2d_0, \qquad \sum_{T \in \mathbb{T}} \mathbf{1}_{Y_1(T)} < 4\mu_{\mathrm{in}} d_0. \end{aligned} \tag{8.2a}$$

In the biased case the initial parent output satisfies

$$\lambda(\widetilde{\mathbb{B}}', Z) \gtrsim C_0^{-(2-\sigma)}(a/b)^{\varepsilon}\kappa_D^{-(2-\sigma)}\mathfrak{L}_0^{-(2-\sigma)}\lambda(\mathbb{T}, Y)^{2-\sigma}; \tag{8.3}$$

in the ordinary endpoint it satisfies

$$\lambda(\widetilde{\mathbb{B}}', Z) \gtrsim C_F^{-1}[\log(2b/a)]^{-1}\kappa_D^{-2}\mathfrak{L}_0^{-2}\lambda(\mathbb{T}, Y)^2. \tag{8.4}$$

and in the coarse/thickened-child endpoint it satisfies

$$\lambda(\widetilde{\mathbb{B}}', Z) \gtrsim C_F^{-1}R_{\mathrm{br}}^{-3}[\log(2b/a)]^{-1}\kappa_D^{-2}\mathfrak{L}_0^{-2}\lambda(\mathbb{T}, Y)^2. \tag{8.4a}$$

A whole-incidence refinement protocol *is a nested sequence*

$$E^{(m)} \subset \cdots \subset E^{(1)} \subset E^{(0)} = E_2, \qquad m \le m_0,$$

where

$$\begin{aligned} Z^{(j)}(B) &= \bigcup_{Q:(B,Q)\in E^{(j)}} Q, \\ Y^{(j)}(T) &= Y_1(T) \cap \bigcup_{Q:(B,Q)\in E^{(j)}} Q \quad (T \in \mathbb{T}_B), \end{aligned} \tag{8.5}$$

and, for $1 \le j \le m$,

$$\sum_B |Z^{(j)}(B)| \ge c_j \sum_B |Z^{(j-1)}(B)|, \qquad 0 < c_j \le 1. \tag{8.6}$$

Such a protocol retains every child incidence of B *in a selected* (B, Q)*-edge. It obeys*

$$\begin{aligned} M(Y^{(m)}) &\ge \mathfrak{L}_0^{-1}2^{-m}\prod_{j=1}^{m} c_j\, M(Y), \\ \mathfrak{L}(\mathbb{T}, \mathbb{B}, h; c_1, \dots, c_m) &:= \mathfrak{L}_0 \prod_{j=1}^{m} \frac{2}{c_j}. \end{aligned} \tag{8.7}$$

Equivalently, $M(Y^{(m)}) \ge \mathfrak{L}(\mathbb{T}, \mathbb{B}, h; c_1, \dots, c_m)^{-1}M(Y)$*. If one additionally knows* $c_j \ge c_{\min} > 0$*, then* $\mathfrak{L} \le \mathfrak{L}_0(2/c_{\min})^{m_0}$*; no such fixed lower bound is built into the theorem. At every stage, exact cellular compatibility, the common inner multiplicity, and the original pointwise multiplicity upper bound survive. Any dyadic right-degree re-regularization is counted as one of the* m *stages and must retain all edges at every selected cell.*

Proof. The proof has three modules: joint regularization, analytic density, and iterated lifting. Run the common inner level, lemma 4.1, right-degree selection, and theorem 4.2. Since $|E_1| \le e_+$, (4.8) is at most $\mathfrak{L}_0$, which proves (8.2). Apply corollary 6.14 with this larger explicit loss, using lemma 6.12 in the third alternative, to obtain (8.3), (8.4), or (8.4a). Iterating theorem 5.1 along (8.5)–(8.6) proves (8.7). In particular a dyadic degree re-regularization may retain only a polylogarithmic fraction; its actual fraction is entered as c_j, rather than compared with a fixed constant. The last assertions are precisely the invariants in theorem 5.1 and corollary 5.2. □

Lemma 8.3 (Source Löwner frame)**.** *Let* $W \subset \mathbb{R}^3$ *be a bounded convex body with nonempty interior. Fix the Löwner (minimum-volume containing) ellipsoid* E_W^{out} *used to define the dimensions of* W*, and order its semiaxes* $\ell_1 \le \ell_2 \le \ell_3$*. When eigenvalues coincide, choose an orthonormal eigenframe by a fixed lexicographic rule against the ambient coordinate frame, use that same rule for both* W *and its rectangular label, and retain the frame as part of the label. There is a rectangle* B_W*, aligned with this ordered outer frame, such that*

$$W \subset B_W, \qquad |B_W| \le C_J|W|, \tag{8.7a}$$

and, under this fixed convention, the labelled tangent plane $TB_W = \mathrm{span}(e_2(W), e_3(W))$ *is the source two-long-axis plane when that plane is uniquely determined. At a repeated eigenvalue*

it is a fixed representative compatible with the source resolution convention, which does not select a lexicographic eigenframe. In particular no comparison with the principal frame of an independently chosen inner John ellipsoid is made.

Proof. The source defines convex dimensions from the axes of its minimum-volume containing ellipsoid and defines plank angle from the two longest of those axes [GWZ26, p. 4 and the plank-angle convention on p. 16]. Take B_W to be the coordinate rectangle with half-side lengths ℓ_1, ℓ_2, ℓ_3. Then $W \subset E_W^{\mathrm{out}} \subset B_W$, and $|B_W| = (6/\pi)|E_W^{\mathrm{out}}|$.

For volume comparability, let c_W be the Löwner centre. The non-symmetric Löwner–John inclusion in dimension n states precisely that the homothetic copy $c_W + n^{-1}(E_W^{\mathrm{out}} - c_W)$ lies in W. Thus, in dimension three,

$$|E_W^{\mathrm{out}}| \le 3^3 |W| = 27|W|;$$

see [Hen12, Theorem 2.1 and equation (2)], together with the original extremal theorem [Joh48]. This proves (8.7a). The deterministic tie rule makes the ordered outer frame a label when an eigenspace is multiple. Using this one chosen outer frame for both W and B_W makes their two tangent planes identical. If the short and middle axes tie, then $\ell_1/\ell_2 = 1$ and the source angle is already unresolved at unit scale; the common tie rule merely chooses one representative of that source ambiguity. If only the middle and long axes tie, their two-dimensional eigenspace is itself unique, so the two-long-axis plane is independent of the basis chosen inside it. If all three axes tie, the source plane is again unresolved at unit scale. Standard background on uniqueness of the Löwner ellipsoid, parallel bodies, and orthogonal choices inside repeated eigenspaces is recorded in [Sch14, Chs. 1, 3, and 10]; only the source convention just cited determines which plane is used in the GWZ call. □

Lemma 8.4 (Transfer to permanent Löwner rectangles)**.** *Suppose a labelled convex parent W has the containing rectangle B_W from lemma 8.3, has a nonempty assigned child family, and every assigned child label is wholly contained in W. Let $p \in \{1, 1+\sigma\}$. If*

$$\frac{\#\{T : T \subset K\}}{\#\{T : T \subset W\}} \le C_0 \left(\frac{|K|}{|W|}\right)^p \qquad (K \subset W \text{ convex}), \tag{8.7b}$$

then the same labelled children, viewed inside B_W, satisfy

$$\frac{\#\{T : T \subset K\}}{\#\{T : T \subset B_W\}} \le C_0 C_J^p \left(\frac{|K|}{|B_W|}\right)^p \qquad (K \subset B_W \text{ convex}). \tag{8.7c}$$

The total child denominator is unchanged. For a labelled parent family,

$$\sum_W |W| \le \sum_W |B_W| \le C_J \sum_W |W|, \qquad \Delta_{\max}(\{B_W\}) \le C_J \Delta_{\max}(\{W\}). \tag{8.7d}$$

The ordered tangent frame is unchanged. The same argument with $p = 1$ transfers the ordinary fine-child and coarse/thickened-child containment conditions, provided the counted carriers are wholly contained in W. Consequently theorem 8.2 applies to the permanent labelled boxes B_W, with its shading and union conclusions interpreted on those boxes. The transfer is one-way: it supplies neither a restriction to the sharp boundary of W nor a bounded-overlap conclusion for the rectangles.

Proof. For a convex $K \subset B_W$, a child already contained in W is contained in K exactly when it is contained in the convex set $K \cap W$. Applying (8.7b) to $K \cap W$, then using $|K \cap W| \le |K|$ and $|B_W|/|W| \le C_J$, proves (8.7c). The label set and child carriers are unchanged, so their denominator is identical.

For the parent concentration, $B_W \subset K$ implies $W \subset K$, and $|B_W| \le C_J|W|$. Summing this implication over labels proves the second part of (8.7d); the first is immediate from containment and volume comparability. The tangent-frame assertion is lemma 8.3. Repeating the same

application	objects	cellular realization	additional check
Section 6.5(A)	$\mathbb{T} = \bigsqcup_W \mathbb{T}_W$ through Frostman planks	P1, P3–P5, (6.10), and $\mu(\mathbb{T}, Y) \lesssim_{\log} \mu(\mathbb{B}, Z)\mu(\mathbb{T}_W, Y_1)$	choose one inner parent attaining the post-refinement average density
Section 6.5(B)	fine tubes grouped through coarse ρ-tubes	same cellular product and quadratic endpoint on the fine labelled multiset	retain a common number of fine descendants per coarse label
biased interface	maximal-density convex parents	permanent labelled rectangles and (6.9)	never restrict the conclusion back to the sharp parent

TABLE 1. The two Section 6 applications of the cellular theorem.

intersection argument for the fine or coarse carrier counted in the relevant condition proves the remaining claims. All labels, including coincident ones, are retained. □

8.1. **Quantitative dependence and iteration budget.** The finite graph makes every loss explicit. For the polynomial-complexity application, $e_+ \le |\mathbb{B}||\mathcal{Q}_h(Y)|$ and

$$\mathfrak{L}_0 \le C(C_{\mathrm{comp}})[\log(2/\delta)]^3. \tag{8.8}$$

In the applications of Sections 9–10, after their displayed bounds have been verified, the additional notation $\delta^{-o(1)}$ abbreviates only the following controlled factors:

$$L_{\mathrm{in}}, \quad 1 + \log(2|E^+|), \quad L_{\deg}, \quad J_\theta, \quad J_d, \quad c_{\mathrm{ang}}^{-1}. \tag{8.9}$$

In theorem 8.2 itself, κ_D remains explicit and no subpolynomial asymptotic bound on it is assumed. An application may append κ_D to (8.9) only after proving such a bound for its denominator level. Section 10 adds the explicit quantities

$$J_v, \quad L_m, \quad L_{a,b}, \quad L_N, \quad L_{\mathrm{reg}}, \quad \kappa_{\mathrm{inc}}, \quad \kappa_{\mathrm{loc}}, \quad \kappa_{\mathrm{amb}}, \quad H_{\mathrm{ang}}. \tag{8.10}$$

Jensen raises every synchronized density loss to $2 - \sigma$, and to two in the Córdoba endpoint; LIFT losses remain separate from density losses.

Remark 8.5. The theorem is not uniform over an arbitrary indexed multiset of super-polynomial cardinality. Without the stated polynomial-complexity bound, the logarithms in (8.1)–(8.2) must remain explicit.

9. TWO SMALL-MIDDLE APPLICATIONS IN GWZ SECTION 6

We first test the cellular theorem on the two small-middle factorizations in the factoring-through-planks argument of [GWZ26]. Both applications need the same three outputs: quantitative retention of child mass, the child–parent multiplicity product, and the ordinary quadratic parent-density endpoint. Neither application needs a ballwise output or a metric reconstruction of the parent shading. Consequently the cellular theorem can be inserted through a short, explicit interface without altering the surrounding proof architecture.

The next four lemmas isolate, in order, fixed-dilate symmetry, conflict coloring, the denominator cost of a mass-selected color, and normalized concentration under bounded ancestry. The application proofs invoke these four roles through their tagged conclusions.

Lemma 9.1 (Fixed-dilate symmetry for source-standard tubes)**.** *Let S, S' be capped source-standard tubes of the same length L and width w, in one fixed bounded direction chart. For every fixed $C \geq 1$,*

$$S \subset CS' \quad \Longrightarrow \quad S' \subset C_* S, \tag{9.0a$'$}$$

where C_* depends only on C and the chart constants.

Proof. If $w \geq cL$ for a fixed structural $c > 0$, both capped carriers have diameter $O(w)$, and the assumed containment places their centres within $O_C(w)$; the conclusion is immediate after a structural dilation. Assume henceforth that $w < cL$. The two endpoint cross-sections of the carrier of S lie within $O_C(w)$ of S'. Their separation is comparable to L, so the corresponding unit directions differ by $O_C(w/L)$, and one endpoint of the carrier of S' is within $O_C(w)$ of the matching endpoint of S. Linear interpolation along the capped carriers gives Hausdorff distance $O_C(w)$ between them. Thickening by width w proves (9.0a$'$), including parallel and coincident carriers. □

Lemma 9.2 (Source-resolution conflict coloring)**.** *Let $\mathcal{F}_0$ be a source-standard, essentially distinct family of w-tubes. Quantitatively, for every fixed $C \geq 1$, assume*

$$\sup_{T_0} \#\{T \in \mathcal{F}_0 : T \subset CT_0\} \leq \kappa_{\mathrm{ed}}(C), \tag{9.0a}$$

where $\kappa_{\mathrm{ed}}(C) = O_C(1)$ in a literal source call. Let I be a labelled representation family with a map $\pi : I \to \mathcal{F}_0$, fibres of size at most κ_{rep}, and carriers T_i contained in and containing fixed enlargements of $\pi(i)$. Join two distinct vertices i, j when

$$T_i \subset C_{\mathrm{src}} T_j \quad \text{or} \quad T_j \subset C_{\mathrm{src}} T_i, \tag{9.0b}$$

where C_{src} is the dilation in the source essential-distinctness convention. Then the conflict graph has maximum degree at most

$$C\kappa_{\mathrm{rep}}\kappa_{\mathrm{ed}}(C'), \tag{9.0c}$$

for fixed structural C, C', and hence has a proper coloring with $H_{\mathrm{src}} \leq 1 + C\kappa_{\mathrm{rep}}\kappa_{\mathrm{ed}}(C')$ colors. Each color is source-essentially-distinct. For arbitrary nonnegative label weights one color carries at least H_{src}^{-1} of their total. If the weights are shading masses and all carrier volumes are comparable, that color has shading density at least a fixed multiple of H_{src}^{-1} times the original density. Passing to a color cannot increase the absolute complete-containment density $\Delta_{\max}$; normalized Frostman constants are treated separately in lemma 9.3. Moreover, a proper color contains at most one vertex in every fibre whose represented carriers are pairwise adjacent in (9.0b), in particular in a fibre of coincident copies. If this condition holds for every fibre, the restriction of π to a color is injective and its absolute $\Delta_{\max}$ is bounded by that of the source family.

Proof. If i and j are adjacent, the fixed two-sided comparison between T_k and $\pi(k)$ turns (9.0b) into $\pi(j) \subset C'\pi(i)$ or the reverse inclusion, with C' structural. For equal-length, equal-width source tubes the reverse inclusion also implies $\pi(j) \subset C''\pi(i)$ by lemma 9.1. Equation (9.0a) leaves at most $C\kappa_{\mathrm{ed}}(C')$ possible geometric source labels, and each has at most κ_{rep} copies. This proves (9.0c), so greedy coloring gives the asserted H_{src}. A fibre whose represented carriers are pairwise adjacent in (9.0b) is a clique, so a proper color contains at most one of its vertices. Coincident copies satisfy this condition.

No conflict remains inside a color, which is precisely the source essential-distinctness condition after the declared fixed enlargement. The largest color by weight proves the mass assertion. Its denominator is no larger than the original denominator and the fixed carrier dilation changes individual volumes only by a structural factor, proving the density assertion. The complete-containment numerator only decreases on passage to a subfamily. Finally a fixed enlargement replaces w by Cw; the scale-variant source statement is applied at this new width. All scale

powers and density thresholds change by fixed constants only, so no hidden power of w is spent. □

Lemma 9.3 (Mass-selected color: absolute versus normalized concentration)**.** *Let a finite labelled family $\mathcal{R}$ of congruent carriers of volume $v > 0$ be partitioned into at most H colors, and let $Y(R) \subset R$ be measurable. Set*

$$D = v|\mathcal{R}|, \qquad M = \sum_{R\in\mathcal{R}} |Y(R)|, \qquad \lambda_{\mathcal{R}} = M/D > 0. \tag{9.0d}$$

If a color $\mathcal{R}_c$ carries at least $H^{-1}M$, then

$$\frac{|\mathcal{R}_c|}{|\mathcal{R}|} \geq \frac{\lambda_{\mathcal{R}}}{H}. \tag{9.0e}$$

Moreover, writing $\mu(\mathcal{R}, Y) = M/|U(\mathcal{R}, Y)|$,

$$\mu(\mathcal{R}, Y) \leq H\mu(\mathcal{R}_c, Y|_{\mathcal{R}_c}). \tag{9.0e$'$}$$

Let $\mathcal{K}$ be any prescribed class of test bodies. Consequently, if

$$\frac{|\mathcal{R}[K]|}{|\mathcal{R}|} \leq C_{\text{test}}\Psi(K) \qquad (K \in \mathcal{K}), \tag{9.0f}$$

then

$$\frac{|\mathcal{R}_c[K]|}{|\mathcal{R}_c|} \leq H\lambda_{\mathcal{R}}^{-1}C_{\text{test}}\Psi(K) \qquad (K \in \mathcal{K}). \tag{9.0g}$$

On the other hand, $\Delta_{\max}(\mathcal{R}_c) \leq \Delta_{\max}(\mathcal{R})$. Thus absolute concentration is monotone under label deletion, whereas normalized concentration must pay the denominator factor in (9.0g). The same statements hold for carrier volumes in $[v, \kappa_v v]$, with one additional factor κ_v.

Proof. The selected shading mass is at most its carrier denominator, so

$$v|\mathcal{R}_c| \geq \sum_{R\in\mathcal{R}_c} |Y(R)| \geq H^{-1}M = H^{-1}\lambda_{\mathcal{R}} v|\mathcal{R}|.$$

This is (9.0e). Since $|\mathcal{R}_c[K]| \leq |\mathcal{R}[K]|$, division by (9.0e) proves (9.0g), pointwise on the prescribed class $\mathcal{K}$; no closure property of that class is used. Also $M \leq HM_c$ and $U(\mathcal{R}_c, Y) \subset U(\mathcal{R}, Y)$, which proves (9.0e$'$). For the absolute assertion, the complete-carrier numerator for every fixed K only decreases. The comparable-volume version follows by using the two endpoints of the comparison interval in the same calculation. □

Lemma 9.4 (Normalized concentration through bounded ancestry)**.** *Let $\mathcal{T}$ and $\mathcal{P}$ be finite nonempty labelled carrier families and let $I \subset \mathcal{P} \times \mathcal{T}$ be a labelled incidence relation such that $T \subset P$ whenever $(P, T) \in I$. Suppose $N > 0$, $R_N, \kappa_I \geq 1$, and*

$$N \leq d_I(P) \leq R_N N \quad (P \in \mathcal{P}), \qquad 1 \leq d_I(T) \leq \kappa_I \quad (T \in \mathcal{T}). \tag{9.0h}$$

Then, for every test body K,

$$\frac{|\mathcal{P}[K]|}{|\mathcal{P}|} \leq \kappa_I R_N \frac{|\mathcal{T}[K]|}{|\mathcal{T}|}. \tag{9.0i}$$

Consequently a normalized complete-containment bound for $\mathcal{T}$ passes to $\mathcal{P}$ with the explicit factor $\kappa_I R_N$. This is an ancestry statement, not monotonicity under taking a color or subfamily.

Proof. Every $P \subset K$ contributes at least N incidences, and all its incident children lie in K. Hence

$$N|\mathcal{P}[K]| \leq \kappa_I|\mathcal{T}[K]|.$$

On the other hand, the lower fine degree and the upper parent degree give $|\mathcal{T}| \leq |I| \leq R_N N|\mathcal{P}|$. Multiplying these two inequalities proves (9.0i). Repeated geometric carriers cause no change because all four counts are labelled counts. □

Interface convention. Each source call is recorded through seven fields: (SC1) family and reference shading, (SC2) essential distinctness, (SC3) normalized scales, (SC4) density, (SC5) concentration, (SC6) affine behavior, and (SC7) source output and pullback. The cited tagged formulas in this table contain the quantitative data used below.

call	(SC1)–(SC2)	(SC3)–(SC5)	(SC6)–(SC7)
Lemma 6.4, child	selected color of the normalized local child family, with its shading	(9.2a), (9.2d0): scales, density, and C_F	affine invariance and (9.I-46), recovered in (9.2d)
Lemma 6.4, parent	selected color of $\widetilde{\mathbb{W}}'$, with cellular parent shading	(9.2c), (9.2e0): scales, density, and C_F	affine invariance and (9.I-45), recovered in (9.2e)
Lemma 6.1, child	complete normalized representation family $\mathcal{F}_{\mathrm{ch}}$, with its local shading	(9.4b), (9.5b), (9.5c0): scales, density, restricted slab tests, and $\Delta_{\max}$	affine invariance and (9.I-49), giving (9.5c)
Lemma 6.1, parent	retained parent family $\mathbb{W}'$, with cellular shading Z	(9.5d0): scales, density, and absolute concentration	affine invariance and (9.I-48), giving (9.5d)

The density field (SC4) is always read on the displayed reference shading; (SC5) uses the corresponding labelled denominator.

Proposition 9.5 (Section 6 source interface)**.** *Fix $0 < \beta \le 1$ and $\varepsilon > 0$. There are $\eta, b_0 > 0$, depending only on these parameters, with the following property. Let $0 < \tau \le s \le t \le b_0$, and let $(\mathcal{P}, A)$ be an essentially distinct labelled family of $s \times t \times 1$ planks in a unit ball, with $\lambda(\mathcal{P}, A) \ge \tau^{\eta}$. The scale-variant convention is the one in* [GWZ26, Remark 3.6]*; thus the losses below are powers of the declared auxiliary scale τ, not powers silently transferred between s and τ.*

If $K_F(\beta)$ holds and, with P_ϑ denoting the source $\vartheta t \times t \times 1$ thickening of P,

$$|\mathcal{P}[P_\vartheta]| \le M\vartheta \quad (P \in \mathcal{P},\ s/t \le \vartheta \le 1), \tag{9.I-F0}$$

then the original statement of [GWZ26, Lemma 6.4, equation (31)] *gives*

$$\mu(\mathcal{P}, A) \lesssim \tau^{-\varepsilon} C_F(\mathcal{P})^{1-\beta/2} M^{\beta/2} \frac{s}{t}\, t^{-2\beta} \big(t^2|\mathcal{P}|\big)^{1-\beta/2}. \tag{9.I-F}$$

Consequently the substitutions

call	s	t	M
parent	a	b	$\lesssim b/a$
child	δ/b	δ/a	$\lesssim (b/a)^2$

(9.I-F1)

give, respectively,

$$\mu_{\mathrm{par}} \lesssim \tau^{-\varepsilon} C_F^{1-\beta/2} \left(\frac{a}{b}\right)^{1-\beta/2} b^{-2\beta} \big(b^2 N_{\mathrm{par}}\big)^{1-\beta/2}, \tag{9.I-45}$$

and

$$\mu_{\mathrm{ch}} \lesssim \tau^{-\varepsilon} C_F^{1-\beta/2} \left(\frac{a}{b}\right)^{1-\beta} \left(\frac{\delta}{a}\right)^{-2\beta} \left(\frac{\delta^2}{a^2} N_{\mathrm{ch}}\right)^{1-\beta/2}. \tag{9.I-46}$$

If instead $K_{KT}(\beta)$ holds, $0 \le \gamma \le 1$, and every $\vartheta \times 1 \times 1$ test slab in the sense of [GWZ26, equation (28)] *satisfies*

$$|\mathcal{P}_S| \le \tau^{-\eta} \vartheta^{\gamma} |\mathcal{P}| \quad (s/t \le \vartheta \le 1), \tag{9.I-K0}$$

where $\mathcal{P}_S$ has exactly the meaning of source equation (28): it counts only those $P \subset S$ for which the planes spanned by the two longest axes of P and of the $\vartheta \times 1 \times 1$ test slab S make angle at

most ϑ, then the original statement of [GWZ26, Lemma 6.1, equation (29)] *gives*

$$\mu(\mathcal{P},A)\lesssim \tau^{-\varepsilon}\Delta_{\max}(\mathcal{P})^{1-\beta}\left(\frac{s}{t}\right)^{\gamma\beta}|\mathcal{P}|^{\beta}. \tag{9.I-K}$$

For $(s,t,\gamma)=(a,b,0)$ and $(\delta/b,\delta/a,1)$, this becomes

$$\mu_{\mathrm{par}}\lesssim \tau^{-\varepsilon}\Delta_{\max}(\mathcal{P}_{\mathrm{par}})^{1-\beta}N_{\mathrm{par}}^{\beta}, \tag{9.I-48}$$

and

$$\mu_{\mathrm{ch}}\lesssim \tau^{-\varepsilon}\Delta_{\max}(\mathcal{P}_{\mathrm{ch}})^{1-\beta}\left(\frac{a}{b}\right)^{\beta}N_{\mathrm{ch}}^{\beta}. \tag{9.I-49}$$

Here $\mathcal{P}_{\mathrm{ch}}$ denotes the local child family supplied to the lemma.

Proof. Apply the scale-variant forms of the two cited lemmas at auxiliary scale τ. Formula (9.I-F) is equation (31) with its parameters renamed $(a,b)=(s,t)$. In the parent row of (9.I-F1), $M^{\beta/2}(s/t)\lesssim(a/b)^{1-\beta/2}$; in the child row it is $\lesssim(a/b)^{1-\beta}$. The remaining powers are obtained by replacing t with b and δ/a, respectively. Formula (9.I-K) is exactly equation (29) with the same renaming. Taking $\gamma=0$ and $\gamma=1$ gives (9.I-48)–(9.I-49), so both conclusions follow directly from the cited source lemmas. □

Proposition 9.6 (Small-middle cellular replacement in GWZ Proposition 6.6(A))**.** *Fix $0<\beta\le 1$, $\varepsilon>0$, and a loss profile $\mathfrak{p}$, and assume $K_F(\beta)$. There exist $\eta=\eta(\varepsilon,\beta)>0$, $b_0=b_0(\varepsilon,\beta)>0$, and $\delta_0=\delta_0(\varepsilon,\beta,\mathfrak{p})>0$ such that the following holds for $0<\delta\le\delta_0$. Here b_0 is chosen to be a sufficiently small structural multiple of the minimum of the original source thresholds, so every fixed enlargement used below of a plank with middle width at most b_0 still lies in the stated range of the corresponding source lemma. Let $(\mathbb{T},Y)$ be uniform in the sense of* [GWZ26, Definition 2.2, p. 4]*, with all source uniformity, comparability, and normalized-interface losses governed by $\mathfrak{p}$, and with $\lambda(\mathbb{T},Y)\ge\delta^{\eta}$, and suppose there is a source coarse scale $\delta\le\rho\le a\le b\le b_0$, with $\delta/a\le b_0$, for which the GWZ Part (A) factoring structure gives $\mathbb{T}=\bigsqcup_{W\in\mathcal{W}}\mathbb{T}_W$, where $\mathcal{W}$ is a nonempty labelled family of $a\times b\times 1$ parents and every $\mathbb{T}_W$ is nonempty. Assume that the normalized parent labels and normalized child representations have maps, with uniformly bounded fibres, to source-essentially-distinct carrier families, with each represented carrier contained in and containing fixed enlargements of its image. Assume also, as part of the factoring data, that their fixed-dilate source conflict graphs have maximum degree $O(1)$. Assume that the numbers $|\mathbb{T}_W|$ are comparable, that every $\mathbb{T}_W$ is Frostman in W, and that after the affine normalizations each parent and child plank family satisfies, with P_ϑ its source thickening,*

$$|\mathcal{P}[P_\vartheta]|\le M\vartheta\quad(P\in\mathcal{P},\ a/b\le\vartheta\le 1),$$

with

$$M_{\mathrm{par}}\lesssim b/a\quad\text{for the parent call,}\qquad M_{\mathrm{ch}}\lesssim(b/a)^2\quad\text{for the normalized child call.}$$

These are exactly the two bounds used in source equations (45)–(46) of [GWZ26]*. Then replacing source Proposition 5.1 by theorem 8.2 gives*

$$\mu(\mathbb{T},Y)\lesssim_{\log}\delta^{-\varepsilon}C_F(\mathbb{T})^{1-\beta/2}\left(\frac{a}{b}\right)^{3\beta/2}\delta^{-2\beta}\big(\delta^2|\mathbb{T}|\big)^{1-\beta/2}. \tag{9.1}$$

Thus the normalization and conclusion are exactly those of source Proposition 6.6(A), up to the explicit cellular logarithms and fixed enlargements.

Proof. Setup and retained density. Let $N_W=|\mathbb{T}_W|$, let M_W be the retained child shading mass in W, and let $N_0\le N_W\le C_NN_0$ be the fixed source count level. P1 gives $\sum_W M_W\gtrsim_{\log}M(Y)$,

and the retained labels form a subfamily of the source partition. Hence

$$\frac{\sum_W N_W \lambda(\mathbb{T}_W, Y_1)}{\sum_W N_W} = \frac{\sum_W M_W}{|T| \sum_W N_W} \gtrsim_{\log} \lambda(\mathbb{T}, Y). \tag{9.2}$$

Choose W_0 attaining this weighted average, so its density is at least $\delta^{\eta+o(1)}$. Take η below the threshold $\eta_{6.4}(\varepsilon, \beta)$ from [GWZ26, Lemma 6.4, p. 15], with strict reserve for the cellular losses and the scale-variant convention of [GWZ26, Remark 3.6, p. 6].

Child source call. The first row of the interface table is realized by the following checks. Normalizing W_0 gives widths

$$a_{\mathrm{ch}} = b^{-1}\delta, \qquad b_{\mathrm{ch}} = a^{-1}\delta, \qquad 0 < a_{\mathrm{ch}} \le b_{\mathrm{ch}} \le 1.$$

Here $b_{\mathrm{ch}} = \delta/a \le b_0$, and the auxiliary scale is $\tau_{\mathrm{src}} = \delta \le a_{\mathrm{ch}}$. After decreasing δ_0, the affine-invariant density from (9.2) exceeds $\delta^{\eta_{6.4}}$. The shading restriction leaves the complete child family, its convex Frostman tests, and the assumed count $|\mathcal{P}[P_\vartheta]| \le M\vartheta$, for every base $P \in \mathcal{P}$ and every $\vartheta \in [a/b, 1]$, unchanged.

Properly color the assumed child source-conflict graph. Its $O(1)$ colors are source-essentially-distinct; the heaviest color is the called family, and complete-containment counts do not increase. If λ_{ch} and $C_{F,\mathrm{ch}}$ are the pre-color density and normalized Frostman constant, lemma 9.3 gives

$$\lambda_{\mathrm{ch},c} \ge H_{\mathrm{src}}^{-1}\lambda_{\mathrm{ch}}, \qquad C_{F,\mathrm{ch},c} \le H_{\mathrm{src}}\lambda_{\mathrm{ch}}^{-1}C_{F,\mathrm{ch}}. \tag{9.2a}$$

Here $\lambda_{\mathrm{ch}} \ge \delta^{\eta+o(1)}$. With source-output allowance $\varepsilon/4$, choose the input exponents and δ_0 so that

$$\left(H_{\mathrm{src}}\lambda_{\mathrm{ch}}^{-1}C_{F,\mathrm{ch}}\right)^{1-\beta/2} \le \delta^{-\varepsilon/8} \tag{9.2a$'$}$$

The fixed enlargement and recovery factor H_{src} is also at most $\delta^{-\varepsilon/8}$. Thus the color and enlargement costs are reserved, and (9.0e$'$) returns the source estimate to the uncolored family. The call data are

$$\begin{gathered} s_{\mathrm{ch}} = \delta/b, \qquad t_{\mathrm{ch}} = \delta/a, \qquad \tau_{\mathrm{src}} = \delta \le s_{\mathrm{ch}}, \qquad \mathcal{F}_{\mathrm{ch}} = \text{the selected child color}, \\ \lambda(\mathcal{F}_{\mathrm{ch}}) \ge H_{\mathrm{src}}^{-1}\lambda_{\mathrm{ch}}, \qquad C_F(\mathcal{F}_{\mathrm{ch}}) \le H_{\mathrm{src}}\lambda_{\mathrm{ch}}^{-1}C_{F,\mathrm{ch}}, \qquad M_{\mathrm{ch}} \lesssim (b/a)^2. \end{gathered} \tag{9.2d0}$$

The normalized thick tests pull back to the source convex class, and replacing W_0 by its source Löwner rectangle costs a dimensional dilation. Applying the child specialization (9.I-46) of proposition 9.5, and then using (9.0e$'$) and (9.2a$'$), gives the following estimate on the uncoloured local family:

$$\mu(\mathbb{T}_{W_0}, Y_1) \lesssim \delta^{-\varepsilon/2} \left(\frac{a}{b}\right)^{1-\beta} \left(\frac{\delta}{a}\right)^{-2\beta} \left(\frac{\delta^2}{a^2}|\mathbb{T}_{W_0}|\right)^{1-\beta/2}. \tag{9.2d}$$

Average multiplicity is affine invariant, and the cardinality is labelled.

Parent source call. Here (8.4) supplies the positive parent density and $b \le b_0$ the source range. Let $\mathbb{T}'$ be the complete child labels retained in the cellular parents. Their shading mass is at least $L_{\mathrm{cell}}^{-1}M(Y)$, where $L_{\mathrm{cell}} = \delta^{-o(1)}$, and is at most their carrier denominator; hence

$$\frac{|\mathbb{T}'|}{|\mathbb{T}|} \ge L_{\mathrm{cell}}^{-1}\lambda(\mathbb{T}, Y). \tag{9.2b}$$

The restricted normalized fine Frostman constant is therefore at most $L_{\mathrm{cell}}\lambda(\mathbb{T}, Y)^{-1}C_F(\mathbb{T})$. Apply lemma 9.4 to $I_{\mathrm{par}} = \{(W, T) : T \in \mathbb{T}'_W\}$. The source assignment is unique on fine labels, and the selected parent-count level lies in $[N_0, C_N N_0]$; hence

$$C_F(\mathbb{W}') \le C_N L_{\mathrm{cell}}\lambda(\mathbb{T}, Y)^{-1}C_F(\mathbb{T}). \tag{9.2c}$$

This factor is $\delta^{-O(\eta)-o(1)}$. Fixed dilation preserves the source count $N(\theta) \lesssim \theta(b/a)$. Properly color the assumed parent source-conflict graph, and let $\widetilde{\mathbb{W}}'_c$ be the shading-mass-heavy color; its color number is $H_{\mathrm{src}} = O(1)$. With the density λ_{par} from (8.4), the parent call data are

$$\begin{gathered} s_{\mathrm{par}} = a, \qquad t_{\mathrm{par}} = b, \qquad \tau_{\mathrm{src}} = \delta \le s_{\mathrm{par}}, \qquad \mathcal{F}_{\mathrm{par}} = \widetilde{\mathbb{W}}'_c, \qquad M_{\mathrm{par}} \lesssim b/a, \\ \lambda(\mathcal{F}_{\mathrm{par}}) \ge H_{\mathrm{src}}^{-1}\lambda_{\mathrm{par}}, \\ C_F(\mathcal{F}_{\mathrm{par}}) \le H_{\mathrm{src}}\lambda_{\mathrm{par}}^{-1} C_N L_{\mathrm{cell}} \lambda(\mathbb{T}, Y)^{-1} C_F(\mathbb{T}). \end{gathered} \tag{9.2e0}$$

This verifies the second row of the interface table, using (9.2c) and lemma 9.3. With source-output allowance $\varepsilon/4$, impose

$$\left(H_{\mathrm{src}}\lambda_{\mathrm{par}}^{-1} C_N L_{\mathrm{cell}} \lambda(\mathbb{T}, Y)^{-1}\right)^{1-\beta/2} H_{\mathrm{src}} \le \delta^{-\varepsilon/4}. \tag{9.2e$'$}$$

Choose the input exponent and δ_0 with strict reserve so that the parent-color density is at least $\delta^{\eta_{6.4}}$. The specialization (9.I-45) of proposition 9.5, with $\tau_{\mathrm{src}} = \delta$, followed by recovery through (9.0e$'$), gives

$$\mu(\widetilde{\mathbb{W}}', Z) \lesssim \delta^{-\varepsilon/2} C_F(\mathbb{T})^{1-\beta/2} \left(\frac{a}{b}\right)^{1-\beta/2} b^{-2\beta} \left(b^2 |\widetilde{\mathbb{W}}'|\right)^{1-\beta/2}. \tag{9.2e}$$

Combination. Finally, P4–P5 give, for the chosen W_0,

$$\mu(\mathbb{T}, Y) \lesssim_{\log} \mu(\widetilde{\mathbb{W}}', Z)\mu(\mathbb{T}_{W_0}, Y_1), \tag{9.3}$$

which is source equation (44) with the cellular loss. Affine changes leave average multiplicity invariant, and the carrier and label denominators in (45)–(46) are the ones displayed above. Since the retained parents form a subfamily of the source partition and $N_W \in [N_0, C_N N_0]$,

$$|\widetilde{\mathbb{W}}'|\,|\mathbb{T}_{W_0}| \le C_N |\mathbb{T}|.$$

This cardinality bound and the scale factors in (9.2d)–(9.2e) combine explicitly as

$$\left(\frac{a}{b}\right)^{1-\beta/2+1-\beta+2\beta} \left(\frac{b}{a}\right)^{2-\beta} = \left(\frac{a}{b}\right)^{3\beta/2}. \tag{9.3a}$$

The factor $\delta^{-2\beta}(\delta^2|\mathbb{T}|)^{1-\beta/2}$ is the remaining child–parent volume product. Substitution in (9.3) proves (9.1), with the two $\varepsilon/2$ allowances adding to ε. □

Proposition 9.7 (Small-middle cellular replacement in GWZ Proposition 6.6(B))**.** *Fix $0 < \beta \le 1$, $\varepsilon > 0$, and a loss profile $\mathfrak{p}$, and assume $K_{KT}(\beta)$. There exist $\eta = \eta(\varepsilon, \beta) > 0$, $b_0 = b_0(\varepsilon, \beta) > 0$, and $\delta_0 = \delta_0(\varepsilon, \beta, \mathfrak{p}) > 0$ such that the following holds for $0 < \delta \le \delta_0$. Here b_0 is chosen to be a sufficiently small structural multiple of the minimum of the original source thresholds, so every fixed enlargement used below of a plank with middle width at most b_0 still lies in the stated range of the corresponding source lemma. Let $(\mathbb{T}, Y)$ be uniform in the sense of* [GWZ26, Definition 2.2, p. 4]*, with all source uniformity, comparability, and normalized-interface losses governed by $\mathfrak{p}$, and with $\lambda(\mathbb{T}, Y) \ge \delta^{\eta}$. Suppose that $\delta \le \rho \le a \le b \le b_0$, $\delta/a \le b_0$, and that the coarse ρ-tubes factor, in the sense of* [GWZ26, Definition 4.2]*, through a nonempty labelled family $\mathcal{W}$ of $a \times b \times 1$ rectangular parents satisfying $\Delta_{\max}(\mathcal{W}) \lesssim 1$, and assume that its normalized parent labels are source-essentially-distinct up to uniformly bounded representation fibres. Assume that the fine tubes are assigned to essentially distinct coarse ρ-tubes, each fine tube has $O(1)$ coarse representations, and the branching numbers lie in $[m, R_{\mathrm{src}}m]$, where $1 \le R_{\mathrm{src}} \le \delta^{-\omega_{\mathrm{br}}(\delta)}$ for a branching modulus ω_{br} controlled by $\mathfrak{p}$. Finally assume that every parent has a nonempty coarse-child family and that the coarse children in every parent $B \in \mathcal{W}$ satisfy*

$$\frac{\#\{S : S \subset K\}}{\#\{S : S \subset B\}} \lesssim_{\log} \frac{|K|}{|B|} \qquad (K \subset B \text{ convex}).$$

Then the coarse/thickened-child alternative of theorem 8.2 gives

$$\mu(\mathbb{T}, Y) \lesssim_{\log} \delta^{-\varepsilon} \Delta_{\max}(\mathbb{T})^{1-\beta} \left(\frac{a}{b}\right)^{\beta} |\mathbb{T}|^{\beta}. \tag{9.4}$$

This is the source Part (B) conclusion. It neither requires nor implies a fine-descendant all-convex Frostman condition.

Proof. Setup and labelled ancestry. Set $\varepsilon_* := \varepsilon/2$. Every source-output and recovery allowance uses ε_*; the remainder pays the final subpower cardinality factor. Source Part (B) factors the coarse ρ-tubes through the rectangular parents, as licensed by Remark 5.3 [GWZ26, proof of Proposition 6.6(B), pp. 24–25]. Its coarse-family input is

$$\frac{\#\{S \in \mathbb{T}_{\rho,B} : S \subset K\}}{|\mathbb{T}_{\rho,B}|} \lesssim_{\log} \frac{|K|}{|B|}, \tag{9.1a}$$

For every source containment $T \subset S$, form the labelled copy (S, T), with the carrier and shading of T, and assign it to S. Source bounded ancestry and branching comparability give $m \geq 1$ and

$$m \leq n_S := \#\{(S, T) : T \subset S\} \leq R_{\mathrm{src}} m, \qquad R_{\mathrm{src}} = \delta^{-o(1)}. \tag{9.1b}$$

Each fine label has between one and $\kappa_{\mathrm{cont}} = O(1)$ representations. Thus labelled mass and denominator change by factors in $[1, \kappa_{\mathrm{cont}}]$, the physical union is unchanged, and the coarse assignment is unique on representation labels.

Source factoring supplies the denominator comparison through $N_\rho \geq 1$ and $C_\rho = \delta^{-o(1)}$:

$$N_\rho \leq |\mathbb{T}_{\rho,B}| \leq C_\rho N_\rho \qquad (B \in \mathcal{W}). \tag{9.1b$'$}$$

If $D_B := |T| \sum_{S \in \mathbb{T}_{\rho,B}} n_S$ is the represented fine-carrier denominator in B, then (9.1b)–(9.1b$'$) give

$$m N_\rho |T| \leq D_B \leq R_{\mathrm{src}} C_\rho m N_\rho |T|. \tag{9.1b$''$}$$

Thus parent denominators are comparable by $R_{\mathrm{src}} C_\rho = \delta^{-o(1)}$, which verifies the ratio condition (6.6a) of lemma 6.13 and hypothesis (F2) of theorem 8.2.

For the later source calls we use the labelled form of source Lemma 6.1 at $0 < \tau \leq s \leq t$. Its low-multiplicity branch and the angular selections, refinements, and pigeonholing in source Lemmas 6.11 and 6.13 run on the label index set, with angles resolved below at s/t. Assign every surviving label to the essentially distinct thick proxy of source Definition 6.10 and dyadically regularize the proxy-fibre size N. Each retained proxy contains $\sim N$ labels and has volume $\asymp (t\vartheta/s)$ times the original plank volume. Fibre counting, together with the monotonicity in source Remark 3.3(B), gives

$$\Delta_{\max}(\mathcal{P}_{\vartheta,S}) \lesssim \frac{t\vartheta}{sN} \Delta_{\max}(\mathcal{P}), \qquad |\mathcal{P}_{\vartheta,S}| \lesssim N^{-1} |\mathcal{P}_S|. \tag{9.1c}$$

Source Lemma 6.13 also gives the proxy shading density $\gtrsim s^{4\eta_{\mathrm{in}}} \geq \tau^{4\eta_{\mathrm{in}}}$; choosing its input exponent with strict reserve verifies the scale-variant density threshold. Only the affine image of the proxy family enters source Lemma 3.7. Using $|\mathcal{P}_S| \leq \tau^{-\eta_{\mathrm{in}}} \vartheta^{\gamma} |\mathcal{P}|$ and $\vartheta \geq s/t$, the complete cancellation is

$$\frac{sN}{t\vartheta} \left(\frac{t\vartheta}{sN} \Delta_{\max}(\mathcal{P})\right)^{1-\beta} \left(N^{-1} |\mathcal{P}_S|\right)^{\beta} \lesssim \tau^{-\eta_{\mathrm{in}}\beta} \Delta_{\max}(\mathcal{P})^{1-\beta} \left(\frac{s}{t}\right)^{\gamma\beta} |\mathcal{P}|^{\beta}.$$

Thus N cancels exactly and the remaining auxiliary-scale power is absorbed by the source-output reserve. This proves equation (29), with all multiplicities, cardinalities, and $\Delta_{\max}$ counted on labels. Thus the analytic inequality in (9.I-49), and its $\gamma = 0$ specialization (9.I-48), hold for the labelled families below; the dyadic losses are governed by $\mathfrak{p}$.

Coarse parent estimate. Since $\rho \le a$, (9.1a) and (9.1b) verify lemma 6.12 with $R_{\mathrm{br}} = R_{\mathrm{src}}$, giving the local estimate (6.6d). Combining it with the denominator comparison (9.1b″) in corollary 6.14 gives the coarse version of (6.10), including arbitrary measurable fine shadings, clipping, and fixed enlargement, with factor $R_{\mathrm{src}}^{-3} = \delta^{o(1)}$.

Child source call. The weighted averaging (9.2), performed on the complete local representation families, chooses a useful inner parent W. Let $\mathcal{R}_W$ be its representation family and put

$$\lambda_W := \frac{\sum_{(S,T)\in\mathcal{R}_W} |Y'(S,T)|}{|\mathcal{R}_W|\,|T|}. \tag{9.4a}$$

Then $\lambda_W \ge \delta^{\eta+o(1)}$, with the stronger normalized-scale lower bound used in the source scale-variant call. We retain the complete labelled representation family. Since each fine label has at most κ_{cont} representations,

$$\lambda(\mathcal{R}_W, Y') = \lambda_W, \qquad \Delta_{\max}(\mathcal{R}_W) \le \kappa_{\mathrm{cont}}\Delta_{\max}(\mathbb{T}). \tag{9.4b}$$

For the restricted slab-test condition, let Φ_W normalize W. For $\vartheta \in [a/b, 1]$ and a source test slab Σ from (9.I-K0), put

$$K_{\vartheta,\Sigma} := \Phi_W^{-1}(\Sigma).$$

Let $\mathcal{K}_{\mathrm{ch}}$ be the pullbacks having nonempty equation-(28) subfamily; the omitted tests are trivial. Fix a counted normalized child plank P, and choose unit normals n_P, n_Σ to its two-long-axis plane and to Σ. In principal coordinates write $D = \Phi_W = \mathrm{diag}(a^{-1}, b^{-1}, 1)$. The plane calculation on pp. 24–25 of [GWZ26], before its final semiaxis comparison, shows that the normalized long plane of P contains a unit vector within $O(a/b)$ of e_1, and hence $|(n_P)_1| \lesssim a/b$. The equation-(28) angular condition permits signs such that

$$|n_\Sigma - n_P| \lesssim \vartheta, \qquad |(n_\Sigma)_1| \lesssim a/b + \vartheta \lesssim \vartheta.$$

Since $a/b \le \vartheta$, $b \le 1$, and $|n_\Sigma| = 1$,

$$a|D^T n_\Sigma| \lesssim \vartheta, \qquad a\|D\| = 1.$$

A fixed inner copy of the Löwner ellipsoid of the normalized $\vartheta \times 1 \times 1$ slab therefore contains a translate of $DB(0, ca)$: its normal support is at most $ca|D^T n_\Sigma| \lesssim c\vartheta$, while its tangential radius is at most $ca\|D\| \le c$. Pulling back gives a translated ca-ball in $K_{\vartheta,\Sigma}$. Writing $a'(K_{\vartheta,\Sigma})$ for its shortest Löwner semiaxis, we consequently have

$$a'(K_{\vartheta,\Sigma}) \gtrsim a \ge \rho.$$

Fix $K = K_{\vartheta,\Sigma}$, and let z_K be the centre of the inball just constructed. For a fine $T \subset K$, the full-carrier conclusion of lemma 6.8 gives

$$S \subset N_{C\rho}(T) \subset N_{C\rho}(K).$$

Thus lemma 3.3 gives

$$K^+ := K + B(0, C\rho) \subset z_K + C_+(K - z_K), \qquad |K^+| \le C_+^3|K|. \tag{9.4c}$$

Every counted coarse child lies in both K^+ and W. Hence (9.1a), branching comparability, and $|K^+| \lesssim |K|$ give

$$\frac{\#\{(S,T) : T \subset K\}}{\#\{(S,T) : T \subset W\}} \le R_{\mathrm{src}} \frac{\#\{S : S \subset K^+ \cap W\}}{\#\{S : S \subset W\}} \lesssim_{\log} R_{\mathrm{src}} \frac{|K|}{|W|}. \tag{9.5}$$

This is the normalized bound for the whole representation family on $\mathcal{K}_{\mathrm{ch}}$; explicitly,

$$\frac{\#\{(S,T) \in \mathcal{R}_W : T \subset K\}}{|\mathcal{R}_W|} \lesssim_{\log} R_{\mathrm{src}} \frac{|K|}{|W|} \qquad (K \in \mathcal{K}_{\mathrm{ch}}). \tag{9.5a}$$

Let $\mathcal{F}_{\mathrm{ch}} := \Phi_W(\mathcal{R}_W)$, and interpret $(\mathcal{F}_{\mathrm{ch}})_\Sigma$ exactly as the angularly filtered source equation (28) subfamily. It lies in the normalized numerator of (9.5a), and the affine Jacobian cancels, so

$$\frac{|K_{\vartheta,\Sigma}|}{|W|} \lesssim \frac{|\Sigma|}{|\Phi_W(W)|} \lesssim \vartheta, \qquad \frac{|(\mathcal{F}_{\mathrm{ch}})_\Sigma|}{|\mathcal{F}_{\mathrm{ch}}|} \lesssim_{\log} R_{\mathrm{src}}\vartheta. \tag{9.5a$'$}$$

This is precisely the restricted class in (9.I-K0), whose necessity was pointed out by Zeraoulia Rafik [Zer26b]. The normalized labelled image has dimensions $(\delta/b) \times (\delta/a) \times 1$; density, multiplicity, and absolute complete-containment density are affine invariant. Put $a_{\mathrm{ch}} = \delta/b$, $b_{\mathrm{ch}} = \delta/a$, and let $\eta_{6.1}$ be the exponent returned by source Lemma 6.1 with an $\varepsilon_*/4$ output allowance. Use its scale-variant form, licensed by [GWZ26, Remark 3.6 and the note after Proposition 6.6], with $\tau_{\mathrm{src}} = \delta \le a_{\mathrm{ch}}$. Choose the input exponent first and then δ_0 so that

$$R_{\mathrm{src}} \log^C(2/\delta) \le \delta^{-\eta_{6.1}}, \qquad \lambda_W \ge \delta^{\eta_{6.1}}. \tag{9.5b}$$

The strict density reserve and $R_{\mathrm{src}} \le \delta^{-\omega_{\mathrm{br}}(\delta)}$ permit (9.5b); moreover $\tau_{\mathrm{src}} = \delta$ and $b_{\mathrm{ch}} = \delta/a \le b_0$.

The seven interface fields for the child call are therefore

$$\begin{gathered} s_{\mathrm{ch}} = \delta/b, \qquad t_{\mathrm{ch}} = \delta/a, \qquad 0 < s_{\mathrm{ch}} \le t_{\mathrm{ch}} \le b_0, \qquad \tau_{\mathrm{src}} = \delta \le s_{\mathrm{ch}}, \qquad \gamma = 1, \\ \mathcal{F}_{\mathrm{ch}} = \Phi_W(\mathcal{R}_W), \qquad \lambda(\mathcal{F}_{\mathrm{ch}}) = \lambda_W \ge \delta^{\eta_{6.1}}, \\ |(\mathcal{F}_{\mathrm{ch}})_\Sigma| \le \delta^{-\eta_{6.1}}\vartheta|\mathcal{F}_{\mathrm{ch}}| \quad (s_{\mathrm{ch}}/t_{\mathrm{ch}} \le \vartheta \le 1,\ \Sigma \text{ as in (9.I-K0)}), \\ \Delta_{\max}(\mathcal{F}_{\mathrm{ch}}) \le \kappa_{\mathrm{cont}}\Delta_{\max}(\mathbb{T}), \qquad s_{\mathrm{ch}}/t_{\mathrm{ch}} = a/b. \end{gathered} \tag{9.5c0}$$

Thus (9.5c0) verifies the child Lemma 6.1 row, including the equation-(28) conditions. Apply the labelled form of source Lemma 6.1 proved at (9.1c), and absorb the fixed factor $\kappa_{\mathrm{cont}}^{1-\beta}$:

$$\mu(\mathcal{R}_W, Y') \lesssim \delta^{-\varepsilon_*/2}\Delta_{\max}(\mathbb{T})^{1-\beta}\left(\frac{a}{b}\right)^{\beta}|\mathcal{R}_W|^{\beta}. \tag{9.5c}$$

The complete original family is used, so there is no color-recovery factor.

Parent source call. For the parent call, $b \le b_0$, and we again retain the complete labelled family. Its call data are

$$\begin{gathered} s_{\mathrm{par}} = a, \qquad t_{\mathrm{par}} = b, \qquad \tau_{\mathrm{src}} = \delta \le s_{\mathrm{par}}, \qquad \gamma = 0, \qquad \mathcal{F}_{\mathrm{par}} = \mathbb{W}', \\ \lambda(\mathcal{F}_{\mathrm{par}}) = \lambda_{\mathrm{par}}, \qquad \Delta_{\max}(\mathcal{F}_{\mathrm{par}}) \lesssim 1. \end{gathered} \tag{9.5d0}$$

Absolute concentration is monotone under the retained parent subfamily, and the cellular density exceeds $\delta^{\eta_{6.1}}$ after decreasing δ_0. Hence the labelled $\gamma = 0$ form proved at (9.1c) gives

$$\mu(\mathbb{W}', Z) \lesssim \delta^{-\varepsilon_*/2}|\mathbb{W}'|^{\beta}. \tag{9.5d}$$

Combination. Finally the cellular product is source equation (47). Since the retained parents form a subfamily and (9.1b)–(9.1b$''$) give comparable represented denominators,

$$|\mathbb{W}'|\,|\mathcal{R}_W| \lesssim \kappa_{\mathrm{cont}}R_{\mathrm{src}}C_\rho|\mathbb{T}| = \delta^{-o(1)}|\mathbb{T}|.$$

Multiplying (9.5c)–(9.5d) through (47) costs $\delta^{-\varepsilon_*}$, together with the displayed cardinality factor. After all parameters and the loss profile $\mathfrak{p}$ are fixed, decrease $\delta_0(\varepsilon, \beta, \mathfrak{p})$ so that

$$(\kappa_{\mathrm{cont}}R_{\mathrm{src}}C_\rho)^{\beta} \le \delta^{-(\varepsilon-\varepsilon_*)}.$$

This is possible because the product is $\delta^{-o(1)}$. Thus the total cost is at most $\delta^{-\varepsilon}$, apart from the declared logarithmic factors, and (9.4) follows. The child call has used the restricted slab-test condition (9.5a$'$), as encoded in (9.5c0). □

Remark 9.8 (Scope of the two source interfaces). Propositions 9.6 and 9.7 cover precisely the two small-middle interfaces for which both the physical parent and normalized child middle widths lie below the returned b_0. The slab estimate (B.2) is used through its stated quadratic form. In Section 10, Lemma 6.4 is invoked in the thick branch, where $a \ge \delta^{1-\tau}$ gives $\delta/a \le \delta^{\tau} < b_0$ after decreasing δ_0. The large-middle branch (S1b) instead uses (10.15)–(10.16) and theorem B.3, while (S1c) uses neither proposition nor a normalized Lemma 6.1/6.4 child call.

Remark 9.9 (Source version). The two calls use Lemmas 6.1 and 6.4 in the forms recorded in proposition 9.5; Appendix D records the corresponding v1 textual comparison.

Within the scope fixed in remark 9.8, these two propositions complete the small-middle applications. Their proof uses the cellular theorem as a self-contained interface and is independent of the multiscale synchronization in the next section.

10. Cellular closure of the very-non-sticky reduction

The very-non-sticky application represents fine tubes by tagged tubelets, factors them through biased parents, regularizes and angularly refines the parent–cell graph, and lifts the survivors to the original labels. The ledger below separates tubelet mass from represented fine mass and keeps angular uniformity on its reference shading. After synchronization and cellularization, five geometric branches close in the parameter order of section 10.9. The final application is conditional on $K_{KT}(\beta)$, $K_F(\beta)$, (10.33), and (I1)–(I4); Appendix D gives the source correspondence.

10.1. **Reference shadings and the object hierarchy.** Fix an r_1-ball B, with $r_1 = \delta^{\varepsilon_{\mathrm{scal}}}$. The objects and their three principal transitions are summarized here; the tagged equations below provide the quantitative definitions.

object	carrier level	creating operation	retained property / later use
$\bar{Y}$	fine tubes	source angular uniformization	fixed reference for cap and direction uniformity
Y^{sc}	fine tubes	all-discrete-scale refinement	source scale shadings and comparisons
$Y^{\mathrm{pre}} \to Y^0$	represented fine tubes	canonical levels and whole-tag level, then good-ball deletion	representation coverage and the mass bound (10.1)
Y_B	canonical tubelets	good-ball restriction of $\widetilde{Y}_B$	reference shading for the local cellular input
$Y_{\mathrm{reg}} \to Y_1, Z$	tubelets / parents	joint cellular factoring, then saturated local-volume selection	compatibility and synchronized multiplicities
$(Y_2, Z^{\angle})$	tubelets / parents	ANGLE-CELL and one lift	resolved angle and retained tubelet mass
$\widehat{Y} \to Y^*$	fine tubes	global fine level and bias, then retained tubelet incidences	represented physical fine mass and final lift

Apply lemma 10.3 to the fixed $Y^{\mathrm{sc}} \subset \bar{Y}$. The induced physical shading Y^{pre} is defined by (10.7X) after the canonical multiplicity and density levels (10.7T)–(10.7W) and the whole-tag outer-ball level (10.7W′). Every (T, x), $x \in Y^{\mathrm{pre}}(T)$, then has a labelled canonical representation (B, T_B) with $T \in \mathbb{T}(T_B)$ and $x \in \widetilde{Y}_B(T_B)$. After the good-ball deletion define

$$Y^0(T) := Y^{\mathrm{pre}}(T) \cap \bigcup_{(B,T_B):\, T \in \mathbb{T}(T_B)} Y_B(T_B). \tag{10.0}$$

Thus Y^0 is the represented fine shading before biased factoring, and

$$\sum_T |Y^0(T)| \ge c_{\mathrm{pre}} \sum_T |\bar{Y}(T)|; \tag{10.1}$$

more explicitly, if L_{tag} is the number of classes in the whole-tag outer-ball level, then (10.7W)–(10.7X), (10.7W′), and (10.1d) permit

$$c_{\mathrm{can}} := \frac{1}{2\kappa_{\mathrm{inc}} L_{\mathrm{mult}} L_{\mathrm{dens}} L_{\mathrm{tag}}}, \qquad c_{\mathrm{pre}} := \frac{c_{\mathrm{sc}} c_{\mathrm{can}}}{4\kappa_{\mathrm{inc}}}. \tag{10.1′}$$

All four level counts are logarithmic under the polynomial-complexity and positive-density hypotheses, so $c_{\mathrm{pre}}^{-1} = \delta^{-o(1)}$. The good-ball restriction of $\widetilde{Y}_B$ defines the fixed tubelet shading Y_B; joint factoring gives $Y_{\mathrm{reg}} \subset Y_B$ and Z, the saturated local-volume selection gives $Y_1 \subset Y_{\mathrm{reg}}$, and ANGLE-CELL gives $Y_2 \subset Y_1$ and $Z^{\angle}$. The global fine-multiplicity level followed by the biased tubelet subset defines $\widehat{Y} \subset Y^0$; if the latter retains represented fraction c_{bias}, set $c_{\mathrm{setup}} = c_{\mathrm{pre}} c_{\mathrm{bias}}$. Finally $Y^* \subset \widehat{Y}$ is represented by the retained tubelet incidences. Cap and direction uniformity remain attached to $\bar{Y}$.

For $x \in Y^{\mathrm{pre}}(T)$, let

$$\mathcal{R}_{\mathrm{pre}}(T,x) := \{(B,T_B) : T \in \mathbb{T}(T_B),\ x \in \widetilde{Y}_B(T_B)\}, \qquad \kappa_{\mathrm{inc}} := \sup_{T,x} |\mathcal{R}_{\mathrm{pre}}(T,x)|. \tag{10.0a}$$

The direct tagged-incidence assignment of lemma 10.3 gives $1 \le |\mathcal{R}_{\mathrm{pre}}(T,x)| \le \kappa_{\mathrm{inc}} = O(1)$. After the good-ball deletion put

$$\mathcal{R}_0(T,x) := \{(B,T_B) : T \in \mathbb{T}(T_B),\ x \in Y_B(T_B)\}. \tag{10.0b}$$

Definition (10.0) gives

$$1 \le |\mathcal{R}_0(T,x)| \le \kappa_{\mathrm{inc}} \qquad (x \in Y^0(T)). \tag{REP}$$

Thus source (97) is read on Y_B, whereas (100)–(102) compare Y^* with $\bar{Y}$; direction uniformity is not assigned to Y^0 or Y^*.

10.2. **Good balls and global fine-multiplicity synchronization.**

Lemma 10.1 (Reordered good-ball selection)**.** *Let $\widetilde{Y}_B(T_B)$ be the canonical tubelet shading immediately after (10.7T)–(10.7W) and the whole-tag outer-ball selection. Suppose $0 < 2\eta_{\mathrm{src}} < \eta$ and*

$$|\widetilde{Y}_B(T_B)| \gtrsim \delta^{2\eta_{\mathrm{src}}} |T_B|, \qquad m(T_B) \le g_{T_B}(x) < 2m(T_B) \quad (x \in \widetilde{Y}_B(T_B)). \tag{10.1a}$$

The source good-δ-ball operation may be performed at this point, before biased parent and dimension selection. It produces a subshading $Y_B(T_B)$, retains a fixed fraction of every tubelet's shaded and represented fine-incidence mass for sufficiently small δ, and preserves the second assertion in (10.1a) on $Y_B(T_B)$. With Y^0 defined by (10.0), it also gives

$$M(Y^0) \ge (4\kappa_{\mathrm{inc}})^{-1} M(Y^{\mathrm{pre}}). \tag{10.1d}$$

All subsequent biased-parent, dimension, and child-count selections may be made by whole tubelet or parent labels, so the same pointwise level persists until cellular factoring.

Proof. Cover each tubelet by a fixed bounded-overlap family of δ-balls. The balls below threshold contribute at most $C_{\mathrm{ov}} c \delta^{\eta} |T_B|$, which is less than half of (10.1a) because $\eta > 2\eta_{\mathrm{src}}$. Since $g_{T_B} \in [m(T_B), 2m(T_B))$ on the old shading, writing $E = \widetilde{Y}_B(T_B)$ and $E' = Y_B(T_B)$ gives

$$\int_{E'} g_{T_B} \ge m(T_B)|E'| \ge \tfrac{1}{2} m(T_B)|E|, \qquad \int_{E} g_{T_B} \le 2m(T_B)|E|, \tag{10.1c}$$

so at least one quarter of represented mass remains. With $\mathcal{D}_{\mathrm{good}}(T_B)$ the passing balls, define

$$Y_B(T_B) := \widetilde{Y}_B(T_B) \cap \bigcup_{D \in \mathcal{D}_{\mathrm{good}}(T_B)} D. \tag{10.1b}$$

Thus every retained point lies in a good ball and keeps its dyadic level. Summing (10.1c) over labelled representations and using the κ_{inc}-to-one physical projection proves (10.1d).

Bias and dimension choices use only carrier data, so lemma 10.15 moves them after this deletion. The global m_{fine}-level must still precede the biased subset; all later child-count operations delete whole labels and preserve its pointwise range. □

Thus, without an occupancy pigeonhole,

$$|D \cap Y_B(T_B)| \gtrsim \delta^{\eta}\delta^3 \quad \text{for every selected ball } D. \tag{10.2}$$

This is the quantified replacement for source equation (94). Accordingly, the global m_{fine}-level is selected before any occupancy class based on unweighted tubelet mass.

For a tubelet T_B, put

$$g^0_{T_B}(x) = \#\{T \in \mathbb{T}(T_B) : x \in Y^0(T)\}. \tag{10.3}$$

Let $g^{\mathrm{pre}}_{T_B}$ denote the canonical count defined in (10.7S), before the good-ball deletion. On the retained tubelet shading one has the identity

$$g^{\mathrm{pre}}_{T_B}(x) = g^0_{T_B}(x) \qquad (x \in Y_B(T_B)). \tag{10.3a}$$

Indeed, if T is counted on the left, then $x \in Y^\circ(T) \cap B$ and $x \in Y_B(T_B) \subset \widetilde{Y}_B(T_B)$; hence (10.7X) places $x \in Y^{\mathrm{pre}}(T)$, and the same retained representation in (10.0) places $x \in Y^0(T)$. Conversely, if T is counted on the right, then $Y^0(T) \subset Y^{\mathrm{pre}}(T) \subset Y^\circ(T)$, while $x \in Y_B(T_B) \subset B$, so T is counted on the left. We write g_{T_B} for this common restriction to $Y_B(T_B)$.

The canonical construction (10.7S)–(10.7U) selects one dyadic level of $g^{\mathrm{pre}}_{T_B}$ and induces the corresponding fine refinement before any biased parent is chosen. Identity (10.3a) transfers that level to $g^0_{T_B}$. Lemma 10.1 performs the one intervening pointwise tubelet operation before our synchronization and proves that its restriction preserves that level. Consequently there is a number $m(T_B) \geq 1$ such that

$$m(T_B) \leq g_{T_B}(x) < 2m(T_B) \quad (x \in Y_B(T_B)). \tag{10.4}$$

What is missing is comparability between different tubelets.

We separate the global incidence constant in (10.0a) from the following local label constant. Uniformly over tagged outer balls B, put

$$\kappa_{\mathrm{loc}} := \sup_{B,T} \#\{T_B \in \mathbb{T}_B : T \in \mathbb{T}(T_B)\}. \tag{10.4b}$$

Bounded overlap of the outer balls and the single-valued half-open canonical assignment, with the fixed neighboring-cell convention, give $\kappa_{\mathrm{inc}} \leq C\kappa_{\mathrm{amb}} = O(1)$ and $\kappa_{\mathrm{loc}} = O(1)$. A fine *label* may cross $\asymp r_1^{-1}$ different outer balls. The uniform bounds concern instead a fine incidence (T, x), which has $O(1)$ tagged representations, and a fine label within a fixed tag, which has $O(1)$ canonical tubelets.

Lemma 10.2 (Global fine-multiplicity synchronization)**.** *There is one dyadic value m_{fine}, selected after the good-ball deletion and before biased factoring, for which the fine mass bound (10.6) holds and every selected tubelet satisfies*

$$m_{\mathrm{fine}} \leq m(T_B) < 2m_{\mathrm{fine}}. \tag{10.4c}$$

On every later whole-tubelet subshading A, the same fixed level obeys the identity (10.7). If a subsequent whole-label selection retains a fraction c of tubelet shading mass, it retains at least $c/(4\kappa_{\mathrm{inc}})$ of represented physical fine mass. No spatial selection made after synchronization is used to redefine m_{fine}.

Proof. Let $N_{\max}$ dominate the fine multiplicities and set $L_m = 1 + \lceil \log_2 N_{\max} \rceil$. For dyadic m, define

$$F_m = \sum_{(B,T_B):\, m \leq m(T_B) < 2m} \int_{Y_B(T_B)} g_{T_B}(x)\, dx. \tag{10.5}$$

By (REP), summing all dyadic layers gives the two-sided comparison

$$M(Y^0) \le \sum_m F_m \le \kappa_{\mathrm{inc}} M(Y^0). \tag{10.5a}$$

Choose $F_{m_{\mathrm{fine}}} \ge M(Y^0)/L_m$ and retain physical incidences having a representation in that layer. By (REP), their mass is at least $(\kappa_{\mathrm{inc}} L_m)^{-1} M(Y^0)$. Biased factoring then retains the represented fraction $c_{\mathrm{bias}} = \delta^{o(1)}$, because tubelet shadings have comparable measure and represented masses are comparable to $m_{\mathrm{fine}}|Y_B(T_B)|$. The induced shading $\widehat{Y}$ satisfies

$$\sum_T |\widehat{Y}(T)| \gtrsim (\kappa_{\mathrm{inc}} L_m)^{-1} c_{\mathrm{setup}} \sum_T |\bar{Y}(T)|. \tag{10.6}$$

For a later tubelet subshading $A(T_B) \subset Y_B(T_B)$, define

$$Y_A(T) := Y^0(T) \cap \bigcup_{(B,T_B):\, T \in \mathbb{T}(T_B)} A(T_B). \tag{10.6a}$$

Its labelled fine-incidence mass is

$$\begin{aligned} M_{\mathrm{lab}}(A) &= \sum_{(B,T_B)} \sum_{T \in \mathbb{T}(T_B)} |Y^0(T) \cap A(T_B)| \\ &= \sum_{(B,T_B)} \int_{A(T_B)} g_{T_B}(x)\, dx, \end{aligned} \tag{10.7}$$

and (10.4), (10.4c) give the explicit two-sided estimate

$$m_{\mathrm{fine}} \sum_{(B,T_B)} |A(T_B)| \le M_{\mathrm{lab}}(A) < 4 m_{\mathrm{fine}} \sum_{(B,T_B)} |A(T_B)|. \tag{10.7a$'$}$$

By (10.6a) and (REP), the physical fine mass satisfies the exact bridge

$$\kappa_{\mathrm{inc}}^{-1} M_{\mathrm{lab}}(A) \le M(Y_A) \le M_{\mathrm{lab}}(A). \tag{10.7$'$}$$

Whole-tubelet, descendant-label, or tag deletion preserves (REP) for the physical shading defined by (10.6a); a common spatial restriction does as well. If A' retains a tubelet-mass fraction c, then

$$M_{\mathrm{lab}}(A') \ge m_{\mathrm{fine}} \sum |A'| \ge c\, m_{\mathrm{fine}} \sum |A| > \frac{c}{4} M_{\mathrm{lab}}(A),$$

and (10.7$'$) supplies $\kappa_{\mathrm{inc}}^{-1}$. This is the claimed tubelet-to-fine mass bridge. □

Multiplicity-expanded tubelet labels. After the global value m_{fine} has been fixed, replace every selected geometric tubelet label T_B by the coincident labelled copies

$$\mathbb{T}_B^{\mathrm{lab}} := \{(T_B, \ell) : T_B \in \mathbb{T}_B,\ 1 \le \ell \le m_{\mathrm{fine}}\}, \qquad Y_B^{\mathrm{lab}}(T_B, \ell) := Y_B(T_B). \tag{10.7a}$$

Then

$$\begin{aligned} M(\mathbb{T}_B^{\mathrm{lab}}, Y_B^{\mathrm{lab}}) &= m_{\mathrm{fine}} \sum_{T_B} |Y_B(T_B)| \asymp M_{\mathrm{lab}}(Y_B), \\ D(\mathbb{T}_B^{\mathrm{lab}}) &= m_{\mathrm{fine}} \sum_{T_B} |T_B|, \qquad U(\mathbb{T}_B^{\mathrm{lab}}, Y_B^{\mathrm{lab}}) = U(\mathbb{T}_B, Y_B). \end{aligned} \tag{10.7b}$$

Thus the expansion changes labelled mass and denominator but not the physical union. The required concentration transfer is the following separate geometric statement.

Lemma 10.3 (Canonical tubelets from tagged fine incidences)**.** *Let $0 < \delta \le r_1 \le 1/10$, let $\mathcal{B}$ be a bounded-overlap cover by radius-r_1 balls, and let Y° be a shading of a labelled family $\mathbb{T}$ of capped unit δ-tubes. Assume that the balls cover $U(\mathbb{T}, Y^\circ)$. All structural constants are fixed before the construction, in the following order: first the cap convention and its ball-enlargement constant C_2; then $C_J > 2C_2 + 2$; then a sufficiently fine half-open net mesh c_{net}; and finally containment dilations C_{can} and $C_\dagger$. Thereafter an unadorned C denotes a fixed maximum of*

these constants. None is changed with the tag, scale, dyadic level, or any later exponent choice. Before any tubelet pigeonhole one can construct, in every tag $B \in \mathcal{B}$, a fixed half-open net $\mathbb{T}_B^{\mathrm{can}}$ of capped $C\delta \times C\delta \times Cr_1$ carriers and a descendant assignment

$$\pi_B : \{T : Y^\circ(T) \cap B \neq \varnothing\} \longrightarrow \mathbb{T}_B^{\mathrm{can}} \tag{10.7P}$$

such that

$$|\mathbb{T}_B^{\mathrm{can}}| \le C(1 + r_1/\delta)^5 \tag{10.7P$''$}$$

and with the following properties.

(C1) *For every active pair (B, T), the carrier $T_B := \pi_B(T)$ contains the solid piece $T \cap B$, is contained in CB, has volume comparable to $\delta^2 r_1$, and contains a subsegment of the carrier of T of length at least cr_1. If B meets an endcap of T, this segment is chosen one-sided.*

No comparison is asserted between $|T_B|$ and $|T \cap B|$, and no inclusion $T_B \subset N_{C\delta}(T \cap B)$ is asserted. In particular this statement remains literal when T only grazes B and $|T \cap B|/(\delta^2 r_1)$ tends to zero.

(C2) *The map π_B has a fixed half-open tie rule and never merges fine labels. In each fixed tag, every active fine label has exactly one canonical representation. Consequently*

$$\kappa_{\mathrm{loc}} = 1 \tag{10.7Q}$$

for this assignment (or $O(1)$ if a fixed set of neighbouring net cells is retained). A physical fine incidence (T, x) belongs to at least one and at most

$$\kappa_{\mathrm{inc}} \le C\kappa_{\mathrm{amb}} \tag{10.7R}$$

tagged representations.

(C3) *Put*

$$\mathbb{T}(T_B) := \{T : \pi_B(T) = T_B\}, \qquad E_{B,T} := Y^\circ(T) \cap B,$$

and

$$g_{T_B}(x) := \sum_{T \in \mathbb{T}(T_B)} \mathbf{1}_{E_{B,T}}(x). \tag{10.7S}$$

Let

$$N_{\mathrm{can}} := \max\{1, \max_{B,T_B} |\mathbb{T}(T_B)|\},$$

where the inner maximum is interpreted as zero if there is no active label, and put $L_{\mathrm{mult}} := 1 + \lceil \log_2 N_{\mathrm{can}} \rceil$. For every canonical label carrying positive represented mass $\sum_{T \in \mathbb{T}(T_B)} |E_{B,T}| > 0$, choose a dyadic number $m(T_B)$ and set

$$\widetilde{Y}_B(T_B) := \{x \in T_B : m(T_B) \le g_{T_B}(x) < 2m(T_B)\} \tag{10.7T}$$

and set $\widetilde{Y}_B(T_B) = \varnothing$ on every remaining zero-mass canonical label, so that

$$\sum_{B,T_B} \int_{\widetilde{Y}_B(T_B)} g_{T_B} \ge L_{\mathrm{mult}}^{-1} \sum_{B,T} |Y^\circ(T) \cap B|. \tag{10.7U}$$

If $\lambda(\mathbb{T}, Y^\circ) =: \lambda^\circ > 0$, labels with

$$\frac{|\widetilde{Y}_B(T_B)|}{|T_B|} < c\lambda^\circ / L_{\mathrm{mult}} \tag{10.7V}$$

carry at most half of the left side of (10.7U), after decreasing the structural c. A further dyadic shading-density selection therefore retains at least

$$(2L_{\mathrm{mult}} L_{\mathrm{dens}})^{-1} \sum_{B,T} |Y^\circ(T) \cap B|, \qquad L_{\mathrm{dens}} := 1 + \left\lceil \log_2 \frac{CL_{\mathrm{mult}}}{\lambda^\circ} \right\rceil, \tag{10.7W}$$

as represented fine-incidence mass, and every surviving canonical label has one common density up to a factor two, bounded below by the right side of (10.7V). The induced physical fine shading

$$Y^{\mathrm{pre}}(T) := Y^{\circ}(T) \cap \bigcup_{(B,T_B):\, T\in\mathbb{T}(T_B)} \widetilde{Y}_B(T_B) \tag{10.7X}$$

satisfies the same lower bound with the additional factor $\kappa_{\mathrm{inc}}^{-1}$.

(C4) *If $T_B \subset K$ for a convex body K, set $K^{\dagger} = K + CB_\delta$. Let $R_{K^\dagger} \supset K^{\dagger}$ be a comparable-volume containing Löwner rectangle with centre z_K, and put*

$$K^{+} = z_K + Cr_1^{-1}(R_{K^\dagger} - z_K). \tag{10.7Y}$$

Every descendant $T \in \mathbb{T}(T_B)$ is contained in this same K^+, and

$$|K^{\dagger}| \le C|K|, \qquad |K^{+}| \le Cr_1^{-3}|K|. \tag{10.7Z}$$

This is the only complete-containment transfer used below; it is one-way from a complete canonical carrier to its fine descendants.

Proof. Write I_T for the unit carrier segment of T, and c_B for the centre of B. Fix a sufficiently large structural C_J, and put

$$\ell_{r_1} := \min\{1, C_J r_1\} \asymp r_1. \tag{10.7P$'$}$$

The set $I_T \cap B(c_B, C_2 r_1)$ is a segment of length at most $2C_2 r_1$. Here C_2 is chosen from the fixed tube-radius convention so that, whenever $T \cap B \neq \varnothing$, this set contains every carrier point within $C\delta$ of $T \cap B$. Choose $C_J > 2C_2 + 2$. If $C_J r_1 < 1$, the displayed segment is contained in a subsegment $J_{B,T} \subset I_T$ of the single length $\ell_{r_1} = C_J r_1$; near an endpoint of I_T, take the extension entirely to the available side. If $C_J r_1 \ge 1$, take $J_{B,T} = I_T$. Thus in all cases every $J_{B,T}$ has the same length ℓ_{r_1}, $N_{C\delta}(J_{B,T})$ contains $T \cap B$, and that neighbourhood lies in CB. This fixed-length window, rather than the possibly very short raw intersection, is the object that is discretized.

Parameterize these equal-length windows by $(u, m) \in \mathbb{RP}^2 \times CB$, where u is the unoriented direction and m the midpoint, and use the scaled product metric

$$d_{r_1}((u,m),(u',m')) := r_1 d_{\mathbb{RP}^2}(u,u') + |m - m'|.$$

Choose a maximal $(c\delta/r_1)$-separated net in $\mathbb{RP}^2$ and a maximal $(c\delta)$-separated net in CB. Order both finite nets once and use the induced half-open Voronoi cells in the product to assign each $J_{B,T}$ to the first nearest net window. Thus the assignment is a genuine function even on Voronoi boundaries. The two-dimensional projective packing bound and three-dimensional midpoint packing bound give

$$\#\mathcal{N}_{\mathrm{dir}} \le C(1 + r_1/\delta)^2, \qquad \#\mathcal{N}_{\mathrm{mid}} \le C(1 + r_1/\delta)^3,$$

which proves (10.7P$''$). A fixed dilation of the chosen net tube contains $N_{C\delta}(J_{B,T})$, so it contains $T \cap B$, remains in CB, has volume comparable to $\delta^2 r_1$, and its carrier segment has Hausdorff distance $O(\delta)$ from $J_{B,T}$. This proves (C1) and the geometric comparison later recorded in (10.7L). Since the assignment acts on fine labels, coincident geometric carriers remain distinct.

Inside one tag the map is single-valued, which proves (10.7Q). If $x \in Y^{\circ}(T)$, every tag used to represent (T, x) contains x. The balls cover the union, so at least one representation exists; bounded overlap gives at most κ_{amb} tags at x. The canonical net is single-valued inside each such tag. Retaining a fixed set of neighbouring net cells changes this by at most a dimensional packing constant. This proves (10.7R): the uniform bound applies to the incidence (T, x), even though a whole unit tube may cross $O(r_1^{-1})$ tags.

For one T_B carrying positive represented mass, layer-cake decomposition of the integer-valued function g_{T_B} gives a level satisfying

$$\int_{\widetilde{Y}_B(T_B)} g_{T_B} \geq L_{\mathrm{mult}}^{-1} \sum_{T \in \mathbb{T}(T_B)} |E_{B,T}|.$$

Zero-mass canonical labels have the declared empty selected shading and contribute zero to both sides. Summing proves (10.7U). The density selection follows from a direct volume-packing estimate for the tag count. The centre of every active radius-r_1 ball lies in $N_{Cr_1}(T)$; bounded overlap and integration over that neighbourhood give

$$\#\{B : Y^\circ(T) \cap B \neq \varnothing\} r_1^3 \lesssim |N_{Cr_1}(T)| \lesssim r_1^2.$$

Thus a unit tube meets only $O(r_1^{-1})$ active tags, and

$$\sum_{B,T_B} |\mathbb{T}(T_B)|\,|T_B| \lesssim |\mathbb{T}|\,|T|, \qquad \sum_{B,T} |Y^\circ(T) \cap B| \geq M(Y^\circ). \tag{10.7U$'$}$$

On the chosen multiplicity level,

$$\int_{\widetilde{Y}_B(T_B)} g_{T_B} < 2m(T_B)|\widetilde{Y}_B(T_B)|, \qquad m(T_B) \leq |\mathbb{T}(T_B)|. \tag{10.7U$''$}$$

Combining (10.7U)–(10.7U$''$) shows

$$\sum_{B,T_B} m(T_B)|\widetilde{Y}_B(T_B)| \gtrsim \frac{\lambda^\circ}{L_{\mathrm{mult}}} \sum_{B,T_B} m(T_B)|T_B|.$$

The labels failing (10.7V) therefore contribute at most half of the selected represented mass when c is sufficiently small. On the remaining range the density lies between $c\lambda^\circ/L_{\mathrm{mult}}$ and one, so there are at most L_{dens} dyadic classes. Selecting a class by represented mass proves (10.7W). Each physical incidence is counted at most κ_{inc} times, which proves (10.7X). Thus the canonical objects carry the two tubelet levels and the induced fine refinement used on pp. 36–37 of [GWZ26].

It remains to prove the one-way convex assertion. If $T_B \subset K$, the net construction supplies, for every descendant T, a carrier subsegment of length at least cr_1 inside $K^\dagger$, with direction within $C\delta/r_1$ of the canonical direction. Because K contains the $C\delta$-thick carrier T_B, it contains a ball of radius $c\delta$. Applying lemma 3.3 with the fixed ratio between the padding and this inradius shows that adding CB_δ changes its volume by only a dimensional factor. Choose the containing Löwner rectangle $R_{K^\dagger}$ and write its half-side lengths as s_1, s_2, s_3. Subtracting the endpoint coordinates of the contained cr_1-segment gives $r_1|u_i| \leq Cs_i$ for the unit direction u of T. From that segment to either endpoint of the unit carrier costs at most $Cr_1^{-1}s_i$ in coordinate i. The solid radius is already absorbed by $K^\dagger$. Hence $T \subset K^+$, simultaneously for all descendants, and

$$|K^+| \lesssim r_1^{-3}|R_{K^\dagger}| \lesssim r_1^{-3}|K|.$$

This proves (10.7Y)–(10.7Z) and the lemma. □

For the application take $Y^\circ = Y^{\mathrm{sc}}$. After (10.7T)–(10.7W), set $\mathsf{F}_B := \sum_{T_B} \int_{\widetilde{Y}_B(T_B)} g_{T_B}(x)\,dx$ for each nonempty tag. Partition the tags dyadically by F_B, retain a class maximizing $\sum_B \mathsf{F}_B$, and restrict all subsequent canonical objects, including the application instance of Y^{pre}, to that class. Let L_{tag} count the nonempty dyadic levels. By (10.7V), the application density lower bound, $m(T_B) \geq 1$, and $|T_B| \asymp \delta^2 r_1$, every nonzero F_B has a fixed-power lower bound; (10.7P$''$) and the input cardinality bound give a fixed-power upper bound. Hence

$$L_{\mathrm{tag}} \leq C \log(2/\delta). \tag{10.7W$'$}$$

This whole-tag level leaves the canonical carriers, descendants, pointwise multiplicity, and tubelet densities fixed.

For the application, fix a structural integer C_{pre} and define the pre-cellular profile

$$\mathcal{E}_{\mathrm{pre}}(\delta) := \max\left\{1, \delta^{-C_{\mathrm{pre}}\omega_0(\delta)}, [\log(2/\delta)]^{C_{\mathrm{pre}}}\right\}, \qquad \omega_{\mathrm{pre}}(\delta) := \frac{\log \mathcal{E}_{\mathrm{pre}}(\delta)}{\log(1/\delta)}. \tag{10.7W$''$}$$

Here C_{pre} absorbs the two input-refinement factors $c_{\mathrm{init}}^{-1}, c_{\mathrm{sc}}^{-1}$, the canonical multiplicity level, and fixed representation constants. Thus $\omega_{\mathrm{pre}}(\delta) \to 0$, and its definition involves neither J_v, an angular factor, nor any post-cellular quantity.

Lemma 10.4 (One-way consequences of canonical ancestry)**.** *Let $0 < \delta \le r_1 \le 1/10$, let B be one fixed tagged r_1-ball, and let $T = N_{c\delta}(I_T)$ be a capped unit δ-tube whose carrier segment I_T has length 1. Use exactly the canonical assignment π_B of lemma 10.3. The following aliases record its one additional local estimate and preserve the downstream interface.*

(G1) *The representation space, tubelet dimensions, containment, and tag localization are those of clause (C1) in lemma 10.3.*

(G2) *If $T \in \mathbb{T}(T_B)$, there is a subsegment $I_{T,B} \subset I_T$ of length at least cr_1 such that*

$$\operatorname{angle}(I_T, I_{T_B}) \le C\delta/r_1, \qquad d_H(I_{T,B}, I_{T_B}) \le C\delta. \tag{10.7L}$$

The assertion remains true when B meets an endcap of T: in that case $I_{T,B}$ is chosen one-sided and CB contains it.

(G3) *The fixed-tag representation bound is clause (C2): $\kappa_{\mathrm{loc}} = 1$ for the half-open assignment and $O(1)$ when the fixed neighboring-cell convention is retained.*

(G4) *If $T_B \subset K$ for a convex body K, set $K^\dagger = K + CB_\delta$. Let $R_{K^\dagger} \supset K^\dagger$ be a comparable-volume containing rectangle with centre z_K, and put*

$$K^+ = z_K + Cr_1^{-1}(R_{K^\dagger} - z_K). \tag{10.7M}$$

Then every descendant $T \in \mathbb{T}(T_B)$ is contained in K^+, and

$$|K^\dagger| \le C|K|, \qquad |K^+| \le Cr_1^{-3}|K|. \tag{10.7N}$$

By clause (C4), these inclusions hold simultaneously for all convex K under the fixed descendant assignment.

Proof. Clauses (C1), (C2), and (C4) give (G1), (G3), and (G4), including (10.7M)–(10.7N); in particular,

$$|K^+| \le Cr_1^{-3}|R_{K^\dagger}| \le Cr_1^{-3}|K|.$$

In the net construction, the carrier of T_B is within $O(\delta)$ of the length-cr_1 axial window of T; division by that length gives the angular bound in (10.7L). The one-sided window in (C1) covers the endcap case. □

Lemma 10.5 (Expanded-tubelet concentration)**.** *Fix one tagged outer ball B. For each selected tubelet in $\mathbb{T}_B$, fix once and for all an injection of its m_{fine} copies into distinct represented fine descendants. If one fine label has at most κ_{loc} canonical tubelet representations inside this fixed tag, including its fixed boundary tie, then*

$$\Delta_{\max}(\mathbb{T}_B^{\mathrm{lab}}) \le C\kappa_{\mathrm{loc}} r_1^{-2} \Delta_{\max}(\mathbb{T}). \tag{10.7c}$$

The same injection is used for every point and every convex test body.

Proof. For each T_B, choose one point $x_{T_B} \in Y_B(T_B)$ and, using the fixed label order, the first m_{fine} descendants supplied by (10.4). This defines the required injection independently of later x and K. Clause (G3) of lemma 10.4, equivalently (C2) of lemma 10.3, bounds the use of any fine label by κ_{loc} in the fixed tag.

For a convex K, use the common test body from (G4), equivalently clause (C4):

$$K^+ = z_K + Cr_1^{-1}(R_{K^\dagger} - z_K). \tag{10.7c$'$}$$

Every injected descendant of a copy contained in K lies in this convex K^+, and (C4) gives

$$|K^+| \lesssim r_1^{-3}|K|, \qquad |T_B| \asymp r_1|T|. \tag{10.7c$''$}$$

The injection multiplicity and the definition of $\Delta_{\max}(\mathbb{T})$ now give

$$\begin{aligned}\sum_{(T_B,\ell)\subset K} |T_B| &\lesssim r_1\kappa_{\mathrm{loc}} \sum_{T\subset K^+} |T| \\ &\le r_1\kappa_{\mathrm{loc}}\Delta_{\max}(\mathbb{T})|K^+| \\ &\lesssim \kappa_{\mathrm{loc}} r_1^{-2}\Delta_{\max}(\mathbb{T})|K|.\end{aligned}$$

Taking the supremum over K proves (10.7c). □

Lemma 10.6 (Labelled denominator from represented fine mass)**.** *Let $A(T_B) \subset Y_B(T_B)$ be any retained tubelet subshading on the fixed global m_{fine}-class, and let Y_A be the physical fine shading in (10.6a). If*

$$M(Y_A) \ge P^{-1}M(\bar{Y}), \tag{10.7d}$$

then the multiplicity-expanded labelled tubelet families obey, with the denominator summed over all retained tags,

$$\begin{aligned}D^{\mathrm{tot}}_{\mathrm{tag}} &:= \sum_{B \text{ retained}} D(\mathbb{T}^{\mathrm{lab}}_B) = m_{\mathrm{fine}} \sum_{(B,T_B)} |T_B| \\ &\gtrsim M_{\mathrm{lab}}(A) \ge M(Y_A) \ge P^{-1}\lambda(\mathbb{T},\bar{Y})|\mathbb{T}|\,|T|.\end{aligned} \tag{10.7e}$$

The factor κ_{inc} is needed when (10.7d) is established from labelled retention, through (10.7′), but is not inserted again in (10.7e).

Proof. Equation (10.7) gives $M_{\mathrm{lab}}(A) \asymp m_{\mathrm{fine}} \sum_{(B,T_B)} |A(T_B)|$, which is at most a fixed multiple of $D^{\mathrm{tot}}_{\mathrm{tag}}$. The middle inequality is the upper half of (10.7′), and the last identity is the definition of the density of the fixed fine reference shading. This proves every displayed comparison. □

The score and Löwner-axis windows required by lemma 8.1 are verified in proposition 10.19. Apply that lemma with $\gamma = \varepsilon_{\mathrm{fact}}$ and $U = CB$ in each tag. The tagged disjoint union retains L_{bias}^{-1}; a later $L_{a,b}$-level selects common widths. Assign coincident copies to one parent. The common duplication factor m_{fine} preserves normalized child fractions. Expansion is used only for the slab denominator; non-slab degree and cap estimates use the unexpanded family, so (10.19e1) counts m_{fine} once.

All dimension, child-count, and regularity levels follow m_{fine}. A global (a,b)-level costs at most

$$L_{a,b} \le \big(1 + \lceil \log_2(r_1/\delta) \rceil\big)^2. \tag{10.8}$$

For each outer-ball tag B, let B^+ be a fixed dilation containing all permanent carriers and active full cells in that tag. Since the outer balls have one common radius and bounded overlap, the family $(B^+)_B$ has overlap $O(\kappa_{\mathrm{amb}})$; absorb this structural factor into κ_{amb}. Run the graph on $\bigsqcup_B \{B\} \times \mathbb{R}^3$, with right vertices (B,Q), and synchronize only after each tagwise output is fixed. The next theorem formalizes this step; physical recombination then costs the single factor κ_{amb} from lemma 3.5.

Theorem 10.7 (Global tagged regularization and synchronization)**.** *Let $\mathcal{A}$ be a finite nonempty family of tags. In tag A, let $(\mathcal{P}_A, \mathcal{T}_A, Y_A)$ be a positive-mass input to theorem 4.2, all supported*

in a fixed dilation of A, and let $\mathcal{Q}_A$ be its finite active half-open cell set. Assume that, for fixed $C_{\mathrm{comp}}, C_{\mathrm{mass}}$,

$$|\mathcal{A}| + \max_{A\in\mathcal{A}}(|\mathcal{P}_A| + |\mathcal{T}_A| + |\mathcal{Q}_A|) \le \delta^{-C_{\mathrm{comp}}}, \qquad \delta^{C_{\mathrm{mass}}} \le M_A(Y_A) \le \delta^{-C_{\mathrm{mass}}} \quad (A \in \mathcal{A}). \tag{10.8A}$$

Run J1, R1, and R2 independently in every tag, write μ_A, ω_A, d_A for the three local dyadic parameters, and denote the local graph edge set by $E_{2,A}$ and its shading output by $Y_{\mathrm{reg},A}$. Put

$$\mathcal{Q}_A^+ = \{Q : \textit{some } (P,Q) \in E_{2,A}\}, \qquad v_A(Q) = |U(\mathcal{T}_A, Y_{\mathrm{reg},A}) \cap N_{Ca}(Q)|.$$

There is a structural $C_v \ge 1$, independent of A, such that every $Q \in \mathcal{Q}_A^+$ satisfies

$$\frac{\omega_A}{2\mu_A} \le v_A(Q) \le C_v a^3. \tag{10.8A$'$}$$

Define the literal number of positive volume levels by

$$J_v(A) := 1 + \max\left\{0, \left\lceil \log_2\left(\frac{2C_v a^3 \mu_A}{\omega_A}\right)\right\rceil\right\}. \tag{10.8A$''$}$$

If $\mathcal{Q}_A^+ = \varnothing$, set $J_v(A) = 1$ and declare the local-volume output empty; the positive-mass hypothesis and P1 exclude this case for every tag retained below. Synchronize μ_A, ω_A, d_A by a whole-tag selection. Only then, independently in each selected tag, partition $\mathcal{Q}_A^+$ into the $J_v(A)$ half-open dyadic intervals covering (10.8A$'$), choose a class maximizing the sum of its incident edge weights, and retain every edge at each chosen right cell. This is the saturated local-volume selection*; call its output $Y_{1,A}$. Put*

$$L_{\mathrm{reg}}^{\max} := \max_A L_{\mathrm{reg}}(A), \qquad J_v^{\max} := \max_A J_v(A). \tag{10.8B}$$

There exist a whole-tag subfamily $\mathcal{A}_1$, dyadic numbers μ_*, ω_*, d_*, and level counts $L_\mu, L_\omega, L_d \le C\log(2/\delta)$ such that

$$\mu_* \le \mu_A < 2\mu_*, \quad \omega_* \le \omega_A < 2\omega_*, \quad d_* \le d_A < 2d_* \qquad (A \in \mathcal{A}_1), \tag{10.8C}$$

every $A \in \mathcal{A}_1$ satisfies the local conclusions P2–P5 and

$$M_A(Y_{1,A}) \ge [J_v(A) L_{\mathrm{reg}}(A)]^{-1} M_A(Y_A), \tag{10.8D}$$

while globally

$$\sum_{A\in\mathcal{A}_1} M_A(Y_{1,A}) \ge [L_\mu L_\omega L_d J_v^{\max} L_{\mathrm{reg}}^{\max}]^{-1} \sum_{A\in\mathcal{A}} M_A(Y_A). \tag{10.8E}$$

Suppose in addition that the common scales determine the same resolved-angle parameters $A_{\mathrm{ang}}, B_{\mathrm{ang}}, K, q_{\mathrm{res}}$ in every tag. Run theorem 7.2 independently in each $A \in \mathcal{A}_1$. There is a whole-tag subfamily $\mathcal{A}_2 \subset \mathcal{A}_1$ and dyadic $\theta_*, d_{\angle,*}$, with

$$\theta_* \le \theta_A < 2\theta_*, \qquad d_{\angle,*} \le d_{\angle,A} < 2d_{\angle,*}, \tag{10.8F}$$

for which all local A1–A4 conclusions hold and

$$\sum_{A\in\mathcal{A}_2} M_A(Y_{2,A}) \ge \frac{c_{\mathrm{ang}}^{\min}}{2L_\theta^{\mathrm{tag}} L_{d_\angle}^{\mathrm{tag}}} \sum_{A\in\mathcal{A}_1} M_A(Y_{1,A}), \quad c_{\mathrm{ang}}^{\min} := \min_{A\in\mathcal{A}_1} \frac{A_{\mathrm{ang}}^{-K}}{J_\theta(A) J_d(A)}, \tag{10.8G}$$

where $L_\theta^{\mathrm{tag}}, L_{d_\angle}^{\mathrm{tag}} \le C\log(2/\delta)$. In particular the common input degree d_* permits one global low/high split, and in the high case the synchronized output degree is $d_{\angle,*}$.

Finally, suppose there are a common $m_* > 0$ and fixed constants $0 < c_- \le c_+ < \infty$ such that a represented fine mass $F_A(V)$ on every whole-tubelet subshading obeys

$$c_- m_* M_A(V) \le F_A(V) \le c_+ m_* M_A(V) \tag{10.8H}$$

then (10.8E) and (10.8G) hold for $\sum_A F_A$, with the additional factor c_+/c_-. Thus the selected tags carry the same polylogarithmic fraction of represented fine mass and, whenever the denominator is comparable to that mass at the fixed density level, of the labelled denominator.

Proof. Fix a tag A. Applying theorem 4.2 only to its own left parents and its own half-open cells gives $M_A(Y_{\mathrm{reg},A}) \ge L_{\mathrm{reg}}(A)^{-1} M_A(Y_A)$. This is a local statement before any tag is discarded.

If $Q \in \mathcal{Q}_A^+$, choose an edge $(P,Q) \in E_{2,A}$. Its weight is at least ω_A, while the inner child multiplicity for this one parent is below $2\mu_A$. Therefore the physical union of those child shadings in Q, and hence the union counted by $v_A(Q)$, has measure at least $\omega_A/(2\mu_A)$. Conversely the Ca-neighbourhood of a cell of side $c_0 a$ has volume at most $C_v a^3$. This proves (10.8A′). The interval between these two positive endpoints is covered by the $J_v(A)$ half-open dyadic intervals in (10.8A″), including the case in which the endpoint ratio is at most one. Thus every symbol in (10.8B) is defined before it is used.

All values in (10.8C) lie in polynomial scale windows. Indeed $1 \le \mu_A \le |\mathcal{T}_A|$ and $1 \le d_A \le |\mathcal{P}_A|$. If $m_A^+ = |E_A^+|$ and $\Omega_A = \sum_{E_A^+} \omega_e$, the proof of lemma 4.1 gives

$$\frac{\Omega_A}{2m_A^+} \le \omega_A \le \Omega_A.$$

The inner level and (10.8A) give $\Omega_A \ge L_{J1}(A)^{-1}\delta^{C_{\mathrm{mass}}}$, while the upper bound in (10.8A) controls the other endpoint. Hence each of μ_A, ω_A, d_A has $O(\log(2/\delta))$ dyadic classes. Partition the *tags* by the triple of classes and choose a class maximizing $\sum_A M_A(Y_{\mathrm{reg},A})$. The selection keeps every edge in each chosen tag. Inside such a tag, the saturated volume class costs $J_v(A)$ and preserves the synchronized local degrees. Summing the local graph bounds, pigeonholing the tags, and then applying the local volume bounds proves (10.8C)–(10.8E), including (10.8D).

For the second stage, the local theorem gives $M_A(Y_{2,A}) \ge (c_{\mathrm{ang},A}/2) M_A(Y_{1,A})$. The possible $\theta_A \in [q_{\mathrm{res}}, 1]$ and integer $d_{\angle,A} \le |\mathcal{P}_A|$ again form only logarithmically many classes. Select a class by its post-angle mass. Summing the local lower bounds and then pigeonholing proves (10.8F)–(10.8G). Since this selection deletes whole tags, every local A1–A4 conclusion persists. Finally (10.8H) converts each coarse sum to the corresponding fine sum on both sides. This proves the last assertion and the theorem. □

Corollary 10.8 (Polynomial-complexity volume-level count)**.** *In the setting of theorem 10.7, assume $0 < a \le 1$. Then*

$$J_v^{\max} \le C(C_{\mathrm{comp}}, C_{\mathrm{mass}}) \log(2/\delta). \tag{10.8I}$$

More generally, without (10.8A) but for finite positive-mass inputs, the local saturated selection and (10.8D) remain valid with the literal integer $J_v(A)$ in (10.8A″); the global proof likewise uses the actual finite numbers of nonempty dyadic tag classes in place of the displayed logarithmic bounds. No logarithmic estimate is part of the abstract local-volume identity. Thus (10.8I) is an application-level complexity consequence, not an implicit hypothesis of the synchronization identity.

Proof. For one tag let $m_A^+ = |E_A^+|$ and $\Omega_A = \sum_{E_A^+} \omega_e$. The inner level and (10.8A) give

$$\Omega_A \ge L_{J1}(A)^{-1}\delta^{C_{\mathrm{mass}}}, \qquad L_{J1}(A) \le C \log(2/\delta),$$

while $m_A^+ \le |\mathcal{P}_A||\mathcal{Q}_A| \le \delta^{-2C_{\mathrm{comp}}}$. The proof of lemma 4.1 therefore gives $\omega_A \ge \Omega_A/(2m_A^+) \ge \delta^{C_1}$ after increasing the fixed exponent C_1. Also $\mu_A \le |\mathcal{T}_A| \le \delta^{-C_{\mathrm{comp}}}$. Since $a^3 \le 1$, the logarithm in (10.8A″) is bounded by $C_2 \log(2/\delta)$, uniformly in A. Taking the maximum proves (10.8I). □

10.3. **Analytic estimates inherited by the application.** The branch estimates below close through two analytic assertions from [GWZ26, Definitions 3.4–3.5, pp. 5–6]. We first state them in their source-standard finite-set form. Fix $0 < \beta \le 1$. The assertion $K_{KT}(\beta)$ means that for every $\varepsilon > 0$ there are $\eta, \delta_0 > 0$ such that every finite set $\mathcal{F}$ of δ-tubes in the unit ball, with

$$\Delta_{\max}(\mathcal{F}) \le \delta^{-\eta}, \qquad \lambda(\mathcal{F}, Y) \ge \delta^{\eta},$$

satisfies

$$\mu(\mathcal{F}, Y) \lesssim_{\varepsilon,\beta} \delta^{-\varepsilon} |\mathcal{F}|^{\beta} \qquad (0 < \delta \le \delta_0). \tag{10.8K}$$

The assertion $K_F(\beta)$ means that, under the same quantifiers, the hypotheses

$$C_F(\mathcal{F}, B_1) \le \delta^{-\eta}, \qquad \lambda(\mathcal{F}, Y) \ge \delta^{\eta}$$

imply

$$|U(\mathcal{F}, Y)| \gtrsim_{\varepsilon,\beta} \delta^{\varepsilon+2\beta} \big(|\mathcal{F}|\,|T_\delta|\big)^{\beta/2}, \tag{10.8J}$$

where T_δ denotes one unit δ-tube. For labelled KKT calls, discard labels whose individual shading density is below half the mean and colour exact repeated carriers into $m_* \le \Delta_{\max}(\mathcal{F})$ carrier-injective layers. Applying the source set assertion to the layers and using $\sum_j |\mathcal{F}_j|^{\beta} \le m_*^{1-\beta} |\mathcal{F}|^{\beta}$ gives the labelled form after reselecting η, δ_0; the random-subset proof of Lemma 3.7 then gives its labelled form. No unrestricted labelled extension of K_F is asserted: its calls in Sections 9–10 use source-standard colours or proxies and charge fibre multiplicity separately. We also use the scale-variant forms in [GWZ26, Remark 3.6, p. 6]. The only consequences for planks are restated, with all parameters used here, in lemmas 10.9 and 10.12. Those restatements fix the complete hypotheses encoded by $K_F(\beta)$ and $K_{KT}(\beta)$.

10.4. **Thick parents.** The first geometric split occurs before cellularization. When the shortest parent width is already large, the inherited Frostman and Katz–Tao estimates close the branch directly; the four cellular alternatives considered later belong only to the complementary thin regime. This branch uses the original labels, without multiplicity expansion.

Lemma 10.9 (The thick-parent alternative)**.** *Assume $a \ge \delta^{1-\tau}$, and assume both $K_F(\beta)$ and $K_{KT}(\beta)$. Let $Y^{\mathrm{sc}} \subset \bar{Y}$ be the all-grid refinement fixed in (S1). Choose the source-output and setup exponents so that*

$$0 < \varepsilon_{\mathrm{thick}}, \eta, \qquad \varepsilon_{\mathrm{thick}} + \eta + 3\eta_{\mathrm{grid}} < \frac{1}{8}\tau\varepsilon_{\mathrm{fact}}\beta \tag{10.8a}$$

and also choose η below $\tau\eta_F(\varepsilon_{\mathrm{thick}}, \beta)$, where η_F is the density threshold in the Frostman plank estimate [GWZ26, Lemma 6.4, p. 15]*. Let r be the already fixed source scale-grid radius with $a \le r \le \delta^{-\eta_{\mathrm{grid}}} a$. For sufficiently small δ, there is a ball B_r of radius comparable to r, centered at a point of the fixed scale-r union, such that*

$$\frac{|U(\mathbb{T}, Y^{\mathrm{sc}}) \cap B_r|}{|B_r|} \gtrsim_{\log} \delta^{-2\nu_{\mathrm{thick}}} \left(\frac{\delta}{r}\right)^{2\beta}, \qquad \nu_{\mathrm{thick}} := \frac{1}{8}\tau\varepsilon_{\mathrm{fact}}\beta > 0. \tag{10.8b}$$

Consequently the source scale-r comparison gives

$$|U(\mathbb{T}, \bar{Y})| \gtrsim_{\log} \delta^{-\nu_{\mathrm{thick}}} |T|\,|\mathbb{T}|^{1-\beta}. \tag{10.8c}$$

Proof. Fix a retained outer ball B. The common parent-density level contains a parent W of dimensions $a \times b \times r_1$ for which $\lambda(\mathbb{T}_{B,W}, Y_B) \gtrsim_{\log} \delta^{\eta}$. The locally proved biased factoring inequality (8.0D), in the exact complete-containment form (10.37b), says that for every convex $K \subset W$,

$$\Delta(\mathbb{T}_{B,W}, K) \lesssim_{\log} \left(\frac{|K|}{|W|}\right)^{\varepsilon_{\mathrm{fact}}} \Delta(\mathbb{T}_{B,W}, W). \tag{10.8d}$$

Taking K to be a child tubelet, for which $\Delta(\mathbb{T}_{B,W}, K) \geq 1$, and using equal child volume gives

$$|\mathbb{T}_{B,W}| \gtrsim_{\log} \left(\frac{|W|}{|T_B|}\right)^{1+\varepsilon_{\mathrm{fact}}} \geq \left(\frac{a}{\delta}\right)^{2+2\varepsilon_{\mathrm{fact}}}. \tag{10.8e}$$

Let L_W be the affine map carrying the labelled source Löwner box of W to a unit box. The complete unexpanded labelled children become planks $\mathcal{P}$ of dimensions

$$s \times t \times 1, \qquad s = \delta/b, \quad t = \delta/a.$$

Fixed Löwner-box and boundary enlargements are absorbed in $\lesssim_{\log}$. Equation (10.8d) gives $C_F(\mathcal{P}) \lesssim_{\log} 1$. For a base label $P \in \mathcal{P}$ and $\vartheta \in [s/t, 1] = [a/b, 1]$, let P_ϑ be its source $\vartheta t \times t \times 1$ thickening. Intersect the pullback of P_ϑ with W, and call the result K_ϑ. This is convex, every completely contained child counted by $\mathcal{P}[P_\vartheta]$ lies in K_ϑ, and

$$\frac{|K_\vartheta|}{|W|} \lesssim \vartheta \left(\frac{\delta}{a}\right)^2.$$

The count form of (10.8d) therefore yields

$$|\mathcal{P}[P_\vartheta]| \lesssim_{\log} \vartheta \left(\frac{\delta}{a}\right)^{2+\varepsilon_{\mathrm{fact}}} |\mathcal{P}|. \tag{10.8f}$$

We use the labelled form of source Lemma 6.4. In its proof, Lemmas 6.11 and 6.13 select and refine labels, with angles resolved below at s/t. Assign every surviving label to the essentially distinct thick proxy chosen in source Definition 6.10 and dyadically regularize the proxy-fibre size N. The bound (10.8f), for every base label, gives $N \lesssim M\vartheta$. On this level the positive-sum proof of source Remark 3.3(A) gives the same Frostman bound with sums over labels, and only the essentially distinct proxy subfamily is passed to source Lemma 3.9. Thus coincident original carriers are retained and counted by M; the logarithmic fibre regularization is absorbed in L_{thick}. Equation (10.8f) gives

$$M = \max\left\{1,\, L_{\mathrm{thick}} \left(\frac{\delta}{a}\right)^{2+\varepsilon_{\mathrm{fact}}} |\mathcal{P}|\right\}, \qquad L_{\mathrm{thick}} = \delta^{-o(1)}.$$

By (10.8e) the second entry is already at least one after enlarging L_{thick}. Also $s \leq t \leq \delta^\tau$, and $\eta < \tau\eta_F$ gives $\lambda(\mathcal{P}, Y_\mathcal{P}) \gtrsim_{\log} s^{\eta_F}$. Hence the scale-variant convention [GWZ26, Remark 3.6, p. 6] applies with call data

$$\begin{gathered} s = \delta/b, \qquad t = \delta/a, \qquad \mathcal{F} = \mathcal{P}, \qquad \lambda(\mathcal{P}, Y_\mathcal{P}) \gtrsim \delta^\eta \geq s^{\eta_F}, \\ C_F(\mathcal{P}) \lesssim_{\log} 1, \qquad M \leq L_{\mathrm{thick}} t^{2+\varepsilon_{\mathrm{fact}}} |\mathcal{P}|, \qquad |\mathcal{P}[P_\vartheta]| \leq M\vartheta \quad (P \in \mathcal{P},\ s/t \leq \vartheta \leq 1). \end{gathered} \tag{10.8f$'$}$$

The labelled form just verified of source Lemma 6.4, equation (31), now gives

$$\mu(\mathcal{P}, Y_\mathcal{P}) \lesssim s^{-\varepsilon_{\mathrm{thick}}} C_F(\mathcal{P})^{1-\beta/2} \frac{s}{t} M^{\beta/2} t^{-2\beta} \left(t^2 |\mathcal{P}|\right)^{1-\beta/2}. \tag{10.8g0}$$

Divide $\lambda|\mathcal{P}|st$ by (10.8g0), insert the bound for M, and pull back by L_W:

$$\begin{aligned} \frac{|U(\mathbb{T}_{B,W}, Y_B) \cap W|}{|W|} &\asymp |U(\mathcal{P}, Y_\mathcal{P})| \\ &\gtrsim \delta^{\eta+\varepsilon_{\mathrm{thick}}+o(1)} \left(\frac{\delta}{a}\right)^{2\beta-\varepsilon_{\mathrm{fact}}\beta/2}. \end{aligned} \tag{10.8g}$$

Here every Y_B-point has positive represented fine multiplicity by (10.3)–(10.4), and the induced descendants belong to the current all-scales fine refinement $Y^{\mathrm{sc}} \subset \bar{Y}$. Hence this coarse union is contained in $U(\mathbb{T}, Y^{\mathrm{sc}})$. A bounded-overlap cover of W by a-balls selects one B_a with the same density lower bound. Choose a shaded point $x \in B_a$ and use the source ball $B_r = B(x, 2r)$;

replacing r by this fixed neighbouring scale is part of the stability fixed in (S1). Since $B_a \subset B_r$, its relative density loses at most $(a/r)^3$, and

$$\left(\frac{a}{r}\right)^3 \left(\frac{\delta}{a}\right)^{2\beta} = \left(\frac{a}{r}\right)^{3-2\beta} \left(\frac{\delta}{r}\right)^{2\beta} \geq \delta^{3\eta_{\mathrm{grid}}} \left(\frac{\delta}{r}\right)^{2\beta}.$$

Since $\delta/a \leq \delta^{\tau}$, (10.8a), and then a choice of δ_0 absorbing the $o(1)$-term within the remaining one-eighth reserve, turn (10.8g) into (10.8b).

To complete the branch, the all-discrete-a multiplicity refinement made immediately after source equation (86) provides $Y^{\mathrm{sc}} \subset \bar{Y}$ and the scale-r shading $(\mathbb{T}_r, Y_r)$ for which

$$\begin{aligned} |U(\mathbb{T}, Y^{\mathrm{sc}})| &\gtrsim_{\log} |U(\mathbb{T}_r, Y_r)| \frac{|U(\mathbb{T}, Y^{\mathrm{sc}}) \cap B_r|}{|B_r|}, \\ |U(\mathbb{T}_r, Y_r)| &\gtrsim \delta^{\nu_{\mathrm{thick}}/9} \left(\frac{r}{\delta}\right)^{2\beta} |T|\,|\mathbb{T}|^{1-\beta}. \end{aligned} \tag{10.8h}$$

These are source equations (87)–(88) on p. 36 at the previously fixed grid radius r. Substitution of (10.8b) gives a power stronger than (10.8c). All later objects used above refine this fixed fine shading, so their local union lies in the union appearing in (10.8h). □

Henceforth we are in the thin case

$$a < \delta^{1-\tau}. \tag{10.8i}$$

10.5. **Local cell volume and the cellular parent output.** In the thin regime apply the first stage of theorem 10.7 tagwise, using (10.7). Before this call, the input density, (10.7V), and the pre-cellular cutoff $\omega_{\mathrm{pre}} < \eta_{\mathrm{src}}$ give

$$\frac{|\widetilde{Y}_B(T_B)|}{|T_B|} \gtrsim \frac{c_{\mathrm{sc}} c_{\mathrm{init}}}{L_{\mathrm{mult}}} \delta^{\eta_{\mathrm{src}}} \geq \delta^{2\eta_{\mathrm{src}}}. \tag{10.8A0}$$

Hence lemma 10.1 applies without any J_v-dependent input. Its fixed retention, one nonempty tubelet of volume $\asymp \delta^2 r_1$, (10.7W′), polynomial complexity, and the fixed-ball upper bound verify (10.8A). Put $\mathcal{B}_{\mathrm{good}} := \mathcal{A}_1$ and

$$\mu_{\mathrm{in}} := \mu_*, \quad \omega_0 := \omega_*, \quad d_0 := d_*, \quad L_{\mathrm{reg}} := L_{\mathrm{reg}}^{\max}, \quad P_{\mathrm{tag}} := L_\mu L_\omega L_d. \tag{10.8k}$$

Every local parameter is within a factor two of these references. By (10.8D)–(10.8E), the retained tags carry at least $(P_{\mathrm{tag}} J_v^{\max} L_{\mathrm{reg}}^{\max})^{-1}$ of represented fine mass.

Fix one such tagged outer-ball component for the next three displays; thus Q means the tagged vertex (B, Q), and all degrees in (10.9)–(10.10) are computed before physical recombination. After its local joint graph regularization, let $w(Q) = \sum_{W:(W,Q)\in E_2} \omega_{WQ}$. If the right degree is $d_B \leq d(Q) < 2d_B$, where $d_0 \leq d_B < 2d_0$, then

$$d_0\omega_0 \leq w(Q) < 16 d_0 \omega_0. \tag{10.9}$$

Property P5 bounds the child multiplicity of Y_{reg} on Q by $16\mu_{\mathrm{in}} d_0$; therefore

$$|U(\mathbb{T}, Y_{\mathrm{reg}}) \cap Q| \geq \frac{\omega_0}{16\mu_{\mathrm{in}}}. \tag{10.10}$$

After projection to physical space, the same pointwise bound and any sum of the local union lower bounds acquire at most the factor κ_{amb}, by lemma 3.5. Thus (10.9)–(10.10) remain tagged identities until this controlled recombination. We may now select one dyadic level of $v(Q) = |U(\mathbb{T}, Y_{\mathrm{reg}}) \cap N_{Ca}(Q)|$. Equations (10.9)–(10.10) and $v(Q) \leq C' a^3$ show that the theorem-level count (10.8A″) is bounded by the common envelope

$$\widehat{J}_v := 1 + \max\left\{0, \left\lceil \log_2\left(\frac{16 C' a^3 \mu_{\mathrm{in}}}{\omega_0}\right)\right\rceil\right\}, \qquad J_v(B) \leq \widehat{J}_v. \tag{10.11}$$

This is logarithmic under polynomial complexity and remains explicit otherwise. Set $J_v := \widehat{J}_v$, padding shorter local partitions by empty intervals. The saturated selection of theorem 10.7 chooses $\mathcal{Q}_v(B)$ with

$$\sum_{Q\in\mathcal{Q}_v(B)} w(Q) \ge J_v^{-1}\sum_{Q} w(Q). \tag{10.11$'$}$$

Retain the saturated edge set

$$E_3(B) = \{(W,Q)\in E_2(B) : Q\in\mathcal{Q}_v(B)\}, \tag{10.11$''$}$$

and define the resulting post-selection shadings, in particular Y_1, by (4.10), discarding empty parents. The saturated conclusions (10.8D), P2–P5, and the local P1 cost J_v now follow directly from theorem 10.7.

Lemma 10.10 (Good-ball local transfer)**.** *Assume every selected tubelet shading $Y_B(T_B)$ is covered by a bounded-overlap family of selected δ-balls D, each of which meets the shading and satisfies*

$$|D\cap Y_B(T_B)| \ge c_{\mathrm{gb}}\delta^{\eta}\delta^3, \tag{10.11a}$$

and assume every physical cell in $U(\widetilde{\mathbb{W}}'_B, Z)$ is a positive parent–cell incidence for the joint factoring output. Then

$$|U(\mathbb{T}_B, Y_B)| \gtrsim c_{\mathrm{gb}}\delta^{\eta}(\delta/a)^3|U(\widetilde{\mathbb{W}}'_B, Z)|. \tag{10.11b}$$

More generally, let $Z'(W)\subset Z(W)$ be any measurable parent subshading, and let B_R be a ball with $R\ge a$. Then, for a fixed dimensional C,

$$|U(\mathbb{T}_B, Y_B)\cap B_{CR}| \gtrsim c_{\mathrm{gb}}\delta^{\eta}(\delta/a)^3|U(\widetilde{\mathbb{W}}'_B, Z')\cap B_R|. \tag{10.11b$'$}$$

Repeated parent labels and cells meeting the parent boundary do not alter the constant.

Proof. The parent union is the disjoint union of its distinct half-open h-cells, with $h = c_0 a$. Color the grid by residue classes modulo a fixed integer $M > 4C/c_0$. A color carrying a maximal number of active physical cells retains a fixed fraction, and two of its cells are separated by more than $3Ca$.

For each retained physical cell Q, choose one labelled positive edge (W,Q). Positivity and the definition of Y_1 give a tubelet T_B and a point $x_Q\in Y_1(T_B)\cap Q$. Choose one of the selected good δ-balls D_Q containing x_Q, using the fixed half-open tie-breaking convention on ball boundaries. Since $\delta\le a$, the balls D_Q belonging to the chosen grid color are pairwise disjoint (enlarging or shrinking all balls by one fixed factor only changes c_{gb}). Consequently

$$\begin{aligned}|U(\mathbb{T}_B, Y_B)| &\ge \sum_Q |D_Q\cap Y_B(T_B(Q))|\\ &\ge c_{\mathrm{gb}}\delta^{\eta}\delta^3\#\{Q\}\\ &\gtrsim c_{\mathrm{gb}}\delta^{\eta}(\delta/a)^3|U(\widetilde{\mathbb{W}}'_B, Z)|.\end{aligned}$$

The count is over physical cells, so coincident parent labels have no effect on the right side. Each measured set lies in the tubelet shading, and the grid coloring costs a dimensional constant. For the localized assertion, run the same argument on all active physical cells that meet $U(\widetilde{\mathbb{W}}'_B, Z')\cap B_R$. Because each such cell has volume h^3, their full cell volumes dominate that union volume. The inclusion $Z'(W)\subset Z(W)$ makes every chosen cell a positive original parent–cell incidence. The point x_Q is within $O(a)$ of B_R, and its good δ-ball is therefore contained in B_{CR} because $R\ge a$. The same residue-class coloring makes those balls disjoint and proves (10.11b$'$). □

The self-contained labelled construction in lemma 8.1 uses the same exponent in (8.0D) and (8.0F)–(8.0G), matching the biased formulation of [GWZ26, Lemma 9.2, p. 35]. We therefore identify

$$\eta_{\mathrm{bias}} := \varepsilon_{\mathrm{fact}}; \tag{10.11c}$$

so the two symbols denote the same parameter. Put

$$\sigma = \eta_{\mathrm{bias}}, \qquad q = 2 - \sigma, \qquad \mathfrak{s} = a/b,$$

and let L_N denote the child-count level and L_{reg} the J1/REG-1/REG-2 product. Define once, for every later branch,

$$P_{\mathrm{pre}} := c_{\mathrm{setup}}^{-1} J_v \kappa_{\mathrm{inc}} L_m L_{a,b}. \tag{10.11d}$$

This collects the setup, representation, dimension, and local-volume losses already exposed above. The rectangular theorem and Jensen give the cellular parent-density estimate

$$\lambda(\widetilde{\mathbb{W}}'_B, Z) \gtrsim C_0^{-q} \mathfrak{s}^{\varepsilon} \kappa_D^{-q} (P_{\mathrm{pre}} L_N L_{\mathrm{reg}})^{-q} \lambda(\mathbb{T}_B, Y_B)^q. \tag{10.12}$$

The relevant scale is a/b, and every synchronized mass loss has power $-q$. This is the parent-density line corresponding to source equation (95). Exact compatibility is

$$(\mathbb{T}_B)_{Y_1}(x) \subset \bigcup_{\widetilde{W}:\, x \in Z(\widetilde{W})} (\mathbb{T}_{B,W})_{Y_1}(x), \tag{10.13}$$

which is the incidence line corresponding to source equation (96). Lemma 10.10 gives

$$|U(\mathbb{T}_B, Y_B)| \gtrsim \delta^{\eta} (\delta/a)^3 |U(\widetilde{\mathbb{W}}'_B, Z)|. \tag{10.14}$$

The left side is evaluated on the reference shading Y_B.

10.6. **Broad middle axes.** When the middle axis is broad, $b \geq \delta^{\varepsilon_{\mathrm{scal}}} r_1$, the labelled slab lemma proved in section B applies to the $a \times b \times r_1$ parent boxes. In this branch use the expanded child family $(\mathbb{T}_B^{\mathrm{lab}}, Y_B^{\mathrm{lab}})$ from (10.7a). Assign all copies of one carrier to the same biased parent and rerun the joint graph. Every parent–cell edge weight is multiplied by the common m_{fine}, so the dyadic edge and right-degree selections, the parent family, and the physical parent shading Z are unchanged. The child shading density and the normalized complete-child fractions are also unchanged by (10.7b). Since the parent boxes have comparable volume, define their labelled parent-shading density by

$$\lambda_B^{\mathrm{par}} := \frac{M(Z)}{|\widetilde{\mathbb{W}}'_B|\,|W|} = \frac{\sum_{W \in \widetilde{\mathbb{W}}'_B} |Z(W)|}{|\widetilde{\mathbb{W}}'_B|\,|W|}. \tag{10.14a}$$

With $L_{\mathrm{slab}} = 1 + \lceil \log_2(r_1/a) \rceil$, the slab lemma gives

$$\frac{|U(\widetilde{\mathbb{W}}'_B, Z)|}{|\widetilde{\mathbb{W}}'_B|\,|W|} \gtrsim \frac{(\lambda_B^{\mathrm{par}})^2}{1 + L_{\mathrm{slab}} \Delta_{\max}(\widetilde{\mathbb{W}}'_B)(r_1/b)^2}. \tag{10.15}$$

This formula retains both normalizations needed in the application. First, union volume follows directly from (10.15), with both powers of λ_B^{par} present. Second, the parent-mass comparison is

$$|\widetilde{\mathbb{W}}'_B|\,|W| \gtrsim_{\log} r_1^2 \delta^{\eta} |\mathbb{T}_B^{\mathrm{lab}}|\,|T_B|, \tag{10.16}$$

and its normalization is as follows. With

$$\Delta_{\max}(\mathbb{T}_B^{\mathrm{lab}}) := \sup_{K \in \mathfrak{K}_3} \frac{\sum_{(T_B, \ell) \subset K} |T_B|}{|K|},$$

complete containment gives

$$\begin{aligned}|\widetilde{\mathbb{W}}'_B|\,|W| &= \sum_{W\in\widetilde{\mathbb{W}}'_B} |W| \\ &\geq \Delta_{\max}(\mathbb{T}_B^{\mathrm{lab}})^{-1} \sum_{W\in\widetilde{\mathbb{W}}'_B}\ \sum_{(T_B,\ell)\in\mathbb{T}_{B,W}^{\mathrm{lab}}} |T_B| \\ &\gtrsim_{\log} \Delta_{\max}(\mathbb{T}_B^{\mathrm{lab}})^{-1}|\mathbb{T}_B^{\mathrm{lab}}|\,|T_B|.\end{aligned} \tag{10.16′}$$

The last inequality is the retained dimension/child-count level; it counts labelled tubelets, including coincident carriers. At this scale, the source thickening comparison gives

$$\Delta_{\max}(\mathbb{T}_B^{\mathrm{lab}}) \leq C\kappa_{\mathrm{loc}} r_1^{-2}\Delta_{\max}(\mathbb{T}) \leq C\kappa_{\mathrm{loc}} r_1^{-2}\delta^{-\eta}, \tag{10.16″}$$

which proves (10.16), since $\kappa_{\mathrm{loc}} = O(1)$. The source complete-containment bound $\Delta_{\max}(\mathbb{T}) \leq \delta^{-\eta}$ contributes the factor δ^{η} in this comparison. Equations (10.14)–(10.16) give the complete local slab inequality. Converting it into a numerical terminal gain uses the common source-to-admissible quantitative bounds proved in proposition 10.19; the resulting exponent is displayed in (10.33a).

10.7. **The angular degree dichotomy.** For every thin branch set

$$\alpha = a/r_1, \qquad q_{\mathrm{ang}} = a/b, \qquad \rho_2 = b/r_1.$$

The common thin range has $\delta^{1-\varepsilon_{\mathrm{scal}}} \leq \alpha \leq \rho_2 \leq 1$ and $q_{\mathrm{ang}} = \alpha/\rho_2 \geq \alpha$. In the non-slab alternative one has the additional upper restriction

$$\delta^{1-\varepsilon_{\mathrm{scal}}} \leq \alpha \leq \rho_2 \leq \delta^{\varepsilon_{\mathrm{scal}}}, \qquad q_{\mathrm{ang}} \geq \alpha. \tag{10.16a}$$

The remaining non-slab argument is organized by the local angular degree: small degree yields a direct multiplicity bound, while large degree passes to the transverse–tangential decomposition. Before splitting by degree, we collect the losses that have already been incurred. With P_{pre} fixed once in (10.11d), and up to one fixed numerical constant, put

$$\begin{aligned}P_{\mathrm{sync}} &= \kappa_{\mathrm{inc}} P_{\mathrm{tag}} P_{\mathrm{pre}} L_N L_{\mathrm{reg}}, \\ P_{\mathrm{angtag}} &:= L_\theta^{\mathrm{tag}} L_{d_\angle}^{\mathrm{tag}}, \\ c_*^{-1} &\leq \frac{64\kappa_{\mathrm{inc}}^2 P_{\mathrm{tag}} P_{\mathrm{angtag}} J_v L_m L_{a,b} L_N L_{\mathrm{reg}}}{c_{\mathrm{setup}} c_{\mathrm{ang}}}.\end{aligned} \tag{10.16b}$$

Define nonnegative exponents by

$$\begin{gathered}P_{\mathrm{sync}} \leq \delta^{-\Lambda_{\mathrm{sync}}}, \qquad P_{\mathrm{den}} := \delta^{-\eta_{\mathrm{src}}} P_{\mathrm{sync}} \leq \delta^{-\Lambda_{\mathrm{den}}}, \qquad C_0\kappa_D(P_{\mathrm{pre}} L_N L_{\mathrm{reg}}) \leq \delta^{-\Lambda_{\mathrm{pre}}}, \\ c_*^{-1} \leq \delta^{-\Lambda_*}, \qquad D_W, D_S \leq \delta^{-\Lambda_D}, \qquad H_{\mathrm{col}} \leq \delta^{-\Lambda_H}.\end{gathered} \tag{10.16c}$$

Here

$$D_W = \max_B \Delta_{\max}(\widetilde{\mathbb{W}}'_B), \qquad D_S = \max_{B,S} \Delta_{\max}(\widetilde{\mathbb{W}}^{\angle}_{B,S}).$$

We next define a uniform envelope independent of the subsequently selected tag. Fix a structural integer C_{env} larger than the total multiplicity with which any primitive factor occurs in (10.16b), (10.19), and (10.25). For every possible retained tag family and every realized thin discrete scale

pair $\delta^{1-\varepsilon_{\mathrm{scal}}} \le \alpha \le \rho_2 \le 1$, set

$$\begin{aligned}
\mathcal{E}_{\mathrm{setup}} &:= c_{\mathrm{setup}}^{-1} C_0 \kappa_D \kappa_{\mathrm{inc}} \kappa_{\mathrm{loc}} J_v L_m L_{a,b} L_N L_{\mathrm{reg}} P_{\mathrm{tag}} P_{\mathrm{pre}} P_{\mathrm{angtag}},\\
\mathcal{E}_{\mathrm{ang}} &:= \max_B c_{\mathrm{ang},B}^{-1},\\
\mathcal{E}_{\mathrm{src}}(\delta) &:= \max\left\{1, \delta^{-C_{\mathrm{env}}\omega_0(\delta)}, [\log(2/\delta)]^{C_{\mathrm{env}}}, \sup_{\mathrm{tags,scales}} \mathcal{E}_{\mathrm{setup}}^{C_{\mathrm{env}}}, \sup_{\mathrm{tags,scales}} \mathcal{E}_{\mathrm{ang}}^{C_{\mathrm{env}}}\right\},\\
\omega_{\mathrm{src}}(\delta) &:= \frac{\log \mathcal{E}_{\mathrm{src}}(\delta)}{\log(1/\delta)}.
\end{aligned} \tag{10.16d}$$

For slab pairs, where ANGLE-CELL is not invoked, the angular-only factors P_{angtag} and $\mathcal{E}_{\mathrm{ang}}$ are interpreted as 1, and neither D_S nor H_{col} is used. The setup suprema range over the finite tagged nets and all thin discrete scales in the fixed polynomial window, including both (S1b) and (S1c) and choices later deleted by $\eta, \tau, \eta_{\mathrm{pl}}$; the angular supremum is restricted to the non-slab window (10.16a). Consequently ω_{src} is defined before, and is independent of, those later exponent choices.

The operation map in lemma 10.18, using the pre-cellular verification (10.8A0) of (10.8A), verifies uniformly that $\omega_{\mathrm{src}}(\delta) \to 0$: all setup, reference-density, and angular entries are bounded or subpower, and the only non-subpower term is inherited complete-containment. The resulting D_W bound holds on the full thin window, while D_S and H_{col} occur only in the non-slab branch:

$$\begin{aligned}
D_W &\le \delta^{-2(1-\varepsilon_{\mathrm{scal}})\varepsilon_{\mathrm{fact}} - \omega_{\mathrm{src}}(\delta)} \quad \text{on all thin pairs,}\\
D_S \le \delta^{-2(1-\varepsilon_{\mathrm{scal}})\varepsilon_{\mathrm{fact}} - \omega_{\mathrm{src}}(\delta)}, &\qquad H_{\mathrm{col}} \le C D_S \quad \text{on non-slab pairs.}
\end{aligned} \tag{10.16e}$$

Consequently we may take

$$\begin{aligned}
&\Lambda_{\mathrm{sync}} = \omega_{\mathrm{src}}(\delta), && \Lambda_{\mathrm{pre}} = \omega_{\mathrm{src}}(\delta), && \Lambda_* = \omega_{\mathrm{src}}(\delta),\\
&\Lambda_D = 2(1-\varepsilon_{\mathrm{scal}})\varepsilon_{\mathrm{fact}} + \omega_{\mathrm{src}}(\delta), && \Lambda_H = \Lambda_D + \omega_{\mathrm{src}}(\delta), && \Lambda_{\mathrm{den}} = \eta_{\mathrm{src}} + \omega_{\mathrm{src}}(\delta).
\end{aligned} \tag{10.16f}$$

The choice of C_{env} absorbs the finite powers in (10.16b), so this single ω_{src} controls every later $o(1)$-term.

In every retained tag, ANGLE-CELL gives a local mass fraction

$$c_{\mathrm{ang},B} \ge \frac{A_{\mathrm{ang}}^{-K}}{J_\theta(B) J_d(B)}. \tag{10.17}$$

In the high-degree case below we invoke the second stage of theorem 10.7, replace the tag family by $\mathcal{A}_2$, and set

$$c_{\mathrm{ang}} := c_{\mathrm{ang}}^{\min}, \qquad \theta := \theta_*, \qquad d_\angle := d_{\angle,*}.$$

These are synchronized reference values. For a retained tag B, keep θ_B and $d_{\angle,B}$ for its local outputs. Then

$$\theta \le \theta_B < 2\theta, \qquad d_\angle \le d_{\angle,B} < 2 d_\angle, \tag{10.17a}$$

and A1–A3 hold with $\theta_B, d_{\angle,B}$. Thus the corresponding common upper bounds for the local cell diameter and degree are 4θ and $4d_\angle$. The whole-tag angle synchronization costs P_{angtag} in global represented mass and preserves the local conclusions. Since the local input degrees are at least d_0, the synchronized output degree satisfies

$$d_\angle \ge \tfrac{1}{4} A_{\mathrm{ang}}^{-K} d_0. \tag{10.18}$$

Moreover $c_{\mathrm{ang}} \ge \alpha^{o(1)}$, because the local level counts are logarithmic. In the low-degree case the angle stage is not run and, for the common bookkeeping only, $P_{\mathrm{angtag}} = c_{\mathrm{ang}}^{-1} = 1$. The factor A_{ang}^{-K} prevents an unbuffered input-degree threshold from implying the output threshold required by the plank-reduction lemma. Define

$$H_{\mathrm{ang}} = 4 A_{\mathrm{ang}}^{K}. \tag{10.19}$$

Low degree. The low-degree estimate passes from the tubelet shading Y_1 to the fine reference shading through represented incidences. Let $\mathcal{R}(T)$ be the retained labelled tubelet representations of a fine tube T, and define

$$Y_1^{\mathrm{f}}(T) = \widehat{Y}(T) \cap \bigcup_{(B,T_B)\in\mathcal{R}(T)} Y_1(T_B). \tag{10.19a}$$

Writing

$$M_{\mathrm{c}}(A) = \sum_{(B,T_B)} |A(T_B)|, \qquad U_{\mathrm{c}}(\mathbb{T}_B, A) = \bigcup_{(B,T_B)} A(T_B), \qquad \mu_{\mathrm{c}}(\mathbb{T}_B, A) = M_{\mathrm{c}}(A)/|U_{\mathrm{c}}(\mathbb{T}_B, A)|,$$

joint regularization and the global m_{fine}-level give

$$\begin{aligned} M_{\mathrm{c}}(Y_1) &\geq (P_{\mathrm{tag}} J_v L_N L_{\mathrm{reg}})^{-1} M_{\mathrm{c}}(Y_B), \\ P_{\mathrm{sync}}^{-1} M(\bar{Y}) &\leq M(Y_1^{\mathrm{f}}) \leq M(\bar{Y}). \end{aligned} \tag{10.19b}$$

Property P5 gives the pointwise and average coarse bounds

$$\sum_{T_B} \mathbf{1}_{Y_1(T_B)}(x) < 16\kappa_{\mathrm{amb}}\mu_{\mathrm{in}} d_0, \qquad \mu_{\mathrm{c}}(\mathbb{T}_B, Y_1) \leq 16\kappa_{\mathrm{amb}}\mu_{\mathrm{in}} d_0, \tag{10.19c}$$

and consequently

$$|U_{\mathrm{c}}(\mathbb{T}_B, Y_1)| \geq \frac{M_{\mathrm{c}}(Y_1)}{16\kappa_{\mathrm{amb}}\mu_{\mathrm{in}} d_0}. \tag{10.19d}$$

Lemma 10.11 (Incident-centred cap comparison)**.** *Assume the reference shading $\bar{Y}$ has the source uniformity property that, whenever $x \in \bar{Y}(T_0)$,*

$$\#\{T : x \in \bar{Y}(T),\ \operatorname{angle}(T, T_0) \leq C_{\mathrm{geo}}\rho_2\} \leq C_{\mathrm{cap}}\mu(\rho_2), \tag{10.19e0}$$

where changing ρ_2 by a fixed dyadic factor changes $\mu(\rho_2)$ by at most a fixed uniformity constant. At a point incident to one retained inner parent,

$$m_{\mathrm{fine}}\mu_{\mathrm{in}} \leq C\kappa_{\mathrm{loc}} C_{\mathrm{cap}}\mu(\rho_2). \tag{10.19e1}$$

Proof. All length-r_1 tubelets contained in one $a \times b \times r_1$ parent have directions in a ball of radius $C_{\mathrm{geo}} b/r_1 = C_{\mathrm{geo}}\rho_2$. At the point under consideration, the μ_{in} tubelet incidences of that parent represent at least $m_{\mathrm{fine}}\mu_{\mathrm{in}}$ labelled fine incidences by (10.4) and (10.7). Inside this fixed tagged outer ball, every physical fine label has at most κ_{loc} canonical tubelet representations, so at least $m_{\mathrm{fine}}\mu_{\mathrm{in}}/\kappa_{\mathrm{loc}}$ distinct incident fine labels remain. Choose one of them as T_0. Every other one lies in the incident-centred angular ball in (10.19e0); applying that upper bound and multiplying by κ_{loc} proves (10.19e1). Thus angular uniformity is invoked on $\bar{Y}$ with an incident-centred cap. "Distinct" means distinct source labels, so coincident geometric carriers remain separate in both counts. □

At a retained physical tubelet point, (10.4), P5 in each tag, and lemma 3.5 bound the represented labelled fine multiplicity by $64\kappa_{\mathrm{amb}} m_{\mathrm{fine}}\mu_{\mathrm{in}} d_0$: the local parent–cell bound contributes 16, the synchronized fine level contributes 4, and tag recombination contributes κ_{amb}. Apply lemma 10.11, and identify $\mu(\rho_2)$ with the source quantity $\mu(\mathbb{T}[T_{\rho_2}], \bar{Y})$ up to the fixed neighbouring-scale constant. Together with the preceding pointwise upper bound, this proves

$$\mu(\mathbb{T}, Y_1^{\mathrm{f}}) \leq 64\kappa_{\mathrm{amb}}\kappa_{\mathrm{loc}} d_0\, \mu(\mathbb{T}[T_{\rho_2}], \bar{Y}). \tag{10.19e}$$

Combining (10.19b) and (10.19e), and using $U(\mathbb{T}, Y_1^{\mathrm{f}}) \subset U(\mathbb{T}, \bar{Y})$, yields the mass–union chain

$$\begin{aligned} |U(\mathbb{T}, Y_1^{\mathrm{f}})| &\geq \frac{M(\bar{Y})}{64\kappa_{\mathrm{amb}}\kappa_{\mathrm{loc}} P_{\mathrm{sync}} d_0 \mu(\mathbb{T}[T_{\rho_2}], \bar{Y})}, \\ \mu(\mathbb{T}, \bar{Y}) &\leq 64\kappa_{\mathrm{amb}}\kappa_{\mathrm{loc}} P_{\mathrm{sync}} d_0 \mu(\mathbb{T}[T_{\rho_2}], \bar{Y}). \end{aligned} \tag{10.19f}$$

Suppose now that $d_0 < H_{\mathrm{ang}}\alpha^{-\eta_{\mathrm{pl}}}$. Cap uniformity and the non-sticky count state

$$\begin{aligned}&\mu(\mathbb{T}[T_{\rho_2}],\bar{Y}) \lesssim \delta^{-\eta_{\mathrm{bias}}}|\mathbb{T}[T_{\rho_2}]|^{\beta},\\ &|\mathbb{T}[T_{\rho_2}]|\,|\mathbb{T}_{\rho_2}| \lesssim_{\log} |\mathbb{T}|, \qquad |\mathbb{T}_{\rho_2}| \ge \rho_2^{-2-\zeta}.\end{aligned} \tag{10.19g}$$

For every fixed $\xi > 0$, the ANGLE-CELL stopping calculation gives $H_{\mathrm{ang}} \le C_\xi \alpha^{-\xi}$. Substitution in (10.19f) gives

$$\mu(\mathbb{T},\bar{Y}) \lesssim \delta^{\nu_{\mathrm{low}}-o(1)}|\mathbb{T}|^{\beta}, \qquad \nu_{\mathrm{low}} = \varepsilon_{\mathrm{scal}}\beta(2+\zeta) - \Lambda_{\mathrm{sync}} - \eta_{\mathrm{bias}} - (1-\varepsilon_{\mathrm{scal}})(\eta_{\mathrm{pl}}+\xi). \tag{10.19h}$$

Thus this alternative closes, without the plank-density lemma, under the strict inequality

$$\Lambda_{\mathrm{sync}} + \eta_{\mathrm{bias}} + (1-\varepsilon_{\mathrm{scal}})(\eta_{\mathrm{pl}}+\xi) < \varepsilon_{\mathrm{scal}}\beta(2+\zeta). \tag{L}$$

If instead $d_0 \ge H_{\mathrm{ang}}\alpha^{-\eta_{\mathrm{pl}}}$, then (10.18) gives the output threshold

$$d_\angle \ge \alpha^{-\eta_{\mathrm{pl}}}. \tag{10.20}$$

The angle mass fraction and (10.12) also imply the parent-density hypothesis

$$c_{\mathrm{ang}} C_0^{-q}(a/b)^{\varepsilon}\kappa_D^{-q}(P_{\mathrm{pre}}L_N L_{\mathrm{reg}})^{-q}\delta^{q\eta} \ge \alpha^{\eta_{\mathrm{pl}}} \tag{10.21}$$

as follows. Denote the left side by $\mathcal{D}$. For each fixed $\xi > 0$, (10.17) gives $c_{\mathrm{ang}} \ge C_\xi \alpha^{\xi}$; hence

$$\mathcal{D} \ge C_\xi \alpha^{E_{\mathrm{high}}}, \qquad E_{\mathrm{high}} = \xi + \varepsilon + \frac{q(\eta+\Lambda_{\mathrm{pre}})}{\varepsilon_{\mathrm{scal}}}. \tag{10.21a}$$

Here $q_{\mathrm{ang}} \ge \alpha$, while $\delta^u \ge \alpha^{u/\varepsilon_{\mathrm{scal}}}$ follows from (10.16a). Choose a fixed $\kappa > 0$ and require

$$\xi + \varepsilon + \frac{q(\eta+\Lambda_{\mathrm{pre}})}{\varepsilon_{\mathrm{scal}}} \le \eta_{\mathrm{pl}} - \kappa. \tag{H}$$

After decreasing δ_0, (H) proves (10.21). The subsequent source call uses (10.20) and (10.21) as its multiplicity and density hypotheses.

One further density comparison is needed in the tangential branch. Given a Katz–Tao loss $\varepsilon_{KT} > 0$, let $\eta_{KT}(\varepsilon_{KT},\beta) > 0$ be its required shading-density exponent. The inequality

$$E_{\mathrm{dens}} := (1-\varepsilon_{\mathrm{scal}})\xi + \varepsilon + q(\eta+\Lambda_{\mathrm{pre}}) < \varepsilon_{\mathrm{scal}}\eta_{KT} \tag{K}$$

implies $\mathcal{D} \ge \rho_2^{\eta_{KT}}$ with a fixed positive power reserve. After decreasing the final scale cutoff, that reserve absorbs the structural carrier-density comparison in (10.29d). Consequently every later proxy colour whose normalized density is at least one half of its ambient proxy-slab density satisfies the density hypothesis of the generalized Katz–Tao estimate.

Lemma 10.12 (Buffered labelled-multiset filling lemma)**.** *Fix $\eta > 0$ and constants $C_{\mathrm{comp}}, C_{\mathrm{box}} \ge 1$. Let $0 < \alpha \le \rho \le 1$ be sufficiently small, and let $(\mathcal{P}, Y)$ be a finite labelled multiset of C_{box}-comparable rectangular $\alpha \times \rho \times 1$ planks in the unit ball, equipped with measurable shadings. Each plank carries an ordered orthonormal short–middle–long frame; coincident carriers with different labels or frames are allowed, and every cardinality, mass, multiplicity, and $\Delta_{\max}$ counted by labels. Assume $|\mathcal{P}| \le \alpha^{-C_{\mathrm{comp}}}$, and suppose*

$$\lambda(\mathcal{P},Y) \ge \alpha^{\eta}, \qquad \mu(\mathcal{P},Y) \ge \alpha^{-\eta},$$

Let $m > 0$, and suppose the labelled multiplicity lies in $[m, 2m)$ on the union. Let $0 < \theta \le 1$, and let $A_{\mathrm{ang}} = A_{\mathrm{ang}}(\alpha), B_{\mathrm{ang}} = B_{\mathrm{ang}}(\alpha) \ge 2$ satisfy $\log A_{\mathrm{ang}} + \log B_{\mathrm{ang}} = o(\log \alpha^{-1})$, and assume that $A_{\mathrm{ang}}/\log^{C}(2/\alpha) \to \infty$ for every fixed C. Assume that at each shaded point all incident two-long-axis planes have raw angular diameter at most $C\theta$, while every incident subcollection of at least $2m/A_{\mathrm{ang}}$ labels has raw angular diameter greater than θ/B_{ang}. Finally assume the buffer

$$\theta \ge 2B_{\mathrm{ang}}\,\alpha/\rho. \tag{10.21b}$$

Then there is a labelled refinement $(\mathcal{P}', Y')$ *with*

$$M(Y') \geq \alpha^{o(1)} M(Y), \tag{10.21c}$$

where this occurrence uses the displayed uniform modulus

$$\omega_{\mathrm{fill}}(\alpha) := \frac{\log(1/c_{\mathrm{fill}})}{\log(1/\alpha)} \longrightarrow 0$$

and $c_{\mathrm{fill}} \in (0,1]$ *is the fixed structural mass-retention product furnished by theorem C.4. In particular, this is a retained labelled shading-mass fraction, not a fraction of distinct carriers or labels. Whenever* $U(\mathcal{P}', Y')$ *meets a ball of radius* $\theta\rho$*, its intersection with a fixed concentric enlargement of that ball has relative volume at least* $\alpha^{4\eta+o(1)}$*. Passing to any labelled subfamily cannot increase its labelled* $\Delta_{\max}$*.*

Proof. This is theorem C.4, proved independently in section C. Its proof uses translated full-piece grids, the labelled denominator in every grid box, and an ordered-pair estimate. The buffer (10.21b) is built into the theorem, whose filling conclusion is the input to the transverse branch below. □

Lemma 10.13 (Transverse plank and filling interface)**.** *Assume (10.20)–(10.21),* $\theta \geq \delta^{-\tau'} q_{\mathrm{ang}}$*, and* $\delta^{-\tau'} \geq 2B_{\mathrm{ang}}$*. Fix a retained outer-ball tag* B*, write* $\theta_B, d_{\angle,B}$ *for its actual outputs in (10.17a), and put* $R_B = \theta_B b$*. There are structural constants* $C_0, C_1 \geq 1$*, a point* y_B*, and a source discrete radius* r *such that*

$$y_B \in U(\mathbb{T}_r, Y_r), \qquad 2C_0 R_B \leq r \leq C_1 \delta^{-\eta_{\mathrm{grid}}} R_B,$$

and

$$\frac{|U(\mathbb{T}, Y^{\mathrm{sc}}) \cap B(y_B, 2C_0 R_B)|}{|B(y_B, 2C_0 R_B)|} \gtrsim_{\log} \delta^{\eta} (\delta/a)^3 \alpha^{4\eta_{\mathrm{pl}}}. \tag{10.22}$$

For this radius,

$$\frac{|U(\mathbb{T}, Y^{\mathrm{sc}}) \cap B(y_B, r)|}{|B(y_B, r)|} \gtrsim_{\log} \delta^{E_{\mathrm{tr}}} \geq \delta^{-2\nu_{\mathrm{tr}}} \left(\frac{\delta}{r}\right)^{2\beta}, \tag{10.23}$$

where

$$E_{\mathrm{tr}} = 3\tau + \eta + 4\eta_{\mathrm{pl}} + 3\eta_{\mathrm{grid}}, \qquad \nu_{\mathrm{tr}} = \tau'\beta - \tfrac{1}{2} E_{\mathrm{tr}} > 0. \tag{10.23a}$$

Consequently the filling-to-multiplicity implication in source equation (89) yields the required very-non-sticky estimate in this alternative.

Proof. Fix the tag B from the statement and scale its r_1-ball to the unit ball. By the permanent-parent convention (3.4d), the dimension-level selection, and the Löwner-rectangle transfer in lemma 8.4, there is one structural $C_{\mathrm{box}} \geq 1$, fixed before the scale cutoff, for which the permanent angular parents passed to Appendix C are C_{box}-comparable rectangular $\alpha \times \rho_2 \times 1$ planks with their inherited ordered frames. Using the actual sum of these carrier volumes, their shading density is at least $\mathcal{D} \geq \alpha^{\eta_{\mathrm{pl}}}$ by (10.21), and their pointwise multiplicity lies in $[d_{\angle,B}, 2d_{\angle,B})$. By (10.17a) and (10.20), $d_{\angle,B} \geq d_{\angle} \geq \alpha^{-\eta_{\mathrm{pl}}}$. Use $m = d_{\angle,B}$, this C_{box}, and the local scale θ_B in lemma 10.12. A2 gives raw angular diameter at most $2\theta_B$. If an incident subcollection has at least $2m/A_{\mathrm{ang}}$ labels, then it has at least the cell degree divided by A_{ang}, so A3 applies. Finally,

$$\theta_B \geq \theta \geq \delta^{-\tau'} q_{\mathrm{ang}} \geq 2B_{\mathrm{ang}} q_{\mathrm{ang}}$$

converts the resolved lower diameter into the required raw one and verifies the buffer. The affine family is a labelled multiset, and its cellular shadings are measurable finite unions of cells; these observations verify the remaining hypotheses of the lemma. Its conclusion supplies a measurable parent subshading $Z_{\mathrm{tr}} \subset Z^{\angle} \subset Z$ whose union has filling fraction $\gtrsim_{\log} \alpha^{4\eta_{\mathrm{pl}}}$ in every intersecting ball of normalized radius $\theta_B \rho_2$, which scales back to $R_B = \theta_B b$. Choose one such ball and call it

$B_0 = B(x_B, C_0R_B)$, increasing the structural constant C_0 once to cover the fixed enlargement in (10.11b′). Applying that localized estimate to Z_{tr} gives

$$\frac{|U(\mathbb{T}_B, Y_B) \cap B_0|}{|B_0|} \gtrsim_{\log} \delta^{\eta}(\delta/a)^3 \alpha^{4\eta_{\mathrm{pl}}}.$$

In particular this intersection has positive measure; choose $y_B \in U(\mathbb{T}_B, Y_B) \cap B_0$. Then $B_0 \subset B(y_B, 2C_0R_B)$, and comparison of the two ball volumes loses only a structural constant. Every point of $Y_B(T_B)$ has positive represented multiplicity in the pre-biased fine shading Y^0, so $U(\mathbb{T}_B, Y_B) \subset U(\mathbb{T}, Y^0) \subset U(\mathbb{T}, \bar{Y})$. The reference ordering (10.32f) more precisely gives $U(\mathbb{T}_B, Y_B) \subset U(\mathbb{T}, Y^{\mathrm{sc}})$, so the recentered-ball comparison proves (10.22). Choose r to be the first source grid radius at least $2C_0R_B$; neighbouring-grid comparability gives $r \le C_1\delta^{-\eta_{\mathrm{grid}}}R_B$. By the definition of the induced scale- r shading, every retained point of Y^{sc} lies in the shading of its scale- r ancestor. Thus $y_B \in U(\mathbb{T}_r, Y_r)$, verifying the centering hypothesis in source equation (87).

The thin case gives $(\delta/a)^3 \ge \delta^{3\tau}$, while $\alpha^{4\eta_{\mathrm{pl}}} \ge \delta^{4\eta_{\mathrm{pl}}}$. Passing from R_B to r loses at most $(R_B/r)^3 \ge \delta^{3\eta_{\mathrm{grid}}}$, proving the first inequality in (10.23). Also

$$R_B = \theta_B b \ge \delta^{-\tau'} a \ge \delta^{1-\tau'}, \qquad r \ge R_B,$$

so $(\delta/r)^{2\beta} \le \delta^{2\tau'\beta}$. The definition of ν_{tr} proves the second inequality, and (10.32c) makes it positive.

We now combine the two terminal source estimates. On the fixed scale shading, source equation (87) says

$$|U(\mathbb{T}, Y^{\mathrm{sc}})| \gtrsim_{\log} |U(\mathbb{T}_r, Y_r)| \frac{|U(\mathbb{T}, Y^{\mathrm{sc}}) \cap B(y_B, r)|}{|B(y_B, r)|}. \tag{10.23b}$$

Source equation (88), with the generalized Katz–Tao allowance $\nu_{\mathrm{tr}}/9$, and the local estimate (10.23), which is the hypothesis labelled (89) in the source, are respectively

$$\begin{aligned} |U(\mathbb{T}_r, Y_r)| &\gtrsim \delta^{\nu_{\mathrm{tr}}/9} \left(\frac{r}{\delta}\right)^{2\beta} |T|\,|\mathbb{T}|^{1-\beta}, \\ \frac{|U(\mathbb{T}, Y^{\mathrm{sc}}) \cap B(y_B, r)|}{|B(y_B, r)|} &\gtrsim_{\log} \delta^{-2\nu_{\mathrm{tr}}} \left(\frac{\delta}{r}\right)^{2\beta}. \end{aligned} \tag{10.23c}$$

Substitution cancels the two scale ratios and gives

$$|U(\mathbb{T}, Y^{\mathrm{sc}})| \gtrsim_{\log} \delta^{-17\nu_{\mathrm{tr}}/9} |T|\,|\mathbb{T}|^{1-\beta}. \tag{10.23d}$$

Since $M(Y^{\mathrm{sc}}) \le |\mathbb{T}|\,|T|$ and $M(Y^{\mathrm{sc}}) \ge c_{\mathrm{sc}} M(\bar{Y})$, while $U(\mathbb{T}, Y^{\mathrm{sc}}) \subset U(\mathbb{T}, \bar{Y})$,

$$\mu(\mathbb{T}, \bar{Y}) \lesssim_{\log} c_{\mathrm{sc}}^{-1} \delta^{17\nu_{\mathrm{tr}}/9} |\mathbb{T}|^{\beta} = \delta^{17\nu_{\mathrm{tr}}/9 - o(1)} |\mathbb{T}|^{\beta}. \tag{10.23e}$$

This is the required very-non-sticky gain and includes the previously fixed all-scale refinement loss. □

10.8. **Tangential parents and the lift to fine tubes.** The transverse high-degree branch has been closed by the preceding interface. We now assume its complementary, tangential alternative

$$\theta < \delta^{-\tau'} q_{\mathrm{ang}} = \delta^{-\tau'} a/b. \tag{10.23f}$$

Equality is assigned to the transverse branch of lemma 10.13. In particular, the weak inequality needed below follows from this strict complementary alternative.

For each retained labelled copy $\widetilde{W}$ of a geometric parent W, use the fixed label order to choose one assigned tubelet and let $v(\widetilde{W})$ be its unoriented unit carrier direction. Every other tubelet assigned to $\widetilde{W}$ makes angle $O(b/r_1) = O(\rho_2)$ with $v(\widetilde{W})$, and each represented full fine tube has an additional direction error $O(\delta/r_1) \le O(\rho_2)$. This is the representative direction

used in the fine-tube cap in source equation (100). It is distinct from $TB_{\widetilde{W}} := TB_W$: the latter is the two-long-axis plane of the source Löwner frame and is used for parent angles and slab normalization. Thus neither object is inferred from the other.

Apply the angle output through one lift. With the same set $\mathcal{R}(T)$ of retained labelled tubelet representations used in (10.19a), define

$$Y^*(T) = \widehat{Y}(T) \cap \bigcup_{(B,T_B)\in\mathcal{R}(T)} Y_2(T_B). \tag{10.24}$$

Equations (10.6)–(10.7), the factoring losses, and the one lift give the explicit fine-refinement fraction

$$c_* := \frac{\sum_T |Y^*(T)|}{\sum_T |\bar{Y}(T)|} \gtrsim \frac{c_{\mathrm{setup}} c_{\mathrm{ang}}}{64\kappa_{\mathrm{inc}}^2 P_{\mathrm{tag}} P_{\mathrm{angtag}} J_v L_m L_{a,b} L_N L_{\mathrm{reg}}}. \tag{10.25}$$

For every point of $U(\mathbb{T}, Y^*)$, compatibility gives the line corresponding to source equation (100):

$$\mathbb{T}_{Y^*}(x) \subset \bigcup_{\widetilde{W}\colon x\in Z^{\angle}(\widetilde{W})} \{T \in \mathbb{T}_{\bar{Y}}(x) : \mathrm{angle}(T, v(\widetilde{W})) \lesssim \rho_2\}. \tag{10.26}$$

The outer degree is less than $4d_{\angle}$ in each tag and less than $4\kappa_{\mathrm{amb}} d_{\angle}$ after physical recombination, while every inner cap is measured in the fixed uniform input. The overlap factor is absorbed in $\lesssim_{\log}$. The refinement-multiplicity estimate [GWZ26, Lemma 5.5, p. 10] and (10.25) therefore give the corresponding multiplicity line:

$$\mu(\mathbb{T}, \bar{Y}) \lesssim_{\log} c_*^{-1} d_{\angle} \mu(\mathbb{T}[T_{\rho_2}], \bar{Y}). \tag{10.27}$$

The Katz–Tao input is

$$\mu(\mathbb{T}[T_{\rho_2}], \bar{Y}) \lesssim \delta^{-\eta_{\mathrm{bias}}} |\mathbb{T}[T_{\rho_2}]|^{\beta}, \tag{10.28}$$

and the resolved-angle statement gives

$$x \in Z_B^{\angle}(W_1) \cap Z_B^{\angle}(W_2) \quad \text{for one retained tag } B \Longrightarrow \mathrm{angle}_{q_{\mathrm{ang}}}(W_1, W_2) < 4\theta, \quad \theta \le \delta^{-\tau'} a/b. \tag{10.29}$$

These are the lines corresponding to source equations (102) and (103). The tangential slab grouping uses the raw angular upper bound, while the labels retain their given multiplicities even when their carriers are parallel.

Slab groups and dense mass. Fix one retained outer-ball tag B for the rest of the slab grouping and conflict coloring, and suppress this tag from the notation. The quantities θ and $d_{\angle}$ are the common reference values in (10.17a), and the local values remain within a factor two. Hence a local degree estimate also controls $d_{\angle}$. All complete-containment and Katz–Tao constants below belong to this fixed tag. Global fine mass was retained in (10.25), and recombination of the resulting pointwise bound contributes the displayed κ_{amb}.

Use a half-open parameter grid to assign every labelled angular parent to one slab S of dimensions comparable to

$$q_{\mathrm{ang}} r_1 \times r_1 \times r_1. \tag{10.29a}$$

Labels are never merged. At a fixed shaded point, (10.29) implies that the incident labels occupy at most

$$L_{\mathrm{grp}} \le \delta^{-2\tau' - o(1)} \tag{10.29b}$$

slab groups. The coefficient 2 is the dimension of the normal net: a slab through a fixed point has $O(1)$ admissible offset cells, while a θ-cap meets $O((\theta/q_{\mathrm{ang}})^2) \le O(\delta^{-2\tau'})$ normal cells.

Write

$$\lambda_S = \frac{M(Z_S^{\angle})}{|\widetilde{\mathbb{W}}_S^{\angle}|\,|W|}, \qquad \lambda = \frac{M(Z^{\angle})}{|\widetilde{\mathbb{W}}^{\angle}|\,|W|}.$$

The half-open slab assignment partitions the labelled parent indices. On the fixed dimension level the definitions use the same carrier-volume proxy $|W|$ for every label, to which the carrier

volumes are structurally comparable. Therefore λ is the denominator-weighted average of the λ_S, with weights $|\widetilde{\mathbb{W}}_S^{\angle}||W|$. Call S dense when $\lambda_S \geq \lambda/2$. Summing $M(Z_S^{\angle}) = \lambda_S|\widetilde{\mathbb{W}}_S^{\angle}||W|$ over the complementary groups shows that they carry less than one half of $M(Z^{\angle})$. Since $\sum_S |U(\widetilde{\mathbb{W}}_S^{\angle}, Z_S^{\angle})| \leq L_{\mathrm{grp}}|U(\widetilde{\mathbb{W}}^{\angle}, Z^{\angle})|$, summing $M(Z_S^{\angle}) \leq \mu(\widetilde{\mathbb{W}}_S^{\angle}, Z_S^{\angle})|U(\widetilde{\mathbb{W}}_S^{\angle}, Z_S^{\angle})|$ gives

$$d_{\angle} \lesssim L_{\mathrm{grp}} \max_{S \text{ dense}} \mu(\widetilde{\mathbb{W}}_S^{\angle}, Z_S^{\angle}). \tag{10.29c}$$

Separating repeated labels. Fix a dense S, and let L_S be its common affine normalization to the unit ball. Write $P_W = L_S(W)$ for the raw image of a labelled permanent parent. In slab-adapted coordinates its three edge vectors have sizes $O(\rho_2), O(\rho_2), O(1)$, their determinant has magnitude $\asymp \rho_2^2$, and the longest one has size $\asymp 1$. Here the fixed side-ratio bounds and the $O(q_{\mathrm{ang}})$ plane tilt give the upper bounds, while the determinant and the first two upper bounds give the remaining lower bounds. Thus, after dividing the first two scale directions by ρ_2, the allowed edge matrices range in one structurally compact subset of GL_3.

Consequently there are structural constants $h_* \in (0, 1]$ and $C_* \geq 1$, common to the whole slab group, such that for every label one may choose a source-standard width-ρ_2 tube $\widehat{T}_W$, with carrier segment of length exactly one, satisfying

$$P_W^* := h_* P_W \subset \widehat{T}_W \subset C_* P_W^*. \tag{10.29d}$$

The last dilation is concentric in the raw parallelepiped coordinates and includes the fixed endpoint caps. Shrinking h_* once makes the first containment and the common ambient unit ball simultaneous; compactness gives the reverse fixed containment. Give $\widehat{T}_W$ the original parent label and the unchanged normalized shading $\widehat{Z}(\widehat{T}_W) := h_* L_S(Z_S^{\angle}(W))$.

The vertices of G_S are these labelled exact source-standard proxies. Join two distinct vertices precisely when the proxies are comparable, meaning that each is contained in a fixed dilate of the other. If $\widehat{T}_{W'}$ is adjacent to $\widehat{T}_W$, then

$$P_{W'}^* \subset \widehat{T}_{W'} \subset C\widehat{T}_W \subset C' P_W^*.$$

The raw parent volumes on the dimension level are comparable, so the complete-containment definition of $\Delta_{\max}$ gives

$$\deg_{G_S}(\widehat{T}_W) + 1 \leq \#\{W' : P_{W'}^* \subset C' P_W^*\} \lesssim D_S.$$

Greedy coloring therefore uses

$$H_{\mathrm{col}} \lesssim \Delta_{\max}(\widetilde{\mathbb{W}}_S^{\angle}) \leq D_S \lesssim \delta^{-O(\varepsilon_{\mathrm{fact}})}. \tag{10.30}$$

Each proxy color is source-essentially distinct. Moreover, if a convex K contains $\widehat{T}_W$, then it contains P_W^*, and $|\widehat{T}_W| \asymp |P_W^*|$; hence

$$\Delta_{\max}(\{\widehat{T}_W\}) \lesssim D_S. \tag{10.30$'$}$$

The common affine map and homothety preserve labels and complete-containment relations and multiply raw carriers, shadings, their union, and convex test bodies by one Jacobian. Passing from P_W^* to $\widehat{T}_W$ leaves every shading, its union, pointwise multiplicity, and average multiplicity unchanged; carrier densities change by only a structural factor, and $\Delta_{\max}$ obeys (10.30$'$).

Let N_S and M_S be the normalized slab cardinality and labelled shading mass, and put

$$\widehat{\lambda}_S := \frac{M_S}{N_S|T_{\rho_2}|} \asymp \lambda_S.$$

Let N_j, M_j, and $\lambda_j = M_j/(N_j|T_{\rho_2}|)$ denote the cardinality, labelled shading mass, and density of color j; write $(\mathcal{F}_{S,j}, A_{S,j})$ for this proxy color and its unchanged shading. Colors with $\lambda_j \geq \widehat{\lambda}_S/2$ retain at least half of the mass of S. Indeed the proper colors partition the labelled indices, their normalized source-standard proxies have one common volume, and $\widehat{\lambda}_S$ is the

$N_j|T_{\rho_2}|$-weighted average of the λ_j; the colors below half the average contribute less than half the denominator-weighted mass. Since every color union is contained in the slab union, this also gives

$$\mu(\widetilde{\mathbb{W}}_S^{\angle}, Z_S^{\angle}) \le 2 \sum_{j \text{ dense}} \mu_j. \tag{10.30a}$$

The strict reserve in condition (K), followed by the final scale cutoff, absorbs the fixed comparison between $\widehat{\lambda}_S$ and λ_S. It therefore verifies the normalized density hypothesis of the generalized Katz–Tao estimate on each such proxy color, and the preceding graph construction verifies its source essential-distinctness and complete-containment hypotheses. Since the proxy and raw families carry the same shadings, the resulting multiplicity estimate pulls back without loss:

$$\mu_j \lesssim \rho_2^{-\varepsilon_{KT}} D_S^{1-\beta} N_j^{\beta}. \tag{10.30b}$$

For $0 < \beta \le 1$, concavity and complete containment give

$$\sum_j N_j^{\beta} \le H_{\mathrm{col}}^{1-\beta} \Big(\sum_j N_j \Big)^{\beta}, \qquad \sum_j N_j \lesssim D_S \rho_2^{-2}. \tag{10.30c}$$

The non-sticky count $\rho_2^{-2} \le \rho_2^{\zeta} |\mathbb{T}_{\rho_2}|$, followed by (10.29c)–(10.30c), yields the full scaling of the source call:

$$d_{\angle} \lesssim_{\log} L_{\mathrm{grp}} \rho_2^{-\varepsilon_{KT}} D_S H_{\mathrm{col}}^{1-\beta} \rho_2^{\beta\zeta} |\mathbb{T}_{\rho_2}|^{\beta}. \tag{10.31}$$

Equivalently,

$$d_{\angle} \lesssim \delta^{-\Lambda_{31} - o(1)} \rho_2^{\beta\zeta} |\mathbb{T}_{\rho_2}|^{\beta}, \quad \Lambda_{31} = 2\tau' + (1 - \varepsilon_{\mathrm{scal}})\varepsilon_{KT} + \Lambda_D + (1-\beta)\Lambda_H. \tag{10.31a}$$

Closing the tangential branch. Insert (10.31a) and (10.28) into the fine lift (10.27), then use

$$|\mathbb{T}[T_{\rho_2}]|\, |\mathbb{T}_{\rho_2}| \lesssim_{\log} |\mathbb{T}|. \tag{10.31b}$$

All cap and coarse-direction factors cancel in this product, leaving

$$\mu(\mathbb{T}, \bar{Y}) \lesssim \delta^{-\Lambda_* - \eta_{\mathrm{bias}} - \Lambda_{31} - o(1)} \rho_2^{\beta\zeta} |\mathbb{T}|^{\beta}. \tag{10.31c}$$

Since $\rho_2 \le \delta^{\varepsilon_{\mathrm{scal}}}$, the available exponent is

$$\nu_{\tan} = \varepsilon_{\mathrm{scal}} \beta\zeta - \Lambda_* - \eta_{\mathrm{bias}} - \Lambda_{31} - o(1). \tag{10.32}$$

In particular, the source leading gain is retained under the explicit condition

$$\Lambda_* + \eta_{\mathrm{bias}} + 2\tau' + (1 - \varepsilon_{\mathrm{scal}})\varepsilon_{KT} + \Lambda_D + (1-\beta)\Lambda_H < \varepsilon_{\mathrm{scal}} \beta\zeta. \tag{T}$$

When the left side is at most half the right side, (10.32) gives $\nu_{\tan} \ge \frac{1}{2} \varepsilon_{\mathrm{scal}} \beta\zeta > 0$ for small δ. Set

$$\underline{\nu}_{\tan} := \tfrac{1}{2} \varepsilon_{\mathrm{scal}} \beta\zeta. \tag{10.32$'$}$$

Thus the tangential conclusion is available with the fixed exponent $\underline{\nu}_{\tan}$, independently of the remaining $o(1)$-term. Replacing a quadratic density loss by $2 - \sigma$ improves only an adjustable $O(\eta)$ term and does not alter this terminal exponent.

10.9. **Parameter hierarchy and closure.** Choose the exponents in proof order. First fix β, ζ, and take the source scale constant $\varepsilon_{\mathrm{scal}}(\beta) > 0$ sufficiently small that

$$8\varepsilon_{\mathrm{scal}} < (1 - \varepsilon_{\mathrm{scal}})\beta(2 + \zeta). \tag{10.32p}$$

Put $G = \varepsilon_{\mathrm{scal}} \beta\zeta$. Choose τ' so that

$$2\tau' < G/16. \tag{10.32a}$$

Next choose $\varepsilon_{KT} > 0$ so that $(1 - \varepsilon_{\mathrm{scal}})\varepsilon_{KT} < G/32$, and let $\eta_{KT}(\varepsilon_{KT}, \beta) > 0$ be the resulting threshold. After fixing this number, choose the single source bias exponent $\varepsilon_{\mathrm{fact}} = \eta_{\mathrm{bias}}$ so small that

$$[1 + 2(1 - \varepsilon_{\mathrm{scal}})(2 - \beta)]\varepsilon_{\mathrm{fact}} < G/64. \tag{10.32a0}$$

This bounds the non-subpower contribution $\eta_{\mathrm{bias}} + \Lambda_D + (1-\beta)\Lambda_H$ in (T), by (10.16f); make $\varepsilon_{\mathrm{fact}}$ smaller once more if needed to fit the strict reserve in (10.32s). All remaining terms from $\Lambda_{\mathrm{sync}}, \Lambda_{\mathrm{pre}}, \Lambda_*$ are copies of the single function $\omega_{\mathrm{src}}(\delta)$ and will be absorbed after the parameters are fixed. Then choose $\tau, \eta_{\mathrm{pl}}, \eta_{\mathrm{grid}}, \xi > 0$, first taking $\xi < \tau'\beta/1024$ and

$$2\xi < \eta_{\mathrm{pl}}, \tag{10.32a$''$}$$

and then making all four numbers small enough that

$$\begin{aligned} 3\tau + 4\eta_{\mathrm{pl}} + 3\eta_{\mathrm{grid}} &< \tfrac{1}{8}\tau'\beta, \\ (1-\varepsilon_{\mathrm{scal}})(\eta_{\mathrm{pl}} + \xi) &< \tfrac{1}{4}\varepsilon_{\mathrm{scal}}\beta(2+\zeta), \\ (1-\varepsilon_{\mathrm{scal}})\xi &< \tfrac{1}{4}\varepsilon_{\mathrm{scal}}\eta_{KT}. \end{aligned} \tag{10.32b}$$

Decrease η_{grid} further, if necessary, so that

$$3\eta_{\mathrm{grid}} < \frac{1}{16}\tau\varepsilon_{\mathrm{fact}}\beta.$$

Then choose $\varepsilon_{\mathrm{thick}} > 0$ smaller than the remaining right-hand reserve in (10.8a), and let $\eta_F(\varepsilon_{\mathrm{thick}}, \beta) > 0$ from the Frostman plank estimate. Let $\eta_{KT}^{\mathrm{cap}} > 0$ be the density exponent supplied by [GWZ26, Lemma 3.7 and Remark 3.6] for output allowance $\varepsilon_{\mathrm{fact}}/2$. Finally choose ε, η, and all remaining setup/refinement exponents so small that (10.8a) holds, $\eta < \frac{1}{8}\tau'\beta$, and their remaining contribution to E_{dens} is less than $\frac{1}{2}\varepsilon_{\mathrm{scal}}\eta_{KT}$, $\eta < \tau\eta_F(\varepsilon_{\mathrm{thick}}, \beta)$, and (L), (H), (K), and (T) hold with strict slack. Using (10.32p), require at the same time

$$2\big[q(\eta + \Lambda_{\mathrm{pre}}) + (1-\varepsilon_{\mathrm{scal}})\varepsilon\big] + \Lambda_D + 4\varepsilon_{\mathrm{scal}} + 2\eta + 3\tau + \eta + \Lambda_{\mathrm{sync}} < \tfrac{1}{2}(1-\varepsilon_{\mathrm{scal}})\beta(2+\zeta). \tag{10.32s}$$

This is possible because, after (10.32a0), the only fixed positive terms on the left are $4\varepsilon_{\mathrm{scal}}$ and the explicitly bounded $2(1-\varepsilon_{\mathrm{scal}})\varepsilon_{\mathrm{fact}}$; (10.32p) and the extra shrinking allowed after (10.32a0) leave strict room. In particular,

$$3\tau + \eta + 4\eta_{\mathrm{pl}} + 3\eta_{\mathrm{grid}} < \tfrac{1}{4}\tau'\beta, \qquad \xi + \varepsilon + \frac{q(\eta + \Lambda_{\mathrm{pre}})}{\varepsilon_{\mathrm{scal}}} < \eta_{\mathrm{pl}}, \qquad E_{\mathrm{dens}} < \varepsilon_{\mathrm{scal}}\eta_{KT}. \tag{10.32c}$$

At this stage the inequalities are imposed with $\omega_{\mathrm{src}} = 0$ and with a fixed positive reserve. Now choose the source input exponent $\eta_{\mathrm{src}} > 0$ so that $2\eta_{\mathrm{src}} < \eta$, $4\eta_{\mathrm{src}} \le \eta_{KT}^{\mathrm{cap}}$, and $2\eta_{\mathrm{src}}$ lies below every other $K_{KT}(\beta)$ density threshold used in proposition 10.19, including the scale-r call with $\nu_{\mathrm{thick}}/10$ in (10.8h), and also require $(1-\beta)\eta_{\mathrm{src}} < \varepsilon_{\mathrm{fact}}/8$. With $\mathfrak{p}$ fixed, first decrease $\delta_0 = \delta_0(\beta, \zeta, \mathfrak{p})$ until $\omega_{\mathrm{pre}}(\delta) < \eta_{\mathrm{src}}$. This cutoff is independent of J_v and gives (10.8A0). The final decrease using ω_{src} is made only after (10.8A) and lemma 10.18 give $J_v = O(\log(1/\delta))$; at that point require $\omega_{\mathrm{src}}(\delta)$ to lie below η_{src} and the minimum of the finitely many reserved gaps in (L), (H), (K), (T), and (10.32s). The gap $\eta - 2\eta_{\mathrm{src}}$ absorbs the good-ball deletion, while the source density gaps absorb the cap and other source-call losses. This absorbs the fixed constants and all $o(1)$-terms uniformly. Decreasing it further also enforces the raw-angle threshold used in the transverse branch. In fact, (10.16a) and (7.3) give

$$\log B_{\mathrm{ang}} = (\log \alpha^{-1})^{3/4} \le \big((1-\varepsilon_{\mathrm{scal}})\log \delta^{-1}\big)^{3/4}.$$

After τ' is fixed, choose δ_0 so that for $0 < \delta \le \delta_0$,

$$\tau' \log \delta^{-1} \ge \big((1-\varepsilon_{\mathrm{scal}})\log \delta^{-1}\big)^{3/4} + \log 2. \tag{10.32d}$$

Then $\delta^{-\tau'} \ge 2B_{\mathrm{ang}}$, hence $\theta \ge 2B_{\mathrm{ang}} q_{\mathrm{ang}}$, and (7.8) gives genuine raw separation even in the presence of repeated parallel labels. This verifies the low, high, dense-color, transverse, and tangential inequalities simultaneously.

The following definition is the input contract for the five terminal branches: it records their common data and quantitative identities, while the branch lemmas supply their conclusions. The closure theorem is conditional on $K_{KT}(\beta)$, $K_F(\beta)$, and the explicit source interface (I1)–(I4).

Under these assumptions, proposition 10.19 derives every admissibility item from the fixed GWZ refinement chain. The resulting admissible data close in theorem 10.20, and corollary 10.21 then returns to the original labelled input.

Admissible input.

Definition 10.14 (Admissible Section 9 input)**.** A fixed reference family $(\mathbb{T}, \bar{Y})$, together with a fixed loss profile $\mathfrak{p}$, with $0 < \beta \le 1$, $\zeta > 0$, and $0 < 2\eta_{\mathrm{src}} < \eta$, at the Section 9 scales is *admissible* if the following data are part of the input.

Geometric admissibility (S1).

(S1) The active fine tubes, tubelets, parents, and cells have total polynomial complexity, and the good-ball occupancy is (10.2). Before the good-ball deletion every active tubelet satisfies (10.1a), and after it

$$\lambda(\mathbb{T}_B, Y_B) \gtrsim \delta^{\eta}. \tag{10.32e}$$

The all-discrete-scale refinement $Y^{\mathrm{sc}} \subset \bar{Y}$ and its scale shadings are fixed before the tubelets, satisfy (10.8h), and obey $M(Y^{\mathrm{sc}}) \ge c_{\mathrm{sc}} M(\bar{Y})$ with $c_{\mathrm{sc}}^{-1} = \delta^{-o(1)}$. The objects are selected in the order (10.1)–(10.8), and the represented fine-mass identities (10.4)–(10.7), with the displayed $\kappa_{\mathrm{inc}}, \kappa_{\mathrm{loc}}, L_m, L_{a,b}$, hold on the same object. At the realized dimension level exactly one of the following three alternatives is designated:

(a) if $a \ge \delta^{1-\tau}$, the fixed data satisfy the hypotheses of lemma 10.9: the inherited assertions $K_F(\beta)$ and $K_{KT}(\beta)$, the bias and density inputs (10.8d)–(10.8f), the scale relation for r, and the already fixed source comparison (10.8h) are available. The conclusions (10.8b)–(10.8c) are not part of the definition;

(b) if $a < \delta^{1-\tau}$ and $b \ge \delta^{\varepsilon_{\mathrm{scal}}} r_1$, the multiplicity-expanded family and bounded-overlap concentration estimate (10.7a)–(10.7c), the complete-parent data (10.12)–(10.16), and the profile denominator bound in the second inequality of (10.34d) hold on the same labels. At $\rho_- := \delta^{1-\varepsilon_{\mathrm{scal}}}$, the source middle-range count and bounded-ancestry comparison

$$|\mathbb{T}| \gtrsim |\mathbb{T}_{\rho_-}| \ge \rho_-^{-2-\zeta}$$

are part of the same input;

(c) if $a < \delta^{1-\tau}$ and $b < \delta^{\varepsilon_{\mathrm{scal}}} r_1$, the scales obey (10.16a).

Reference-shading admissibility (S2)–(S3).

(S2) The incident-centred pointwise upper bound (10.19e0) holds on the reference shading $\bar{Y}$, uniformly under fixed dyadic dilation of ρ_2. With the source notation $\mu(\rho_2)$,

$$\mu(\rho_2) \lesssim \delta^{-\eta_{\mathrm{bias}}} |\mathbb{T}[T_{\rho_2}]|^{\beta}, \qquad |\mathbb{T}[T_{\rho_2}]|\,|\mathbb{T}_{\rho_2}| \lesssim_{\log} |\mathbb{T}|, \qquad |\mathbb{T}_{\rho_2}| \ge \rho_2^{-2-\zeta}.$$

The same constants and dyadic levels apply to all incident-centred direction balls. No fixed-cap boundary lower bound is assumed. This condition, and (S3), are required only under the thin non-slab alternative (S1c).

(S3) For each chosen $\varepsilon_{KT} > 0$, there is $\eta_{KT}(\varepsilon_{KT}, \beta) > 0$ such that every essentially distinct normalized ρ_2-tube family $(\mathcal{F}, A)$ with Katz–Tao constant D and shading density at least $\rho_2^{\eta_{KT}}$ satisfies

$$\mu(\mathcal{F}, A) \lesssim_{\varepsilon_{KT}, \beta} \rho_2^{-\varepsilon_{KT}} D^{1-\beta} |\mathcal{F}|^{\beta}.$$

Quantitative-loss admissibility (S4).

(S4) Under the thin non-slab alternative (S1c), the setup fraction, bias constant, denominator level, incidence and fixed-tag representation numbers, dyadic level counts, angular fraction, parent Katz–Tao constant, and conflict-color number satisfy the individual bounds (10.16d)–(10.16f). Every selection after the positive graph is a saturated right-cell selection, a complete-tag synchronization from theorem 10.7, or a whole-incidence selection as in

(10.11″) and (8.5). Consequently the mass identities (10.19b) and (10.25) are controlled by the conservative uniform envelope (10.16d) and the operation map in lemma 10.18. All fixed constants and convergence moduli occurring in these bounds are controlled by the previously fixed profile $\mathfrak{p}$. Under the thin slab alternative (S1b), the required loss data are instead (10.7b), (10.12)–(10.16), and the second inequality in (10.34d); the relevant exponents $\Lambda_{\mathrm{pre}}, \Lambda_D, \Lambda_{\mathrm{den}}$ obey (10.16f) and the strict slab reserve (10.32s). Under the thick alternative (S1a), they are $c_{\mathrm{pre}}, c_{\mathrm{bias}}$, and L_{thick} in (10.8d)–(10.8h), after which the construction stops.

These fixed source inputs are used on their stated reference objects; later angular or spatial refinements enter only through the listed saturated operations.

Source verification.

Lemma 10.15 (Carrier–shading commutation rule)**.** *Let $\mathcal{L}$ be a finite labelled carrier family and let $Y^B(\ell) \subset Y(\ell)$ satisfy*

$$|Y^B(\ell)| \geq c_B|Y(\ell)| > 0 \qquad (\ell \in \mathcal{L}) \tag{10.32g}$$

for one $c_B > 0$. Let A be a deterministic operation whose output depends only on the complete labelled carriers and on carrier-only data (complete containment, dimensions, a fixed label order, or a partition into finitely many such classes), and not on the location or mass of the shading. Then applying A before or after the restriction $Y \mapsto Y^B$ selects exactly the same labels, parents, and carrier classes. For every class $\mathcal{L}_A$ selected by A,

$$\sum_{\ell\in\mathcal{L}_A} |Y^B(\ell)| \geq c_B \sum_{\ell\in\mathcal{L}_A} |Y(\ell)|. \tag{10.32h}$$

If a carrier-only partition has at most L classes and a class is chosen only after the restriction by shaded mass, the chosen class instead retains at least $(c_B/L)M(Y)$; equality of its label set with a class chosen before the restriction is not asserted.

There are three noncommutation cautions.

(C1) *A pointwise lower-multiplicity or angular statement remains attached to the shading on which it was proved. Mere inclusion does not guarantee transfer; below we transfer it only when every counted incidence is kept.*

(C2) *For a mass-scored operation, (10.32h) does not identify the outputs before and after restriction. Below it is rerun unless its scoring data are separately unchanged, and any new pigeonhole loss is recorded.*

(C3) *Arbitrary internal deletion from a positive parent–cell graph need not preserve its identities. Deleting whole right cells, tags, or positive incidences while keeping every child incidence on each surviving edge is the sufficient commutation rule used below.*

Proof. The complete label set and every carrier-only relation are unchanged by $Y(\ell) \mapsto Y^B(\ell)$; determinism therefore gives the same output of A. Summing (10.32g) over a selected class proves (10.32h). If the class is chosen afterwards, sum (10.32h) over all at most L classes and take a largest summand. Clause (C1) follows because deleting even one of several incidences through a point can destroy a lower bound there. Clause (C2) is the distinction between carrier data and shading scores. For (C3), the edge identity in theorems 4.2 and 5.1 shows that internal cutting may change the weight or child multiplicity; the saturated rule avoids this failure. □

The three noncommuting classes in lemma 10.15(C1)–(C3) govern the concrete source order recorded next in (R1)–(R4).

Lemma 10.16 (Reference objects and permitted reordering)**.** *For the fixed source refinement chain associated with a uniform input, the source and corrected operations may be performed in the following order:*

$$\begin{aligned} Y \longrightarrow \bar{Y} \longrightarrow Y^{\mathrm{sc}} \longrightarrow \textit{canonical windows} \longrightarrow (Y^{\mathrm{pre}}, Y_B) \longrightarrow Y^0 \longrightarrow m_{\mathrm{fine}} \\ \longrightarrow \textit{biased label subset} \longrightarrow (a,b,N)\textit{-levels} \longrightarrow \textit{tagged cellular graph.} \end{aligned} \tag{10.32f}$$

Source equation numbers refer to the archived version identified in Appendix D. Here $\bar{Y}$ is the output of source equation (86), Y^{sc} is the all-discrete-scale output used in (87)–(88), the canonical windows and their two levels are constructed by lemma 10.3, Y_B is their good-ball restriction, and m_{fine} is the global level of lemma 10.2. Every arrow after Y^{sc} is an explicit induced fine-incidence restriction, a deletion of complete canonical tubelet labels or tags, or a saturated tagged incidence deletion. In particular:

(R1) *the cap statement is read only on $\bar{Y}$, not transferred to a later shading;*
(R2) *(87)–(88) are read only on Y^{sc}, whose later local unions are subsets of that fixed union;*
(R3) *the good-ball operation precedes m_{fine}, and m_{fine} precedes every biased or dimension subset;*
(R4) *every operation after the positive graph is saturated on right cells, deletes complete outer-ball tags during one of the two synchronization stages of theorem 10.7, or is the one whole-incidence lift of theorem 5.1.*

By lemma 10.15, the good-ball deletion commutes with the biased-parent and dimension selections; lemma 10.1 supplies uniform per-label retention. Every other arrow is used in the displayed order and on its stated reference object. The product of the retained fractions before the graph is exactly $c_{\mathrm{pre}}c_{\mathrm{bias}}/(\kappa_{\mathrm{inc}}L_mL_{a,b}L_N)$, up to the fixed source comparison constants already named in those symbols.

Proof. In the v1 order, (86) precedes (87)–(88), which precede the tubelet levels. The canonical construction (10.7U)–(10.7X) gives (R1)–(R2). By lemmas 10.1 and 10.15, the good-ball deletion crosses the carrier-only bias and dimension choices while retaining (10.1c); lemma 10.2 then gives (R3). Saturation of the tagwise positive graph, the whole-tag deletions in theorem 10.7, and the single lift in theorem 5.1 give (R4). Multiplying the named representation and dyadic factors yields the stated retained fraction. □

Lemma 10.17 (The reference cap and the generalized KKT estimate)**.** *Assume $K_{KT}(\beta)$. Under the thin non-slab alternative, source equation (86) on p. 36 supplies the incident-centred cap relation (10.35). Assume also the separate coarse-average interface clause (10.35a) in (I2). Then* [GWZ26, Definition 2.2, p. 4; Lemma 3.7, p. 7; Remark 3.6, p. 6] *and these two relations imply (S2)–(S3) on the same reference shading $\bar{Y}$. More precisely, (10.35)–(10.37) hold for every incident-centred direction ball, and every essentially distinct normalized family of density at least $\rho_2^{\eta_{KT}}$ obeys the estimate in (S3). Both conclusions are read on the fixed reference shading $\bar{Y}$.*

Proof. Source equation (86) gives, for every $x \in \bar{Y}(T_0)$,

$$\#\{T : x \in \bar{Y}(T),\ \mathrm{angle}(T, T_0) \le \rho\} \asymp \mu(\rho). \tag{10.35}$$

The additional interface clause (10.35a) states separately that, for every coarse label S,

$$\mu\big(\mathbb{T}[S], \bar{Y}|_{\mathbb{T}[S]}\big) \asymp \mu(\rho). \tag{10.35a}$$

Thus the incident-centred cap count and the coarse-child average multiplicity are two distinct data on the same reference shading $\bar{Y}$. Neighbouring-scale stability and the common branching number give

$$N_\rho |\mathbb{T}_\rho| \lesssim_{\log} |\mathbb{T}|. \tag{10.36}$$

Write $M_S = \sum_{T\in\mathbb{T}[S]} |\bar{Y}(T)|$. Bounded representation and uniform branching give

$$\sum_S M_S \asymp M(\bar{Y}), \qquad \sum_S |\mathbb{T}[S]|\,|T| \asymp |\mathbb{T}|\,|T|.$$

Hence some S_0 has density $\gtrsim_{\log} \lambda(\mathbb{T},\bar{Y})$, while (10.35a) gives its multiplicity $\asymp \mu(\rho)$. Its Katz–Tao constant is at most $\delta^{-\eta_{\mathrm{src}}}$. The parameter closure already gives $(1-\beta)\eta_{\mathrm{src}} < \varepsilon_{\mathrm{fact}}/8$ and $\eta_{\mathrm{src}} \le \eta_{KT}^{\mathrm{cap}}/4$; reserve another quarter for the finitely many uniformity losses. Write $0 \le \omega_{\mathrm{cap}}(\delta) \le \omega_{\mathrm{src}}(\delta) \to 0$ for those losses. The profile cutoff $\omega_{\mathrm{src}}(\delta) < \eta_{\mathrm{src}}$ also absorbs the combined loss in the preceding density selection. The cap call uses the interface

$$\begin{gathered} s_\rho = \delta/\rho, \qquad \tau_{\mathrm{src}} = \delta \le s_\rho, \qquad \mathcal{F}_\rho = L_{S_0}(\mathbb{T}[S_0]), \\ A_\rho = L_{S_0}(\bar{Y}|_{\mathbb{T}[S_0]}), \qquad \lambda(\mathcal{F}_\rho, A_\rho) \gtrsim \delta^{2\eta_{\mathrm{src}}}, \\ \Delta_{\max}(\mathcal{F}_\rho) \le \delta^{-\eta_{\mathrm{src}}}, \qquad |\mathcal{F}_\rho| = N_\rho, \qquad \mu(\mathcal{F}_\rho, A_\rho) \asymp \mu(\rho). \end{gathered} \tag{10.36a}$$

No essential-distinctness assertion is needed here: the labelled form derived after (10.8J) from [GWZ26, Lemma 3.7] applies using the displayed density and $\Delta_{\max}$ bounds. The affine normalization preserves density and average multiplicity, and $2\eta_{\mathrm{src}} < \eta_{KT}^{\mathrm{cap}}$ verifies the source density threshold. After decreasing δ_0 so that $\omega_{\mathrm{cap}}(\delta) < \varepsilon_{\mathrm{fact}}/8$, Lemma 3.7 and Remark 3.6 give

$$\begin{aligned} \mu(\rho) &\lesssim \delta^{-\varepsilon_{\mathrm{fact}}/2}\Delta_{\max}(\mathcal{F}_\rho)^{1-\beta}N_\rho^{\beta} \\ &\le \delta^{-\varepsilon_{\mathrm{fact}}/2-(1-\beta)\eta_{\mathrm{src}}-\omega_{\mathrm{cap}}(\delta)}N_\rho^{\beta} \le \delta^{-\varepsilon_{\mathrm{fact}}}N_\rho^{\beta}, \end{aligned} \tag{10.37}$$

This proves the first part of (S2); the non-sticky hypothesis supplies its remaining count.

The later dense-color call uses

$$\begin{gathered} s = \rho_2, \qquad \mathcal{F} = \mathcal{F}_{S,j}, \qquad A = A_{S,j}, \qquad |\mathcal{F}| = N_j, \\ \mathcal{F}_{S,j} \text{ is one proper source-proxy conflict color}, \qquad \Delta_{\max}(\mathcal{F}_{S,j}) \lesssim D_S, \\ \lambda(\mathcal{F}_{S,j}, A_{S,j}) = \lambda_j \ge \rho_2^{\eta_{KT}}. \end{gathered} \tag{10.37a0}$$

The proxy containment comparison preserves the required convex-test bound, and the unchanged shading preserves multiplicity, so Lemma 3.7 gives (10.30b) and pulls it back to the raw normalized family. For every dense color, condition (K) and its fixed reserve verify the last line because

$$\lambda_j \ge \tfrac{1}{2}\widehat{\lambda}_S \gtrsim \lambda_S \ge \tfrac{1}{2}\lambda(Z^{\angle}) \ge \tfrac{1}{4}c_{\mathrm{ang}}\lambda_B^{\mathrm{par}} \gtrsim \mathcal{D}, \tag{10.37a}$$

the strict inequality in (K), after the final cutoff, makes the last comparison at least $\rho_2^{\eta_{KT}}$ despite its fixed structural constant. □

Lemma 10.18 (Quantitative loss envelope and property map)**.** *On the chain (10.32f), the operation-level losses and the properties they preserve are bounded by the following exchange table.*

operation	*deletion type*	*retained factor*	*property kept for later use*
angular uniformization	*fine incidences*	c_{init}	*cap counts fixed on* $\bar{Y}$
all-scale level	*fine incidences*	c_{sc}	*(87)–(88) fixed on* Y^{sc}
canonical window/net	*labelled assignment*	*no deletion*	*(10.7P)–(10.7R), one fixed ancestry*
canonical multiplicity/density/tag levels	*induced fine incidences and whole tags*	c_{can}	*(10.7T)–(10.7X), canonical shading density*
good-ball level	*induced fine incidences*	$(4\kappa_{\mathrm{inc}})^{-1}$	*(10.2), (10.4), representation coverage*
fine-multiplicity synchronization	*whole tubelet labels*	$(\kappa_{\mathrm{inc}}L_m)^{-1}$	*one* m_{fine}*, identity (10.7)*
labelled bias, dimension, count levels	*whole tubelet labels*	$c_{\mathrm{bias}}/(L_{a,b}L_N)$	*(8.0D)–(8.0G), common denominator*
REG-1, REG-2, local volume	*saturated tagged edges*	$(J_v L_{\mathrm{reg}})^{-1}$	*P2–P5 in each tag*
first whole-tag synchronization	*whole outer-ball tags*	P_{tag}^{-1}	*common* $\mu_{\mathrm{in}}, \omega_0, d_0$*, local P1 unchanged*
ANGLE-CELL plus one LIFT	*whole incidences inside each tag*	$c_{\mathrm{ang}}/2$	*local resolved angle, P3–P5*
angle whole-tag synchronization	*whole outer-ball tags*	P_{angtag}^{-1}	*common* $\theta, d_\angle$*, (10.25)*
conflict coloring	*partition, no deletion*	*estimate costs* H_{col}	*essential distinctness in each color*

The table records a conservative operation-to-factor envelope. The products in (10.16b) are upper bounds, and (10.16d) is the single uniform envelope used later. Equation (10.25) records the actual angular lift and tag retention, while the fixed physical recombination factor is κ_{amb}*. Hence (10.16d)–(10.16f) follow without an additional refinement, so (S4) holds.*

Proof. The rows follow respectively from lemmas 10.2 and 10.16, the saturated definition (10.11″), (10.8E), (8.5), (10.8G), and theorems 5.1 and 7.2. Coloring partitions labels; physical recombination costs κ_{amb} by lemma 3.5.

For the quantitative bounds, lemma 8.1 and equal canonical-child volume give, for convex $K \subset W$,

$$\frac{\#\mathbb{T}_{B,W}[K]}{\#\mathbb{T}_{B,W}} = \frac{\Delta(\mathbb{T}_{B,W}, K)|K|}{\Delta(\mathbb{T}_{B,W}, W)|W|} \lesssim \delta^{-o(1)} \left(\frac{|K|}{|W|}\right)^{1+\varepsilon_{\mathrm{fact}}}. \tag{10.37b}$$

The fixed Löwner rectangle and (8.0F)–(8.0G) give

$$C_0 \le \delta^{-o(1)}, \qquad \Delta_{\max}(\mathbb{W}_B) \lesssim (\alpha\rho_2)^{-\varepsilon_{\mathrm{fact}}}\delta^{-o(1)}. \tag{10.38}$$

Common child counts give $D \le D_B \le \kappa_D D$, $\kappa_D = O(1)$; the canonical net and ball cover bound $\kappa_{\mathrm{loc}}, \kappa_{\mathrm{inc}}$.

For the local-volume count, let Ω and $m_+ = |E^+|$ be the pre-REG-1 weight and edge count. The application-level condition (10.8A), verified from (10.8A0) before any J_v-dependent envelope,

gives, after enlarging the fixed mass exponent if necessary,

$$\Omega = M(Y_0^{\mathrm{joint}}) \geq \delta^{C_{\mathrm{mass}}}. \tag{10.38a}$$

Polynomial complexity bounds m_+, μ_{in}, and lemma 4.1 gives $\omega_0 \geq \Omega/(2m_+)$. Hence corollary 10.8 yields $J_v = O(\log(1/\delta))$; the other levels are polylogarithmic. On the non-slab window, (10.17) gives $c_{\mathrm{ang}}^{-1} = \alpha^{-o(1)}$; angular rows are absent in the slab case.

Complete-containment density is monotone under label deletion and affine invariant. Thus (10.38) and $\alpha, \rho_2 \geq \delta^{1-\varepsilon_{\mathrm{scal}}}$ give, in the applicable thin branches,

$$\begin{aligned} D_W, D_S &\leq \delta^{-\Gamma_D - o(1)}, \\ \Gamma_D &:= 2(1-\varepsilon_{\mathrm{scal}})\varepsilon_{\mathrm{fact}}. \end{aligned} \tag{10.39}$$

The greedy conflict coloring gives $H_{\mathrm{col}} \leq C D_S$. Substitution in the conservative products is bounded by (10.16d)–(10.16f), proving the lemma. □

Proposition 10.19 (Conditional verification of the GWZ Section 9 interface)**.** *Fix $0 < \beta \leq 1$, $\zeta > 0$, and a loss profile $\mathfrak{p}$, and assume $K_{KT}(\beta)$ and $K_F(\beta)$. Choose the parameters in the order of section 10.9. Then there exist $0 < 2\eta_{\mathrm{src}} < \eta$ and $\delta_0 = \delta_0(\beta, \zeta, \mathfrak{p}) > 0$ such that, for every $0 < \delta \leq \delta_0$, the following holds. Let $(\mathbb{T}, Y)$ be a uniform input satisfying the formal hypotheses displayed in* [GWZ26, Lemma 9.1]*:*

$$\Delta_{\max}(\mathbb{T}) \leq \delta^{-\eta_{\mathrm{src}}}, \qquad \lambda(\mathbb{T}, Y) \geq \delta^{\eta_{\mathrm{src}}}, \qquad |\mathbb{T}_\rho| \geq \rho^{-2-\zeta} \tag{10.33}$$

for the source middle range of ρ. All fixed constants and moduli in the input below are governed by $\mathfrak{p}$. In addition, assume the following explicit conditional interface:

- (I1) *$\mathbb{T}$ is a finite, source-essentially-distinct labelled family of unit-length δ-tubes in a fixed ball, $|\mathbb{T}| \leq \delta^{-C_{\mathrm{comp}}}$, with bounded neighbouring-scale ancestry;*
- (I2) *a reference refinement $\bar{Y} \subset Y$ exists with $M(\bar{Y}) \geq c_{\mathrm{init}} M(Y)$ and $c_{\mathrm{init}}^{-1} \leq \delta^{-\omega_0(\delta)}$ for one fixed $\omega_0(\delta) \to 0$; it satisfies the incident-centred cap and local average-multiplicity statements (10.35)–(10.35a) at every source scale;*
- (I3) *an all-discrete-scale refinement $Y^{\mathrm{sc}} \subset \bar{Y}$ exists, in the order (10.32f), with retained factor $c_{\mathrm{sc}}^{-1} \leq \delta^{-\omega_0(\delta)}$, and the scale- r shadings and comparisons are exactly the source (87)–(88) interface named in (R2); the common coarse branching numbers obey (10.36); and*
- (I4) *all carrier ancestry and representation maps are labelled and have bounded fibres. Specifically,*

  $$\max\{\kappa_{\mathrm{inc}}, \kappa_{\mathrm{loc}}, \kappa_{\mathrm{amb}}\} = O(1)$$

  with the roles displayed below, and subsequent operations are restricted to the canonical-window and saturated whole-incidence protocol in (R3)–(R4).

Under (I1)–(I4) there is one fixed function $\omega_{\mathrm{ref}}(\delta) \to 0$, uniform over all admissible tags and scales, such that the source refinements followed by the corrected cellular preprocessing equip the fixed reference refinement $(\mathbb{T}, \bar{Y})$ with all admissible data in the sense of definition 10.14, while retaining the quantified labelled fine mass

$$M(\bar{Y}) \geq \delta^{\omega_{\mathrm{ref}}(\delta)} M(Y). \tag{10.33R}$$

Thus "subpower refinement" means the signed lower bound (10.33R). More precisely, the reference angular shading obeys

$$M(\bar{Y}) \geq c_{\mathrm{init}} M(Y), \qquad c_{\mathrm{init}}^{-1} \leq \delta^{-\omega_0(\delta)}. \tag{10.33′}$$

Appendix D records the complete source correspondence and version convention. For the proof itself, the following matrix collects the proof-bearing analytic calls and the reference family on which each hypothesis is verified.

source	scale and actual family	checks on called family	output and pullback
GWZ (86)	ρ; $(\mathbb{T}, \bar{Y})$	reference shading fixed before all later deletions; incident-centred caps	(10.35), with neighbouring-scale stability
GWZ (87)–(88)	r; $(\mathbb{T}, Y^{\mathrm{sc}})$, $(\mathbb{T}_r, Y_r)$	all-scale level fixed before tubelets; retained factor c_{sc}	(10.8h), (10.23b)–(10.23d); scale ratios cancel
GWZ Lemma 6.4 (31)	$s = \delta/b, t = \delta/a$; $(\mathcal{P}, Y_{\mathcal{P}})$	(10.8f′): $\lambda \geq s^{\eta_F}$, $C_F \lesssim_{\log} 1$, the displayed M-bound; Definition 6.10 proxy family essentially distinct	(10.8g0); original labels retained and charged in M, then pullback gives (10.8g)
GWZ Lemma 3.7, cap	$s_\rho = \delta/\rho$; $(\mathcal{F}_\rho, A_\rho)$	(10.36a): reference density $\gtrsim \delta^{2\eta_{\mathrm{src}}}$, $\Delta_{\max} \leq \delta^{-\eta_{\mathrm{src}}}$; derived labelled form, no essential-distinctness hypothesis	(10.37); average multiplicity is affine invariant
GWZ Lemma 3.7, color	ρ_2; exact source-tube proxy color $(\mathcal{F}_{S,j}, A_{S,j})$	proper proxy color, $\lambda_j \geq \rho_2^{\eta_{KT}}$, $\Delta_{\max} \lesssim D_S$	(10.30b); two-sided proxy containment transfers convex tests and the unchanged shading pulls multiplicity back
GWZ Lemma 5.5	ρ_2; $Y^* \subset \bar{Y}$	compatibility (10.26), outer degree $d_\angle$, refinement fraction c_*	equation (10.27) on the fixed reference cap shading

Proof. Geometric input (S1). Follow lemma 10.16: freeze $\bar{Y}$ at (86), then form $Y^{\mathrm{sc}} \subset \bar{Y}$ at (87)–(88), with retained fraction c_{sc} and $c_{\mathrm{sc}}^{-1} = \delta^{-o(1)}$. Clause (C3) of lemma 10.3, the density level (10.7W), the tag level (10.7W′), and the physical projection (10.7X) then give

$$M(Y^{\mathrm{pre}}) \geq c_{\mathrm{sc}} c_{\mathrm{can}} M(\bar{Y}), \qquad c_{\mathrm{can}}^{-1} = 2\kappa_{\mathrm{inc}} L_{\mathrm{mult}} L_{\mathrm{dens}} L_{\mathrm{tag}}. \tag{10.33P}$$

Now lemma 10.1 and (10.0)–(10.1d) give (10.1) with $c_{\mathrm{pre}} = c_{\mathrm{sc}} c_{\mathrm{can}}/(4\kappa_{\mathrm{inc}})$. Thus $c_{\mathrm{pre}}^{-1} = \delta^{-o(1)}$. Moreover $\lambda(\mathbb{T}, Y^{\mathrm{sc}}) \geq c_{\mathrm{sc}} c_{\mathrm{init}} \delta^{\eta_{\mathrm{src}}}$, so (10.7V) and the pre-cellular cutoff $\omega_{\mathrm{pre}} < \eta_{\mathrm{src}}$ give (10.1a) with exponent $2\eta_{\mathrm{src}}$. The gap $\eta - 2\eta_{\mathrm{src}}$ then gives (10.1)–(10.2).

Testing $\Delta_{\max}(\mathbb{T})$ on a containing ball gives

$$|\mathbb{T}|\delta^2 \lesssim \delta^{-\eta_{\mathrm{src}}}, \tag{10.34}$$

so $|\mathbb{T}| \leq \delta^{-C}$. The bounded-overlap cover and canonical net give at most $C|\mathbb{T}|r_1^{-1}$ representations; biased parents partition them, and the tags contain $O(a^{-3})$ cells. Hence all later graphs have polynomial size.

For the windows in lemma 8.1, every maximizing hull $K_{\mathcal{A}}$ contains a canonical $\delta \times \delta \times r_1$ tubelet and lies in $C_{\mathrm{amb}}B$. If $\ell_1(K_{\mathcal{A}}) \leq \ell_2(K_{\mathcal{A}}) \leq \ell_3(K_{\mathcal{A}})$ are its Löwner semiaxes, then

$$c\delta \leq \ell_1 \leq \ell_2 \leq \ell_3 \leq Cr_1, \qquad cr_1 \leq \ell_3, \tag{10.34a}$$

and hence

$$c\delta^2 r_1 \leq |K_{\mathcal{A}}| \leq Cr_1^3. \tag{10.34b}$$

The contained tubelet, ambient ball, and Löwner–John comparison prove (10.34a)–(10.34b). The score numerator lies between $c\delta^2 r_1$ and $C|\mathcal{A}|\delta^2 r_1 \leq C\delta^{-C_{\mathrm{comp}}}\delta^2 r_1$. Since $r_1 = \delta^{\varepsilon_{\mathrm{scal}}}$, (10.34b) therefore gives a fixed C_{win} such that

$$\delta^{C_{\mathrm{win}}} \leq \Phi_{\mathcal{A}}(K_{\mathcal{A}}) \leq \delta^{-C_{\mathrm{win}}}, \qquad \delta^{C_{\mathrm{win}}} \leq \ell_j(K_{\mathcal{A}}) \leq \delta^{-C_{\mathrm{win}}}. \tag{10.34c}$$

Thus the score and three axis windows have $L_{\mathrm{bias}} \le C\log^4(2/\delta) = \delta^{-o(1)}$ classes.

By (R3), perform (10.3)–(10.7) before biased factoring, so $M_{\mathrm{lab}}(A) \sim m_{\mathrm{fine}} \sum |A(T_B)|$. Apply lemma 8.1 tagwise. Its (8.0C), the density class (10.7W), and (10.7) retain the represented fraction c_{bias}, including the single physical/labelled conversion:

$$c_{\mathrm{bias}}^{-1} \le C\kappa_{\mathrm{inc}} L_{\mathrm{bias}} = \delta^{-o(1)}, \qquad c_{\mathrm{setup}} = c_{\mathrm{pre}} c_{\mathrm{bias}}. \tag{10.33Q}$$

The separately charged (a,b) and child-count levels cost $L_{a,b}L_N$ and delete complete labels. The thick branch stops after dimensions; thin branches continue through REG and saturated local volume. Thus (10.4)–(10.7) remain on one object, and lemma 10.1 supplies (10.32e). In the thin slab branch, (10.6) and P_{sync} verify (10.7d) with $P = P_{\mathrm{sync}}$. Since $\lambda(\mathbb{T}, \bar{Y}) \ge c_{\mathrm{init}}\delta^{\eta_{\mathrm{src}}}$, lemma 10.6 gives

$$\sum_{B \text{ retained}} |\mathbb{T}_B^{\mathrm{lab}}|\,|T_B| \gtrsim c_{\mathrm{init}} P_{\mathrm{den}}^{-1} |\mathbb{T}|\,|T| \gtrsim \delta^{\Lambda_{\mathrm{den}}+\omega_0(\delta)} |\mathbb{T}|\,|T|. \tag{10.34d}$$

Here (10.7a) is used, and κ_{inc} is already included in P_{sync}; hence $P_{\mathrm{den}} = \delta^{-\eta_{\mathrm{src}}} P_{\mathrm{sync}}$. At $\rho_- = \delta^{1-\varepsilon_{\mathrm{scal}}}$, (10.33) and (I1) give

$$|\mathbb{T}| \gtrsim |\mathbb{T}_{\rho_-}| \ge \rho_-^{-2-\zeta}.$$

Apply lemma 10.18. Its proof now gives $J_v = O(\log(1/\delta))$ from the independently verified condition (10.8A), so the full envelope satisfies $\omega_{\mathrm{src}}(\delta) \to 0$. Make the deferred final decrease of δ_0 until ω_{src} is below η_{src} and every reserved gap named in the parameter closure. Together with (10.7a)–(10.7e), (10.12)–(10.16), and the bounds from that lemma, this is (S1b); its strict reserve is (10.32s).

If $a \ge \delta^{1-\tau}$, the assumed $K_F(\beta)$ and $K_{KT}(\beta)$, (10.32e), (10.8h), and lemma 10.18 verify lemma 10.9, hence (S1a), before cellularization. In the thin non-slab branch the source bounds $\delta \le a \le b \le \delta^{\varepsilon_{\mathrm{scal}}} r_1$, with $r_1 = \delta^{\varepsilon_{\mathrm{scal}}}$, imply

$$\delta^{1-\varepsilon_{\mathrm{scal}}} \le a/r_1 \le b/r_1 \le \delta^{\varepsilon_{\mathrm{scal}}}, \qquad a/b \ge a/r_1,$$

which is (10.16a), hence (S1c). Thus (S1) holds in all three cases; (S2)–(S3) and the non-slab part of (S4) are needed only in (S1c).

Reference cap input (S2). Take $\rho = \rho_2$ in (10.35)–(10.37) of lemma 10.17; (10.36) and (10.33) give (S2).

Dense-color input (S3). The color clause of lemma 10.17, the conflict graph, and (K) give respectively the estimate, essential distinctness, and density in (S3).

Loss envelope (S4). By lemma 10.18, (10.37b)–(10.39), (10.19b), and (10.25) verify (S4) on the stated reference objects. □

Admissible closure.

Theorem 10.20 (Very-non-sticky estimate for admissible families)**.** *Fix $0 < \beta \le 1$, $\zeta > 0$, and a loss profile $\mathfrak{p}$, choose the parameters as in section 10.9, and let $(\mathbb{T}, \bar{Y})$ be admissible in the sense of definition 10.14, with all its losses governed by $\mathfrak{p}$. There are $\nu_{\mathrm{adm}} = \nu_{\mathrm{adm}}(\beta,\zeta) > 0$ and $\delta_0 = \delta_0(\beta,\zeta,\mathfrak{p}) > 0$ such that, whenever $0 < \delta \le \delta_0$,*

$$\mu(\mathbb{T}, \bar{Y}) \le \delta^{\nu_{\mathrm{adm}}} |\mathbb{T}|^{\beta}. \tag{10.40}$$

Proof. Under alternative (S1a), lemma 10.9 and $M(\bar{Y}) \le |\mathbb{T}|\,|T|$ give

$$\mu(\mathbb{T}, \bar{Y}) \lesssim_{\log} \delta^{\nu_{\mathrm{thick}}} |\mathbb{T}|^{\beta}. \tag{10.8j}$$

Thus the thick case is complete. For the rest of the branch verification assume the complementary thin condition (10.8i).

The source-call matrix identifies (10.2), (10.12)–(10.14), (10.22)–(10.23), and (10.26)–(10.29) with source blocks (94)–(103). The bridge (10.4)–(10.7), used in (10.19a)–(10.19f), converts tubelet retention to physical fine mass without identifying the two shadings.

Under alternative (S1b), the slab estimate follows from section B and (10.15)–(10.16). More explicitly, (10.12), the common bounds $\delta \le a \le b \le r_1 = \delta^{\varepsilon_{\mathrm{scal}}}$, and hence $a/b \ge \delta^{1-\varepsilon_{\mathrm{scal}}}$, give

$$\lambda_B^{\mathrm{par}} \ge \delta^{\Lambda_{\mathrm{par}}+o(1)}, \qquad \Lambda_{\mathrm{par}} := q(\eta+\Lambda_{\mathrm{pre}}) + (1-\varepsilon_{\mathrm{scal}})\varepsilon.$$

In the slab range, $(r_1/b)^2 \le \delta^{-2\varepsilon_{\mathrm{scal}}}$, so (10.15), (10.14), and (10.16) give

$$|U(\mathbb{T}_B, Y_B)| \gtrsim \delta^{\Lambda_{\mathrm{slab}}+o(1)} |\mathbb{T}_B^{\mathrm{lab}}|\,|T_B|, \quad \Lambda_{\mathrm{slab}} = 2\Lambda_{\mathrm{par}} + \Lambda_D + 4\varepsilon_{\mathrm{scal}} + 2\eta + 3\tau. \tag{10.33a}$$

Here Λ_D controls the pre-angle parent constant D_W, as well as its later descendants. The r_1-ball cover has bounded overlap, and $U(\mathbb{T}_B, Y_B) \subset U(\mathbb{T}, Y^0)$. Summing (10.33a) and using (10.34d) therefore gives

$$|U(\mathbb{T}, \bar{Y})| \ge |U(\mathbb{T}, Y^0)| \gtrsim \delta^{\Lambda_{\mathrm{slab}}+\Lambda_{\mathrm{den}}+o(1)} |\mathbb{T}|\,|T|. \tag{10.33b}$$

At $\rho_- = \delta^{1-\varepsilon_{\mathrm{scal}}}$, the S1b count and ancestry data in definition 10.14 give

$$|\mathbb{T}| \gtrsim |\mathbb{T}_{\rho_-}| \ge \delta^{-(1-\varepsilon_{\mathrm{scal}})(2+\zeta)}.$$

Consequently (10.33b) implies

$$\mu(\mathbb{T}, \bar{Y}) \lesssim \delta^{\nu_{\mathrm{slab}}-o(1)} |\mathbb{T}|^{\beta}, \qquad \nu_{\mathrm{slab}} := (1-\varepsilon_{\mathrm{scal}})\beta(2+\zeta) - \Lambda_{\mathrm{slab}} - \Lambda_{\mathrm{den}} > 0. \tag{10.33c}$$

The last strict inequality follows from (10.32p) and the later small choices in section 10.9. Thus the local slab branch has been summed and converted to the source multiplicity target. Under the remaining alternative (S1c), the threshold (10.19) divides all inputs. Below it, (10.19g) and cap uniformity give (10.19h), and condition (L) makes its exponent positive. Above it, (10.18) gives (10.20), while (10.21a) and (H) prove the parent-density hypothesis (10.21). Condition (K) separately verifies the density of every color to which the Katz–Tao estimate is applied.

In the transverse alternative lemma 10.13 gives (10.22)–(10.23), including the filling-to-multiplicity implication. In the tangential alternative, (10.29a)–(10.29c) retain dense slab mass, the explicitly defined conflict graph gives (10.30), and (10.30a)–(10.31a) prove the full scaled degree bound. The single fine lift (10.27), cap estimate (10.28), and uniformity (10.31b) then give (10.31c)–(10.32). The choices (10.32a)–(10.32c) verify (L), (H), (K), and (T) simultaneously. The loss envelope (10.16b), (10.16d)–(10.16f), and (10.25) applies throughout this chain. By the fixed zero-profile reserves in section 10.9 and the final cutoff of ω_{src}, fix constants $\underline{\nu}_{\mathrm{low}}, \underline{\nu}_{\mathrm{slab}} > 0$, independent of δ and of the loss profile, such that uniformly for $0 < \delta \le \delta_0(\beta, \zeta, \mathfrak{p})$,

$$\nu_{\mathrm{low}}(\delta) \ge \underline{\nu}_{\mathrm{low}}, \qquad \nu_{\mathrm{slab}}(\delta) \ge \underline{\nu}_{\mathrm{slab}}. \tag{10.39a}$$

Let

$$\nu_{\mathrm{br}} := \min\{\nu_{\mathrm{thick}}, \underline{\nu}_{\mathrm{slab}}, \underline{\nu}_{\mathrm{low}}, \nu_{\mathrm{tr}}/2, \underline{\nu}_{\mathrm{tan}}\} > 0.$$

The pre-cellular thick branch, the thin slab branch, and the three thin non-slab alternatives just proved are mutually exhaustive and give $\mu(\mathbb{T}, \bar{Y}) \le \delta^{\nu_{\mathrm{br}}-o(1)} |\mathbb{T}|^{\beta}$. Because $\mathfrak{p}$ was fixed before the cutoff, decrease $\delta_0(\beta, \zeta, \mathfrak{p})$ so that the remaining $o(1)$-loss is at most $\delta^{-\nu_{\mathrm{br}}/2}$. This proves (10.40) with $\nu_{\mathrm{adm}} = \nu_{\mathrm{br}}/2$. □

Corollary 10.21 (Conditional GWZ Section 9 replacement)**.** *Fix $0 < \beta \le 1$, $\zeta > 0$, and a loss profile $\mathfrak{p}$, and assume $K_{KT}(\beta)$ and $K_F(\beta)$. There exist $\eta_{\mathrm{src}} = \eta_{\mathrm{src}}(\beta, \zeta) > 0$, $\delta_0 = \delta_0(\beta, \zeta, \mathfrak{p}) > 0$, and $\nu = \nu(\beta, \zeta) > 0$ such that, for every $0 < \delta \le \delta_0$, each uniform labelled input $(\mathbb{T}, Y)$ satisfying (10.33) and the conditional interface (I1)–(I4) of proposition 10.19, with all fixed constants and moduli governed by $\mathfrak{p}$, obeys*

$$\mu(\mathbb{T}, Y) \le \delta^{\nu} |\mathbb{T}|^{\beta}. \tag{10.41}$$

Proof. Proposition 10.19 supplies an admissible reference shading $\bar{Y}$ with $M(\bar{Y}) \geq c_{\mathrm{init}} M(Y)$ and $c_{\mathrm{init}}^{-1} \leq \delta^{-\omega_0(\delta)}$. Apply theorem 10.20. Monotonicity of the union and the definition of average multiplicity give

$$\mu(\mathbb{T}, Y) = \frac{M(Y)}{|U(\mathbb{T}, Y)|} \leq c_{\mathrm{init}}^{-1} \frac{M(\bar{Y})}{|U(\mathbb{T}, \bar{Y})|} = c_{\mathrm{init}}^{-1} \mu(\mathbb{T}, \bar{Y}).$$

Since ω_0 is controlled by the fixed profile $\mathfrak{p}$, decrease $\delta_0(\beta, \zeta, \mathfrak{p})$ once more so that $\omega_0(\delta)$ is below $\nu_{\mathrm{adm}}/2$, and hence c_{init}^{-1} costs at most $\delta^{-\nu_{\mathrm{adm}}/2}$. Then (10.41) holds with $\nu = \nu_{\mathrm{adm}}/2$. □

11. Sharpness of the rectangular density exponent

The rectangular theorem converts the planar Furstenberg incidence exponent of Dąbrowski–Orponen–Villa [DOV22] into the shading power $q = 2 - \sigma$. The construction below shows that this power is optimal when one asks for a bound uniform over every allowed aspect ratio. This is a uniform statement: a fixed parent shape may admit a stronger dependence on the shading density.

Proposition 11.1 (Uniform rectangular sharpness)**.** *For every fixed $0 < \sigma \leq 1$, the exponent $2 - \sigma$ cannot be replaced, uniformly over $\delta \leq a \leq b \leq r$, by a smaller exponent of the shading density. Precisely, if $q' < 2 - \sigma$ and $0 < \varepsilon < 2 - \sigma - q'$, there do not exist constants $c > 0$ and $\delta_0 > 0$ such that every family satisfying (6.3) with a uniformly bounded constant C_0, at every $0 < \delta < \delta_0$, obeys*

$$\frac{|B \cap N_a(U(\mathbb{T}_B, Y))|}{|B|} \geq c(a/b)^{\varepsilon} \lambda_B^{q'}. \tag{11.0}$$

The constants in this negation may depend on the fixed σ, q', ε, but not on δ, a, b, r.

Lemma 11.2 (Two-line convex-hull volume)**.** *For $p = (p_0, p_1) \in [0,1]^2$, let*

$$\ell_p = \{(x, (1-x)p_0 + xp_1) : 0 \leq x \leq 1\}.$$

If $\ell_p(c\delta)$ denotes its $c\delta$-neighbourhood in a fixed bounded rectangle, then

$$\left|\mathrm{conv}\left(\ell_p(c\delta) \cup \ell_{p'}(c\delta)\right)\right| \gtrsim \delta + |p - p'|. \tag{11.1a}$$

Proof. At the vertical slice x, the convex hull contains the interval between the two affine graph values, enlarged by $c\delta$. Hence its area is at least a fixed multiple of

$$\delta + \int_0^1 |(1-x)(p_0 - p_0') + x(p_1 - p_1')| \, dx.$$

For an affine function f on $[0,1]$, $\int_0^1 |f| \geq c(|f(0)| + |f(1)|)$; this follows by splitting at its possible zero. The last display is therefore bounded below by $c(\delta + |p - p'|)$. □

Proof of proposition 11.1. Planar extremal data. Put $t = 1 + \sigma$ and fix $u \in [0,1]$. Proposition 5.2 of [DOV22] gives a δ-separated (δ, t, C)-set $P \subset [0,1]^2$, with $|P| \asymp \delta^{-t}$, and a $c\delta$-separated bounded-chart line family $\mathcal{L}_{u,t}$ such that every $p \in P$ is δ-incident to at least δ^{-u} lines and

$$|\mathcal{L}_{u,t}| \lesssim \delta^{-2u-(1-u)(t-1)}. \tag{11.1b}$$

Apply bounded point–line duality. A source line becomes a δ-separated point in a fixed parameter rectangle, while a source point $p = (p_0, p_1)$ becomes, up to one fixed affine change of coordinates, the graph segment ℓ_p in lemma 11.2. Incidence means that the dual point lies in a fixed $C\delta$-neighbourhood of that graph.

For each p, choose exactly $\asymp \delta^{-u}$ incident source lines and let Z_p be the union of fixed $C\delta$-squares about their dual points. The separation in (11.1b) gives

$$|Z_p| \asymp \delta^{2-u}, \qquad \frac{|Z_p|}{|\ell_p(C\delta)|} \asymp \delta^{1-u} =: \lambda. \tag{11.1c}$$

Although the same dual square may occur in several labelled shadings, their physical union is contained in the $C\delta$-neighbourhood of the dual point set of $\mathcal{L}_{u,t}$. Thus

$$\left|\bigcup_{p\in P} Z_p\right| \lesssim \delta^2 |\mathcal{L}_{u,t}| \lesssim \delta^{(1-u)(2-\sigma)} = \lambda^{2-\sigma}. \tag{11.1d}$$

Three-dimensional realization and the all-convex condition. Let

$$B = [-C\delta, 1 + C\delta] \times [-C, C] \times [-C\delta, C\delta],$$

permuting its axes to the convention $a = \delta$, $b \asymp r \asymp 1$. For $p \in P$, let T_p be a fixed solid $\delta \times \delta \times 1$ tube about $\ell_p \times \{0\}$, and set

$$Y(T_p) = Z_p \times [-c\delta, c\delta] \subset T_p$$

after changing the fixed tube constants if necessary. These are labelled children even when two geometric pieces overlap, and the enlarged endpoint interval in B makes every capped solid tube T_p a complete child. Set

$$E = U(\{T_p : p \in P\}, Y) = \bigcup_{p\in P} Y(T_p).$$

The δ-neighbourhood of E only replaces each dual δ-square and the transverse interval by fixed dilates. To make this step literal, let q_ℓ be the dual parameter point of $\ell \in \mathcal{L}_{u,t}$. The points q_ℓ are $c\delta$-separated in a fixed parameter rectangle, so the fixed dilates $C'Q(q_\ell, \delta)$ have overlap bounded by a constant depending only on C'. Every square occurring after the neighbourhood operation is contained in one such $C'Q(q_\ell, \delta)$; no new parameter point is created. Hence

$$\left|\bigcup_{\ell\in\mathcal{L}_{u,t}} C'Q(q_\ell, \delta)\right| \le C\delta^2 |\mathcal{L}_{u,t}|, \tag{11.1d$'$}$$

and (11.1d) still controls its volume up to an absolute constant. Equations (11.1c)–(11.1d) give $\lambda_B \asymp \lambda$ and

$$\frac{|B \cap N_a(E)|}{|B|} \lesssim \lambda^{2-\sigma}. \tag{11.1}$$

It remains to verify the hypothesis for every convex $K \subset B$. Put $P_K = \{p : T_p \subset K\}$. If $P_K \neq \varnothing$, convexity and the common transverse δ-interval imply

$$\frac{|K|}{|B|} \gtrsim \delta + \operatorname{diam}(P_K) \tag{11.1e}$$

by applying lemma 11.2 to a diametral pair (and by the volume of one tube when P_K is a singleton). Since P is a (δ, t, C)-set, P_K lies in a ball of radius $\operatorname{diam}(P_K) + \delta$, and hence

$$\frac{\#P_K}{\#P} \lesssim (\delta + \operatorname{diam}(P_K))^t \lesssim \left(\frac{|K|}{|B|}\right)^{1+\sigma}. \tag{11.1f}$$

This is exactly the all-convex complete-child condition (6.3), uniformly in K, rather than only a parameter-ball estimate.

Finally let $q' < 2 - \sigma$, choose $0 < \varepsilon < 2 - \sigma - q'$, take $u = 0$, and let $\delta \downarrow 0$. Then $\lambda = \delta$, so a uniform lower bound $\delta^\varepsilon \lambda^{q'}$ is asymptotically larger than the upper bound $\lambda^{2-\sigma}$ in (11.1), a contradiction. Varying u gives the whole resolution range $\delta \le \lambda \le 1$, up to harmless integer rounding. The proof includes $\sigma = 1$; constants are allowed to depend on the fixed σ. □

Uniform sharpness does not imply aspectwise sharpness. If $a = b \ll r$, one shaded tube already gives

$$|B \cap N_a(Y(T))| \gtrsim \lambda |B|. \tag{11.2}$$

Indeed, the longitudinal projection of $Y(T)$ has measure $\gtrsim \lambda r$, and each selected slice fills a fixed portion of the $a \times a$ parent cross-section after a-thickening. Conversely, consider the model in which a sufficiently dense transverse family fills the parent cross-section over one common longitudinal interval I of length $|I| = \lambda r$, and take its shading to be exactly that filled portion. Fubini's theorem, with the two endpoint layers created by a-thickening, gives

$$\frac{|B \cap N_a(E)|}{|B|} \asymp \lambda + a/r. \tag{11.3}$$

This is only an aspect-ratio model calculation, not another family satisfying all hypotheses of proposition 11.1. It explains why density power one is the natural target above the a/r resolution plateau in the square cross-section regime. At $a \asymp r$, a single a-ball already occupies a parent-scale volume.

Several natural degenerations do not improve (11.1). A fixed union of translated bushes changes only constants. Sparse multi-bushes produce a planar union of size $\lambda/N + \lambda^2$, never smaller than the quadratic obstruction in their valid range. Parallel stacks have relative union size $\lambda + \delta$. Exact projection fibres are charged either by the all-convex constant or by the atom weight in the weighted planar argument. Sharp wedges expose a boundary-localization problem, but the elementary hidden-ray constructions violate the absolute all-convex hypothesis before they violate the desired conclusion.

Remark 11.3. The DOV construction covers $a = \delta, b = r$, the endpoints $\lambda = \delta, 1$, and $\sigma = 1$. It does not prove aspectwise optimality for $b = r, a \gg \delta$, nor does it describe $\lambda < \delta$, where the δ-neighbourhood has a resolution plateau. Constants are not asserted uniform as $\sigma \downarrow 0$.

12. Conclusion: stability and the sharp-convex boundary

The method rests on a stable unit of refinement: a positive incidence between a labelled permanent parent and a half-open cell at the shortest parent scale. Selecting these incidences as whole units lets one edge set control child mass, parent mass, both multiplicities, angular selection, and the final lift. An arbitrary refinement inside a cell could instead retain parent mass where no child mass remains. The fixed sequence of whole-incidence lifts has an explicit product of retention fractions, and labelled counting retains the distinct child partitions and multiplicity contributions of coincident parents.

This mechanism yields the unconditional cellular theorem, the two Section 6 applications, and the admissible very-non-sticky closure. The weighted planar input is the layer-cake consequence of [DOV22], the quadratic endpoint is the Córdoba mechanism [Cór77], and the terminal gain is the one from [GWZ26]. Assuming both $K_{KT}(\beta)$ and $K_F(\beta)$, and under the explicit interface (I1)–(I4), proposition 10.19 and corollary 10.21 construct the admissible data and supply the corresponding conditional input to the reduction toward Main Lemma 2.

The principal geometric boundary is the sharp-convex-parent analogue of the rectangular theorem. A convex parent W with a sharp boundary can be placed inside the comparable-volume source Löwner box B_W, and complete-child counts transfer monotonically to the box. The induced shading does not. For a triangular prism, a box neighbourhood may extend substantially beyond the slanted boundary while the numerator is clipped to the prism. The complete-child hypothesis counts tubes contained in a convex test body, while a quantitative comparison of the two neighbourhood measures would require additional control of the induced shading. This boundary interface is where the Löwner-box reduction stops.

The elementary wedge models considered here do not disprove the sharp-convex statement: concentrating a fraction of the tube family in a subwedge worsens the hypothesis constant by a reciprocal power, stronger than the direct boundary loss. The remaining question is whether one can replace the permanent rectangle by an equally stable unit inside W, perhaps through boundary-sensitive cellularization, a weighted coarea argument that retains the transverse factor, or a hypothesis controlling induced shadings rather than complete containment alone.

Appendix A. Weighted planar incidence estimates

This appendix supplies the analytic input isolated in section 6. Starting from the unweighted superlevel estimate, two layer-cake reductions produce the weighted statement for repeated atoms and arbitrary measurable shadings. Let $1 < t \le 2$, $q = 3 - t$, $0 < h \le 1$, and $L \ge 1$, and let

$$\nu = \sum_i \alpha_i \delta_{p_i}, \qquad \sum_i \alpha_i = 1, \qquad \nu(B(p,\rho)) \le L\rho^t \quad (h \le \rho \le 1). \tag{A.1}$$

The points lie in a bounded planar parameter chart and may repeat. Let Y_i be arbitrary measurable shadings of constant-width h-tubes and put

$$\Lambda = h^{-1} \sum_i \alpha_i |Y_i|. \tag{A.2}$$

Lemma A.1 (Resolution cutoff for weighted superlevels)**.** *Let $0 < h \le 1/2$, let $F : (0,1] \to [0,\infty)$, and suppose*

$$F(u) \le \min\{A/u, B\}, \qquad B/A \le Ch^{-C_0}, \tag{A.2a}$$

with the convention that the second condition is void when $A = 0$. Then

$$\int_0^1 F(u)\,du \le C'(C, C_0) A \log(2/h). \tag{A.2b}$$

In particular the bound is independent of any smallest positive superlevel.

Proof. If $A = 0$ there is nothing to prove. Put $u_0 = \min\{1, A/B\}$. If $B \le A$, the integral is at most $B \le A$. Otherwise split at $u_0 = A/B$:

$$\int_0^1 F(u)\,du \le Bu_0 + A\int_{u_0}^1 \frac{du}{u} = A(1 + \log(B/A)).$$

Condition (A.2a) bounds the last logarithm by $O_{C,C_0}(\log(2/h))$. No atom weight enters the split. □

Lemma A.2 (Quantitative physical–parameter duality)**.** *Fix bounded intervals I_x, I_s, I_t. For $p = (s,t) \in I_s \times I_t$, let*

$$\Gamma_p = \{(x, sx + t) : x \in I_x\}, \qquad T_p(Ah) = N_{Ah}(\Gamma_p),$$

with endpoint truncation to a fixed enlargement of I_x. For a physical point $q = (x_0, y_0)$, define the dual parameter line

$$\ell_q := \{(s,t) : t = y_0 - sx_0\}. \tag{A.2c}$$

For every fixed $A \ge 1$ there are constants $A_1(A), A_2(A)$, depending only on A, the three fixed bounded intervals, and the fixed tube-enlargement convention, such that the following hold for $0 < h \le 1$.

(D1) *If an Ah-tube about Γ_p meets the half-open physical h-square with centre q, then*

$$\operatorname{dist}(p, \ell_q) \le A_1(A)h. \tag{A.2c′}$$

Conversely, if this distance is at most Ah, then the $A_2(A)h$-tube about Γ_p meets a fixed enlargement of that square.

(D2) *The assignment $q = (x_0, y_0) \mapsto \ell_q$ identifies the usual slope–intercept coordinates of ℓ_q with $(-x_0, y_0)$. Consequently centres of distinct squares in an h-grid give an h-separated family of dual lines (and a bounded union of translated grids gives a separated family after only a bounded colouring).*

(D3) *If C is a half-open parameter h-square with centre p_C, then every $T_p(Ah)$, $p \in C$, is contained in $T_{p_C}(A_2(A)h)$. Thus parameter-cell aggregation introduces only the fixed incidence tolerance in (D1), not a scale-dependent constant.*

All statements include physical endcaps and half-open cell boundaries.

Proof. Write $p = (s, t)$. The Euclidean distance from p to ℓ_q is

$$\frac{|t - y_0 + sx_0|}{\sqrt{1 + x_0^2}}. \tag{A.2d}$$

If (x, y) lies both in the physical square about q and within Ah of Γ_p, then $|x - x_0| + |y - y_0| \le Ch$ and $|y - sx - t| \le C_A h$. Boundedness of s and (A.2d) prove the first implication. Conversely, (A.2d) bounds $|sx_0 + t - y_0|$ by $C_A h$; the graph point with first coordinate x_0, or the nearest endpoint when x_0 lies in an endpoint square, belongs to a fixed enlargement and proves the reverse implication.

A line $t = as + b$ has slope–intercept coordinates (a, b); hence ℓ_q has coordinates $(-x_0, y_0)$. This signed coordinate swap is an isometry, proving (D2). Finally, for $p \in C$ and $x \in I_x$,

$$|(s - s_C)x + (t - t_C)| \le C(I_x)h,$$

which proves (D3). Closed endcaps only add the same fixed error, while the half-open convention provides a unique cell on every grid boundary. □

Theorem A.3 (Weighted arbitrary-shading union bound)**.** *For every $\varepsilon > 0$,*

$$\left|\bigcup_i Y_i\right| \gtrsim_{\varepsilon,t} L^{-q} h^{\varepsilon} \Lambda^q. \tag{A.3}$$

The constant is independent of the number of atoms, the least positive weight, repeated parameters, and the individual shading densities.

Proof. Let $E = \bigcup_i Y_i$. We first dispose of the compact-scale range $1/2 < h \le 1$. Since $\sum_i \alpha_i = 1$,

$$|E| \ge \sum_i \alpha_i |Y_i| = h\Lambda \ge \tfrac{1}{2}\Lambda. \tag{A.3a}$$

All graph tubes lie in a fixed bounded physical chart, so $|Y_i| \le C_{\mathrm{ch}} h$ and $\Lambda \le C_{\mathrm{ch}}$. Because $1 \le q < 2$, $L \ge 1$, and $h^{\varepsilon} \le 1$, (A.3a) implies (A.3), with a constant depending only on the fixed chart and ε, t. Hence assume from now on that $0 < h \le 1/2$; lemma A.1 is then applicable.

Partition parameter space into half-open h-squares C, write $m_C = \sum_{p_i \in C} \alpha_i$, and place one atom of mass m_C at the cell center. The aggregated measure remains t-Frostman with constant $O(L)$. By lemma A.2(D3), one fixed enlargement of the graph tube at the cell centre contains every tube with parameter in that cell. Take the cell super-shading $Z_C = \bigcup_{p_i \in C} Y_i$. Then

$$\sum_C m_C |Z_C| \ge h\Lambda. \tag{A.4}$$

We first derive the weighted incidence estimate strictly from [DOV22, Theorem 1.12] (restated and proved as Theorem 4.1 there). We use the specialization $d = 2, n = 1$. If $\mu = \sum_j m_j \delta_{p_j}$ is h-separated and $\mathcal{L}$ is an h-separated family of dual lines, put

$$I_{Ah}(\mu, \mathcal{L}) = \sum_j m_j \#\{\ell \in \mathcal{L} : \operatorname{dist}(p_j, \ell) \le Ah\}.$$

For $u > 0$, the superlevel set $P_u = \{p_j : m_j \geq u\}$ is an $(h, t, L/(u|P_u|))$-set. That theorem with $d = 2, n = 1$ gives

$$I_{Ah}(P_u, \mathcal{L}) \lesssim_{A,\eta,t} \frac{L}{u} h^{-\eta} |\mathcal{L}|^{1/q} h^{(t-1)/q}. \tag{A.5}$$

The exact layer-cake identity is

$$I_{Ah}(\mu, \mathcal{L}) = \int_0^1 I_{Ah}(P_u, \mathcal{L})\, du. \tag{A.5a}$$

The singularity at $u = 0$ is cut off by the trivial separated-cardinality bound $I_{Ah}(P_u, \mathcal{L}) \lesssim h^{-2}|\mathcal{L}|$. Apply lemma A.1 with

$$A_0 = CLh^{-\eta}|\mathcal{L}|^{1/q} h^{(t-1)/q}, \qquad B_0 = Ch^{-2}|\mathcal{L}|.$$

Since $|\mathcal{L}| \lesssim h^{-2}$, L may be replaced by $\max\{1, L\}$, and $1 \leq q < 2$, the ratio B_0/A_0 is bounded by a fixed power of h^{-1}. Therefore

$$I_{Ah}(\mu, \mathcal{L}) \lesssim_{A,\eta,t} Lh^{-\eta} \log(2/h) |\mathcal{L}|^{1/q} h^{(t-1)/q}. \tag{A.6}$$

The logarithm depends on the resolution, not on the smallest atom weight; this is exactly the conclusion of lemma A.1.

Let $E = \bigcup_C Z_C$, partition physical space into h-squares Q, and set

$$w_Q = |E \cap Q|/h^2, \qquad W = \sum_Q w_Q = h^{-2}|E|,$$

$$\iota_Q = \sum_{C : \widetilde{T}_C \cap Q \neq \varnothing} m_C.$$

Equation (A.4) implies $\sum_Q w_Q \iota_Q \geq \Lambda/h$. For $\mathcal{Q}_u = \{Q : w_Q \geq u\}$, lemma A.2(D1)–(D2) sends the cell centres to an h-separated family of dual lines and converts every condition $\widetilde{T}_C \cap Q \neq \varnothing$ into an incidence with one fixed Ah tolerance. Therefore (A.6) gives

$$\sum_{Q \in \mathcal{Q}_u} \iota_Q \lesssim Lh^{-\eta} \log(2/h) |\mathcal{Q}_u|^{1/q} h^{(t-1)/q}.$$

Layer cake in w_Q, followed by concavity of $x^{1/q}$, yields

$$\frac{\Lambda}{h} \lesssim Lh^{-\eta} \log(2/h) h^{(t-1)/q} W^{1/q}.$$

Substituting $W = h^{-2}|E|$ cancels the geometric powers of h and gives

$$|E| \gtrsim L^{-q} [\log(2/h)]^{-q} h^{q\eta} \Lambda^q.$$

Taking $\eta = \varepsilon/(2q)$ and absorbing the logarithm proves (A.3). When $t = 2$, $q = 1$, the concavity step is equality; thus the endpoint is included. □

The ordinary one-dimensional marginal required at $\sigma = 0$ has a different proof. We record the exact labelled form, including atomic repetitions and nonuniform measurable shadings.

Lemma A.4 (Capped graph-tube intersection)**.** *Let $T_p, T_{p'}$ be two constant-width h-tubes whose carriers are graphs over the same fixed bounded interval and whose slope–intercept parameters lie in one fixed bounded chart. Then*

$$|T_p \cap T_{p'}| \lesssim \frac{h^2}{h + |p - p'|}. \tag{A.9}$$

The implicit constant is uniform over the chart, and the estimate includes coincident parameters.

Proof. Write γ for the slope separation and d for the intercept separation. If $\gamma \geq h$ and the two carrier graphs cross in a fixed enlargement of the graph interval, slicing parallel to the first carrier gives intersection area $O(h^2/\gamma)$. If they do not cross there, the intercept gap gives the stronger bound $O(h^2/d)$, with empty intersection once the gap exceeds the fixed enlargement. In this crossing regime the parameter distance is comparable, up to chart constants, to the larger of γ and the corresponding crossing-scale separation. If $\gamma < h$, either the tubes are disjoint or their intersection has area $O(h)$; moreover a nonempty intersection forces $d = O(h)$, hence $|p - p'| = O(h)$. Thus every case is bounded by the right-hand side of (A.9). When $p = p'$, it reduces to the trivial $O(h)$ tube-area bound. □

Theorem A.5 (Weighted labelled Córdoba endpoint)**.** *Let*

$$\nu = \sum_i \alpha_i \delta_{p_i}, \qquad \sum_i \alpha_i = 1, \qquad \nu(D(p,\rho)) \leq L\rho \quad (h \leq \rho \leq 1), \tag{A.7}$$

where $L \geq 1$, the parameters lie in one fixed bounded slope–intercept chart, and repetitions are allowed. If Z_i is an arbitrary measurable subshading of the corresponding constant-width h-tube and

$$\Lambda = h^{-1} \sum_i \alpha_i |Z_i|,$$

then

$$\left| \bigcup_i Z_i \right| \gtrsim L^{-1} [\log(2/h)]^{-1} \Lambda^2. \tag{A.8}$$

The constant depends only on the fixed chart and tube-enlargement constants, not on the number of labels or the least positive weight.

Proof. Put $F = \sum_i \alpha_i \mathbf{1}_{Z_i}$ and $E = \bigcup_i Z_i$. By the preceding estimate lemma A.4, the containing tubes satisfy (A.9).

For each fixed p_i, decompose parameter space into $D(p_i, 2h)$ and the annuli $2^k h < |p - p_i| \leq 2^{k+1} h$. Condition (A.7) gives

$$\sum_j \frac{\alpha_j}{h + |p_i - p_j|} \lesssim L \log(2/h). \tag{A.10}$$

This also controls the full mass of repeated atoms at p_i, since that mass is at most Lh. Equations (A.9)–(A.10) imply

$$\int F^2 \leq \sum_{i,j} \alpha_i \alpha_j |T_{p_i} \cap T_{p_j}| \lesssim L h^2 \log(2/h).$$

On the other hand $\int F = h\Lambda$. Cauchy–Schwarz on E now gives $(h\Lambda)^2 \leq |E| \int F^2$, which is (A.8). Since only tube containment was used to replace $Z_i \cap Z_j$ by $T_i \cap T_j$, the argument applies to arbitrary measurable shadings. □

An arbitrary representative from a parameter cell is not valid: a heavy atom may have empty shading while a light atom carries all the mass. The super-shading above is exact. Alternatively, choosing a maximum-density representative preserves the weighted cell mass up to a factor two. No bounded projection fibre or positive lower atom weight is needed.

Appendix B. A labelled slab L^2 estimate

The broad-middle-axis branch needs an L^2 estimate that counts repeated geometric slabs by label and allows measurable shadings. We prove that form directly here. Let $\mathbb{S}$ be a finite nonempty labelled family of $a \times b \times c$ convex slabs in a ball $B_R \subset \mathbb{R}^3$, where $0 < a \leq b \leq c \leq R$. Repeated geometric slabs with different labels are allowed. Here "dimensions $a \times b \times c$" means that each label comes with an ordered orthonormal frame and a centred rectangle R_S of those

side lengths such that $c_J R_S \subset S \subset R_S$ after a common fixed dilation convention, for one dimensional constant $c_J > 0$. In particular $|S| \asymp abc$, and all intersection and containment comparisons below have constants depending only on c_J. Put

$$D := \Delta_{\max}(\mathbb{S}), \qquad L = 1 + \lceil \log_2(R/a) \rceil . \tag{B.1}$$

For measurable $Y(S) \subset S$, write

$$A = \sum_S |Y(S)|, \qquad U = \bigcup_S Y(S), \qquad \lambda = A / \sum_S |S|.$$

When $A > 0$, also put $\mu = A/|U|$; then $|U| > 0$. For every label choose the middle plane H_S of its declared rectangle, normal to the shortest axis, and choose a unit normal n_S. The sign of the normal is immaterial; angles below are projective and lie in $[0, \pi/2]$.

Lemma B.1 (Two thickened affine planes in a ball)**.** *Let $H_i = \{x : n_i \cdot x = t_i\}$, $i = 1, 2$, be affine planes with projective normal angle $\phi \in [0, \pi/2]$. If $E_i \subset B_R \cap N_{Ca}(H_i)$, where $0 < a \le R$, then*

$$|E_1 \cap E_2| \le C' \frac{a^2 R}{\max\{a/R, \phi\}}. \tag{B.1a}$$

The estimate is uniform in the offsets t_i, and remains valid for capped or one-sided subsets of the two thickened planes.

Proof. If $\phi \le a/R$, then Fubini in the n_1-direction gives the trivial bound $|E_1 \cap E_2| \le |B_R \cap N_{Ca}(H_1)| \le C' a R^2$, which is (B.1a). Suppose $\phi > a/R$, and put $\tau = (n_1 \times n_2)/|n_1 \times n_2|$. The affine coordinate map

$$x \longmapsto (n_1 \cdot x - t_1, n_2 \cdot x - t_2, \tau \cdot x) \tag{B.1b}$$

has Jacobian $|\det(n_1, n_2, \tau)| = \sin\phi$. On the intersection, the first two coordinates range over intervals of length $O(a)$, while the third ranges over an interval of length $O(R)$ because $x \in B_R$. Consequently

$$|E_1 \cap E_2| \le C \frac{a^2 R}{\sin\phi} \le C' \frac{a^2 R}{\phi}.$$

Offsets merely translate the first two coordinate intervals. Endcaps and one-sided truncations only decrease the set being measured. □

Lemma B.2 (Intersecting near-parallel slabs lie in one test slab)**.** *Fix $S \in \mathbb{S}$, let $a/R \le \theta \le 1$, and suppose $S' \cap S \ne \varnothing$ and* $\operatorname{angle}(n_S, n_{S'}) \le C_0\theta$*. Then*

$$S' \subset K_{S,\theta} := B_R \cap N_{C_1\theta R}(H_S), \qquad |K_{S,\theta}| \le C_2 \theta R^3. \tag{B.1c}$$

In particular

$$\#\{S' : S' \cap S \ne \varnothing,\ \operatorname{angle}(n_S, n_{S'}) \le C_0\theta\} \le C_3 \frac{D\theta R^3}{abc}. \tag{B.1d}$$

All counts are labelled counts.

Proof. Choose $z \in S \cap S'$, and choose the sign of $n_{S'}$ so that $|n_S - n_{S'}| \le C\theta$. Write $H_S = \{n_S \cdot x = t_S\}$ and similarly for S'. Since both declared rectangles have shortest side comparable to a,

$$|n_S \cdot z - t_S| + |n_{S'} \cdot z - t_{S'}| \le Ca.$$

For $y \in S'$, add and subtract the two affine plane equations at z:

$$\begin{aligned} \operatorname{dist}(y, H_S) &\le \operatorname{dist}(y, H_{S'}) + |(n_S - n_{S'}) \cdot (y - z)| \\ &\quad + |n_S \cdot z - t_S| + |n_{S'} \cdot z - t_{S'}| \\ &\le C(a + \theta R) \le C'\theta R. \end{aligned} \tag{B.1e}$$

Here $y, z \in B_R$ and $\theta \geq a/R$. This proves containment in $K_{S,\theta}$; Fubini normal to H_S gives its volume bound. Finally every counted S' is wholly contained in this convex body, so (B.1) and $|S'| \geq c'_J abc$ give (B.1d). Coincident carriers with different labels are counted separately and are paid for by D. □

Theorem B.3 (Labelled rectangular slab lemma)**.** *For every $\lambda \geq 0$, one has*

$$|U| \gtrsim \frac{\lambda^2 |\mathbb{S}| abc}{1 + LDR^4/(b^2c^2)}. \tag{B.3}$$

If $\lambda > 0$, this is equivalently

$$\mu \lesssim \lambda^{-1} \left(1 + LD\frac{R^4}{b^2c^2}\right), \tag{B.2}$$

No separation or pointwise-constant multiplicity hypothesis is required.

Proof. If $A = 0$, then $\lambda = 0$ and (B.3) is immediate. Hence assume $A > 0$, so that $|U| > 0$ and all divisions below are legitimate. For intersecting slabs define

$$\vartheta(S, S') = \max\{a/R, \operatorname{angle}(n_S, n_{S'})\}.$$

Lemma B.1 gives

$$|S \cap S'| \lesssim \frac{a^2 R}{\vartheta(S, S')}. \tag{B.4}$$

Lemma B.2 gives the labelled count

$$\#\{S' : S' \cap S \neq \varnothing,\ \operatorname{angle}(n_S, n_{S'}) \lesssim \theta\} \lesssim \frac{D\theta R^3}{abc}. \tag{B.5}$$

For $f = \sum_S \mathbf{1}_{Y(S)}$, divide the pairs into the L dyadic ϑ-classes. Equations (B.4)–(B.5), with the diagonal added, give

$$\int f^2 \lesssim |\mathbb{S}| abc + |\mathbb{S}| L \frac{DaR^4}{bc}. \tag{B.6}$$

Cauchy–Schwarz and $\int f = A \asymp \lambda |\mathbb{S}| abc$ imply

$$|U| \geq \frac{A^2}{\int f^2} \gtrsim \frac{\lambda^2 |\mathbb{S}| abc}{1 + LDR^4/(b^2c^2)},$$

which proves both forms. □

The special case of the source statement [GWZ26, Lemma 6.9, p. 16] follows by taking $b = c = R$; no unlisted textual correction of that lemma is used. For Section 9, set $c = R = r_1$; then (B.3) is exactly (10.15). Notice that the proof uses global shading density only, so it applies to the arbitrary cellular parent shading and to fixed comparable enlargements.

Appendix C. A labelled-multiset filling lemma

The transverse branch requires a filling statement on a labelled multiset, so the argument is formulated on the index set and never identifies coincident carriers. Boundary cells are the only delicate point. We admit only cells containing a full $c\alpha \times c\rho \times c\rho$ piece, and all densities are computed with the resulting true geometric denominator. We do not assert the stronger thickened-proxy conclusions of [GWZ26, Lemma 6.13, p. 18], which are not needed for the transverse branch.

Throughout this appendix a *rectangular* $\alpha \times \rho \times 1$ *plank* means a labelled box

$$P = x_P + \{t_s e_s + t_m e_m + t_\ell e_\ell : |t_s| \leq \alpha/2,\ |t_m| \leq \rho/2,\ |t_\ell| \leq 1/2\}, \tag{C.0}$$

with a fixed ordered orthonormal frame (e_s, e_m, e_ℓ). Its two-long-axis plane is $\operatorname{span}\{e_m, e_\ell\}$. The frame is part of the label even when side lengths agree. For $C_{\mathrm{box}} \geq 1$, a C_{box}-*comparable rectangular* $\alpha \times \rho \times 1$ *plank* has the same form, with ordered side lengths $a_P \leq b_P \leq \ell_P$ satisfying

$$C_{\mathrm{box}}^{-1}\alpha \leq a_P \leq C_{\mathrm{box}}\alpha, \qquad C_{\mathrm{box}}^{-1}\rho \leq b_P \leq C_{\mathrm{box}}\rho, \qquad C_{\mathrm{box}}^{-1} \leq \ell_P \leq C_{\mathrm{box}}. \tag{C.0$'$}$$

The exact class (C.0) is the case $C_{\mathrm{box}} = 1$. Thus the full-piece claims below concern rectangular boxes with controlled side lengths, not a general convex body specified only by Löwner dimensions.

Lemma C.1 (Full-piece translated grids for rectangular planks)**.** *Fix $C_{\mathrm{box}} \geq 1$. Let S be a $\theta \times 1 \times 1$ slab and let P be a C_{box}-comparable rectangular $\alpha \times \rho \times 1$ plank in the sense of (C.0$'$), whose two-long-axis plane makes angle $O(\theta)$ with the plane of S. Suppose $\theta\rho \geq 4\alpha$. There is a structural finite family of translated, half-open grids, in coordinates adapted to S, whose cells have dimensions comparable to*

$$\theta\rho \times \rho \times \rho, \tag{C.1}$$

with the following property. For every $x \in P$, at least one grid has a cell $Q = Q(x)$ for which

$$c\alpha\rho^2 \leq |P \cap Q| \leq C\alpha\rho^2, \tag{C.2}$$

and $P \cap Q$ contains a $c\alpha \times c\rho \times c\rho$ subplank having x as one of its points (possibly on its boundary). The constants and the number of grids are independent of $\alpha, \rho, \theta, P, S$, with structural dependence on C_{box}. For each fixed grid G, the set

$$\mathcal{I}_{\mathrm{full}}(P, G) := \{x \in P : Q_G(x) \text{ has the displayed full-piece property at } x\}$$

is measurable; here $Q_G(x)$ is the unique half-open cell selected by the fixed boundary order.

Proof. Straighten the middle plane of S. In its two tangential coordinates use mesh $C_0\rho$, and in the normal coordinate use mesh $C_0\theta\rho$, where C_0 is a sufficiently large structural constant. Take three equally spaced translates in each coordinate. For every point one translate places it a fixed proportion of the mesh away from all cell faces. Inside a tangential disk of radius $c\rho$, the middle plane of P drifts by at most $C\theta\rho$ in the normal coordinate. Choosing c small relative to C_0, and using $\alpha \leq \theta\rho/4$, leaves a $c\alpha \times c\rho \times c\rho$ subplank in the cell. If x lies on, or within $c\rho$ of, a middle or long endpoint face of the rectangular carrier, choose the corresponding $c\rho$-interval one-sided toward the interior, shortening on the endpoint side and extending by the same amount on the interior side. Since both tangential side lengths are at least $C_{\mathrm{box}}^{-1}\rho$, choosing c below a structural multiple of C_{box}^{-1} gives these intervals even at a corner. In the short coordinate use a one-sided interval of length $c\alpha$ containing x; this fits because the actual short side has length at least $C_{\mathrm{box}}^{-1}\alpha$. Hence the lower volume in (C.2) is uniform at all faces, edges, and corners. Conversely, the projection of $P \cap Q$ to the middle plane of P has area $O_{C_{\mathrm{box}}}(\rho^2)$; Fubini across its $O_{C_{\mathrm{box}}}(\alpha)$ short thickness gives the upper bound in (C.2). Half-open cell faces and a fixed order of all endpoint faces assign all boundary points deterministically. For a fixed grid, membership in $\mathcal{I}_{\mathrm{full}}(P, G)$ is described by finitely many weak and strict affine inequalities in the coordinates of x: the half-open inequalities select $Q_G(x)$, and the remaining inequalities assert that the prescribed one-sided intervals fit in both P and that cell. It is therefore a finite union of Borel polyhedral pieces, hence measurable. □

Lemma C.2 (Bounded packing of slab parameters)**.** *Work in a fixed bounded ball. For $0 < \theta \leq 1$, there is a finite half-open parameter net $\mathfrak{S}_\theta$ of $\theta \times 1 \times 1$ slabs with the following two properties. Every such slab is contained in a fixed enlargement of some $S \in \mathfrak{S}_\theta$ whose middle plane is at projective-plane distance at most $C\theta$. Conversely, for every point x and every plane cap $\mathcal{C}$ of radius $C_0\theta$,*

$$\#\{S \in \mathfrak{S}_\theta : x \in CS,\ P_S \in \mathcal{C}\} \leq C(C_0, C). \tag{C.2a}$$

The same estimate holds for labelled candidates after first grouping equal net parameters; labels within one group are not discarded.

Proof. Parameterize an unoriented affine plane by $(n,t)\in\mathbb{RP}^2\times[-C_1,C_1]$, where its equation is $n\cdot y=t$. Take a $c\theta$-separated net in $\mathbb{RP}^2$ and, for every net normal, a half-open $c\theta$-mesh in t. Associate to a parameter pair the portion, in the fixed ambient ball, of the $C\theta$-neighbourhood of that plane. Nearest-net assignment, with the half-open order on Voronoi cells and on the offset intervals, gives the covering assertion.

If $P_S\in\mathcal{C}$, separation and two-dimensional packing on $\mathbb{RP}^2$ leave $O_{C_0,C}(1)$ possible normals. For one normal, the condition $x\in CS$ restricts t to an interval of length $O_C(\theta)$, which contains only $O_C(1)$ offset-net points. Their product proves (C.2a). Grouping labels by their net parameter changes neither the physical union nor this count of geometric parameter groups. □

Lemma C.3 (Capped two-plank intersection inside a cell)**.** *Let $Q\subset\mathbb{R}^3$ have diameter at most $C_Q\rho$, and let $P_i\subset N_{C_P\alpha}(H_i)$, $i=1,2$, where H_i are affine planes whose projective normal angle is $\phi\in(0,\pi/2]$. Then*

$$|P_1\cap P_2\cap Q|\le C\min\left\{\alpha\rho^2,\frac{\alpha^2\rho}{\sin\phi}\right\}\le C\min\left\{\alpha\rho^2,\frac{\alpha^2\rho}{\phi}\right\}. \tag{C.2f}$$

The estimate is uniform in the affine offsets and in all endpoint-cap or one-sided truncations. Consequently, if $\phi\ge\theta/(2B_{\mathrm{ang}})$ and $\theta\rho\ge 2B_{\mathrm{ang}}\alpha$, then

$$|P_1\cap P_2\cap Q|\le CB_{\mathrm{ang}}\frac{\alpha^2\rho}{\theta}\le C\alpha\rho^2. \tag{C.2g}$$

Proof. The first, trivial bound follows by enclosing Q in a ball of radius $C_Q\rho$ and slicing $N_{C_P\alpha}(H_1)$ normal to H_1. For the transverse bound choose unit normals n_i, write $H_i=\{x: n_i\cdot x=t_i\}$, and put $\tau=(n_1\times n_2)/|n_1\times n_2|$. The affine map

$$x\longmapsto(n_1\cdot x-t_1,n_2\cdot x-t_2,\tau\cdot(x-x_Q)) \tag{C.2h}$$

has Jacobian $\sin\phi$, where $x_Q\in Q$ is arbitrary. On $P_1\cap P_2\cap Q$, the first two coordinates range over intervals of length $O(\alpha)$, and the last over an interval of length $O(\rho)$. The change-of-variables formula gives $C\alpha^2\rho/\sin\phi$. Since $\sin\phi\ge(2/\pi)\phi$ on $[0,\pi/2]$, (C.2f) follows. Under the two assumptions in (C.2g), the transverse bound is at most $CB_{\mathrm{ang}}\alpha^2\rho/\theta$, and the buffer makes this at most the trivial scale $C\alpha\rho^2$. Changing an offset translates a coordinate interval, while caps or cell faces only remove points. □

Theorem C.4 (Self-contained labelled rectangular-plank filling)**.** *Fix $\eta>0$ and constants $C_{\mathrm{comp}},C_{\mathrm{box}}\ge 1$. Let $0<\alpha\le\rho\le 1$ be sufficiently small, and let $(\mathcal{P},Y)$ be a finite labelled multiset of C_{box}-comparable rectangular $\alpha\times\rho\times 1$ planks in the sense of (C.0′), contained in a fixed unit ball and equipped with measurable shadings, and allow coincident geometric carriers or coincident carriers with different ordered frames. Assume*

$$|\mathcal{P}|\le\alpha^{-C_{\mathrm{comp}}},\qquad\lambda(\mathcal{P},Y)\ge\alpha^{\eta},\qquad\mu(\mathcal{P},Y)\ge\alpha^{-\eta}, \tag{C.2b}$$

Let $m>0$, and suppose the labelled multiplicity belongs to $[m,2m)$ at every point of the union. Let $0<\theta\le 1$, and let $A_{\mathrm{ang}}=A_{\mathrm{ang}}(\alpha),B_{\mathrm{ang}}=B_{\mathrm{ang}}(\alpha)\ge 2$ satisfy

$$\log A_{\mathrm{ang}}+\log B_{\mathrm{ang}}=o(\log\alpha^{-1}),\qquad\frac{A_{\mathrm{ang}}}{\log^{C}(2/\alpha)}\longrightarrow\infty\quad\text{for every fixed } C. \tag{C.2c}$$

Suppose that at each shaded point the incident two-long-axis planes have raw angular diameter at most $C\theta$, while every incident subcollection of at least $2m/A_{\mathrm{ang}}$ labels has raw angular diameter greater than θ/B_{ang}. Finally assume

$$\theta\rho\ge 2B_{\mathrm{ang}}\alpha. \tag{C.2d}$$

Then there is a labelled refinement $(\mathcal{P}',Y')$ satisfying

$$M(Y')\ge\alpha^{o(1)}M(Y). \tag{C.2d′}$$

In this occurrence the uniform modulus required by the convention after (1.3) is

$$\omega_{\mathrm{fill}}(\alpha) := \frac{\log(1/c_{\mathrm{fill}})}{\log(1/\alpha)} \longrightarrow 0,$$

where $c_{\mathrm{fill}} \in (0,1]$ *is the fixed structural product of the retention fractions in the proof. This notation always refers to retained labelled shading mass; it does not mean a fraction of distinct carriers or a fraction of label indices. Whenever its union meets a ball* B_0 *of radius* $\theta\rho$*,*

$$\frac{|U(\mathcal{P}', Y') \cap CB_0|}{|CB_0|} \geq \alpha^{4\eta+o(1)}. \tag{C.2e}$$

Passing to a labelled subfamily cannot increase labelled $\Delta_{\max}$*. Every deletion below is a labelled deletion or a pointwise shading restriction, and every occurrence of cardinality, multiplicity, and mass counts labels.*

Proof. Write the multiset as a finite index set I, a carrier map $i \mapsto P_i$, and shadings $Y_i \subset P_i$. All planks have volume comparable to $\alpha\rho$, and all sums below run over indices.

Choose the half-open slab net of lemma C.2. Assign each P_i to the first slab $S(i)$ whose fixed enlargement contains it and whose middle plane is $O(\theta)$-parallel to that of P_i. At any shaded point, the upper angular-diameter hypothesis and (C.2a) show that the incident labels belong to only C_1 slab-parameter groups. Coincident labels remain together inside their group and therefore do not invalidate the geometric packing count. Assign the point, with a fixed tie rule, to a group having maximal incident labelled multiplicity, and retain at that point only the labels in this group. This pointwise restriction retains at least $C_1^{-1}M(Y)$, and its multiplicity is between m/C_1 and $2m$.

For each group use lemma C.1. Averaging over the structural finite set of grids, choose one translated grid per group for which the labelled mass of points lying in a full piece is at least a fixed fraction of the group mass. Write this structural fraction as $g_0 > 0$, and restrict to those points. Let E_0 be the set on which the retained grid multiplicity is below $c_1 m$. Before this restriction the multiplicity was at least m/C_1, so the discarded incidences contribute at least $(C_1^{-1} - c_1)m|E_0|$, whereas the retained mass on E_0 is below $c_1 m|E_0|$. Since the total discarded mass is at most the pre-grid mass,

$$M_{\mathrm{grid}}(E_0) \leq \frac{c_1}{C_1^{-1} - c_1} M_{\mathrm{pre-grid}}.$$

Choose c_1 so that the factor on the right is at most $g_0/2$. Deleting E_0 then leaves at least $g_0/2$ of the pre-grid labelled mass and gives the asserted lower multiplicity.

At this stage define the present per-label average exactly by

$$\overline{m}_Y := \frac{M(Y)}{|I|} = \frac{1}{|I|}\sum_{i\in I}|Y_i|. \tag{C.2e$'$}$$

We next discard precisely the labels for which $|Y_i| < \overline{m}_Y/2$. Their total mass is strictly less than $|I|\overline{m}_Y/2 = M(Y)/2$. This label deletion can again be followed by deleting the points of multiplicity below $c_2 m$, with $c_2 > 0$ structural. Here is the integral comparison. Immediately before label deletion the pointwise multiplicity is at least $c_1 m$. If E_{bad} is the set on which the surviving label multiplicity is below $c_2 m$, then the deleted labels contribute at least $(c_1 - c_2)m|E_{\mathrm{bad}}|$. They contribute less than half of the pre-deletion mass. The surviving mass on E_{bad} is therefore at most

$$c_2 m|E_{\mathrm{bad}}| \leq \frac{c_2}{2(c_1 - c_2)} M_{\mathrm{pre}}.$$

Taking, for example, $c_2 \le c_1/5$ leaves a fixed positive fraction after both operations. Denote the resulting family again by (I, Y). It satisfies

$$M(Y) \ge c_3 M(Y_{\mathrm{in}}), \qquad f(x) := \sum_{i \in I} \mathbf{1}_{Y_i}(x) \ge c_2 m \quad (x \in U(I,Y)), \tag{C.3}$$

for a structural constant $c_3 > 0$. Since the two pointwise deletions and the label deletion together retain a fixed fraction of labelled mass while the surviving label denominator can only decrease,

$$\frac{M(Y)}{\sum_{i \in I} |P_i|} \ge \alpha^{\eta + o(1)}. \tag{C.4}$$

The displayed $o(1)$ in (C.4) uses the uniform modulus obtained by writing the finitely many structural constants as powers of α. The assumed growth of A_{ang} ensures $c_2 m > 4m/A_{\mathrm{ang}}$ for small α.

From this point on, a grid cell is a tagged pair consisting of its slab group and one half-open cell in the grid selected for that group; we continue to write it as Q. Different tagged cells may overlap as physical boxes, but the preceding pointwise group assignment places every surviving shaded incidence in exactly one tag and then in exactly one half-open cell. Thus the tagged shading pieces, unlike the underlying physical boxes, form a measurable partition of the labelled mass. For such a tagged cell, let

$$\begin{aligned} I_Q &:= \{i : Y_i \cap Q \ne \varnothing\}, \quad D_Q := \sum_{i \in I_Q} |P_i \cap Q|, \\ M_Q &:= \sum_{i \in I_Q} |Y_i \cap Q|, \qquad E_Q := \bigcup_{i \in I_Q} (Y_i \cap Q), \\ \lambda_Q &:= M_Q / D_Q. \end{aligned} \tag{C.5}$$

Only full pieces survived, so lemma C.1 gives

$$D_Q \asymp |I_Q| \alpha \rho^2. \tag{C.6}$$

This is the denominator that is lost if "nonempty" intersections are silently treated as full pieces. Because the cells are half-open, $\sum_Q M_Q = M(Y)$; because their geometric intersections are disjoint, $\sum_Q D_Q \le \sum_i |P_i|$. Thus (C.4) implies that the D_Q-weighted average of λ_Q is at least $\alpha^{\eta + o(1)}$. Retaining all cells with

$$\lambda_Q \ge \tfrac{1}{2} \alpha^{\eta + o(1)} \tag{C.7}$$

removes at most half the labelled mass. This is a whole-cell restriction, so the pointwise lower multiplicity in (C.3) is unchanged on retained cells.

Fix such a cell and put $f_Q = \sum_{i \in I_Q} \mathbf{1}_{Y_i \cap Q}$. If x belongs to its support and $n = f_Q(x)$, then for each incident label i fewer than $2m/A_{\mathrm{ang}}$ incident labels can have raw plane angle at most $\theta/(2B_{\mathrm{ang}})$ from i; otherwise those labels would form a subcollection of angular diameter at most θ/B_{ang}, contrary to the hypothesis. Since $n \ge c_2 m$ and $A_{\mathrm{ang}} \to \infty$, fewer than half of the ordered pairs are close. Hence, pointwise,

$$f_Q(x)^2 \le 2 \sum_{\substack{i,j \in I_Q \\ \mathrm{angle}(P_i, P_j) > \theta/(2B_{\mathrm{ang}})}} \mathbf{1}_{Y_i \cap Q}(x) \mathbf{1}_{Y_j \cap Q}(x). \tag{C.8}$$

This direct domination includes the diagonal issue automatically: diagonal pairs are close and are paid for by the strict majority of separated ordered pairs.

Each Q has diameter $O(\rho)$, and every capped plank lies in an $O(\alpha)$-neighbourhood of its two-long-axis plane. Therefore lemma C.3, with $\phi > \theta/(2B_{\mathrm{ang}})$, gives

$$|P_i \cap P_j \cap Q| \le C B_{\mathrm{ang}} \frac{\alpha^2 \rho}{\theta}.$$

The buffer (C.2d) verifies the second inequality in (C.2g), so this estimate is uniform down to the smallest admitted angle and is compatible with the trivial full-piece volume $O(\alpha\rho^2)$. Endcaps and the cell truncation only decrease the intersection. Integrating (C.8) therefore gives

$$\int_Q f_Q^2 \le CB_{\mathrm{ang}}|I_Q|^2\frac{\alpha^2\rho}{\theta}. \tag{C.9}$$

By (C.5)–(C.7), Cauchy–Schwarz, and $|Q| \asymp \theta\rho^3$,

$$\begin{aligned} |E_Q| &\ge \frac{M_Q^2}{\int_Q f_Q^2} && \gtrsim B_{\mathrm{ang}}^{-1}\lambda_Q^2|Q| \\ &\ge \alpha^{3\eta+o(1)}|Q|. \end{aligned} \tag{C.10}$$

Here $B_{\mathrm{ang}} = \alpha^{-o(1)}$; the displayed exponent is a convenient weakening of the resulting $2\eta + o(1)$. Fix the uniform exponent function in this occurrence and write

$$\Xi(\alpha) := \alpha^{3\eta+\omega_C(\alpha)}, \qquad \omega_C(\alpha) \longrightarrow 0,$$

so that (C.10) reads $|E_Q| \ge c_0\Xi(\alpha)|Q|$ for one structural $c_0 > 0$.

Finally partition every retained Q into half-open subcells R of diameter $O(\theta\rho)$ and volume comparable to $(\theta\rho)^3$. Call R dense for the tagged cell Q if

$$|E_Q \cap R| \ge c\Xi(\alpha)|R|, \tag{C.11}$$

with c below the constant in (C.10). The tag-specific union E_Q in non-dense subcells has the following quantified bound. Let $c_0 > 0$ be the structural constant implicit in (C.10), fix $0 < \epsilon_{\mathrm{sub}} \le c_2/4$, and choose the constant in (C.11) so that $c \le c_0\epsilon_{\mathrm{sub}}$. Since the subcells partition Q,

$$\left| E_Q \cap \bigcup_{R \text{ non-dense}} R \right| \le c\Xi(\alpha)|Q| \le \epsilon_{\mathrm{sub}}|E_Q|. \tag{C.11a}$$

On the support of f_Q, one has $c_2 m \le f_Q \le 2m$. Hence the labelled mass deleted in (C.11a) is at most

$$2m\epsilon_{\mathrm{sub}}|E_Q| \le \frac{2\epsilon_{\mathrm{sub}}}{c_2}M_Q \le \tfrac{1}{2}M_Q. \tag{C.11b}$$

Thus dense-subcell selection retains at least one half of the labelled mass in every retained tagged cell. Define Y_i' by retaining, for every tagged Q, all incidences $Y_i \cap Q \cap R$ in its dense subcells R, and call the resulting refinement (I', Y'). In particular

$$E_Q \cap R \subset U(I', Y') \quad \text{for every dense pair } (Q, R). \tag{C.11$'$}$$

Let B_0 be any radius-$\theta\rho$ ball meeting $U(I', Y')$, and choose one retained incidence at a meeting point x. Its unique tagged cell Q and dense subcell R determine the relevant tag-specific support. The subcell R lies in a fixed enlargement of B_0, and its volume is comparable to $(\theta\rho)^3$. Equations (C.11)–(C.11$'$), weakened once more, give

$$\frac{|U(I', Y') \cap CB_0|}{|CB_0|} \ge \alpha^{4\eta+o(1)}. \tag{C.12}$$

All choices were made on the label index set, so coincident carriers were counted with their full multiplicity. Label deletion can only decrease labelled $\Delta_{\max}$. The fixed fractions retained at the grouping, grid, multiplicity, density-level, and dense-subcell steps have a fixed product $c_{\mathrm{fill}} > 0$. With $\omega_{\mathrm{fill}}(\alpha) = \log(1/c_{\mathrm{fill}})/\log(1/\alpha)$, this product is $\alpha^{\omega_{\mathrm{fill}}(\alpha)}$, so the final refinement satisfies (C.2d$'$) with the displayed uniform modulus. This proves the theorem. □

Appendix D. Quantitative constants, source map, and version convention

This appendix gathers the quantitative hierarchy, the exact interfaces with the cited sources, and the archival convention used for those comparisons. Keeping these records together separates provenance from the mathematical narrative while fixing the versions on which every source-dependent statement is based.

Version convention and reproducibility. The frozen source identifier for this release line is `CMDF-R7-20260827`. The manuscript is self-contained at the TeX level: the bibliography is embedded, and three clean `pdflatex` runs resolve the contents and cross-references. The external files checked for the source-dependent comparisons have the following fixed records.

GWZ arXiv:2601.14411v1 (20 January 2026); PDF byte object at `https://arxiv.org/pdf/2601.14411v1`; `application/pdf`; 566167 bytes; retrieved 2026-09-01T09:12:57Z; SHA-256
`1bc7df83c394cb74038af673c4a25e98`
`b3e018039c75f3b26501ed496028dce9`

Cohen, 18-page notes dated 7 April 2026; object `KakeyaRigorous_4-7-26.pdf`; `application/pdf`; 384469 bytes; retrieved 2026-09-01T09:12:58Z; SHA-256
`071cadd886f9730f458bcc49caad160a1`
`02d1d85b50594fa9f1614cafc1b1379`

Zeraoulia Rafik, Zenodo record 22067052, version 1.0 (23 August 2026); record file `Zeraoulia_Kakeya_Corrections_2026_Public_Revision.pdf`; `application/pdf`; 491215 bytes; retrieved 2026-09-01T09:13:06Z; SHA-256
`41619b18aaaed0dd225eff4963e8bdfd`
`de7e087b4b690b3b8fbf055552d78229`

Zeraoulia Rafik, Zenodo record 22122679, version 1.0 (27 August 2026); record file `Hua_Yang_v3_Correction_Note.pdf`; `application/pdf`; 288054 bytes; retrieved 2026-09-01T09:13:14Z; SHA-256
`1c16cd0b5a574d1e7dc6251c7db5fb49`
`e505063d8c2fa6a0d76d885a356c6656`

These archival identifiers are intentionally separate from the conventional bibliography.

All source-specific comparisons in the paper use these dated versions. In particular, "the archived GWZ manuscript" means arXiv:2601.14411v1 (20 January 2026). The symbol $\gtrapprox$ is reserved for a literal source quotation and is not a third asymptotic convention in the proofs. The GWZ phrase "outer John ellipsoid" denotes the minimum-volume containing Löwner ellipsoid used in lemma 8.3, rather than the maximum-volume inscribed John ellipsoid; the repeated-eigenvalue tie rule in this paper fixes the labelled frame and is not attributed to the source.

Section 6 version comparison. Branching comparability is used in (6.6e)–(6.6f), (9.1b′)–(9.1b″), and (9.5), and the arguments allow highly concentrated fine descendants. The pinned v1 proof of Proposition 6.6 prints three negative density exponents incompatible with densities bounded by one and an unintroduced left member in equation (49). The present derivation instead obtains (9.2d)–(9.2e) from Lemma 6.4 through proposition 9.5 and (9.5c)–(9.5d) from the labelled Lemma 6.1 argument at (9.1c). These documentary discrepancies therefore do not enter the logical interface.

The hierarchy is read from left to right: fixed structural exponents and the terminal positive gain are chosen before the small factoring, grid, shading, and discretization exponents. The notation below records logical roles, not a new optimization of the source parameters.

For comparison with the source notation, the complete interface chain is

$$\text{fixed (94)} \longrightarrow \text{global fine level} \longrightarrow \text{joint cellular factoring} \longrightarrow (95\text{–}97)_{\text{new}}$$
$$\longrightarrow \text{resolved ANGLE-CELL} \longrightarrow \text{one LIFT and } d_{\angle} \longrightarrow (98\text{–}103)_{\text{new}}.$$

The chain terminates at rectangular carriers. Its sharp-parent extension is the open problem isolated in section 12.

GWZ equations	formulation in this paper	mathematical role
(94)	Y_B, (10.2)	removal of low-occupancy outer balls
(95)	Z, (10.12)	the scale a/b and Jensen exponent q
(96)	Y_1, Z, (10.13)	compatibility on positive parent–cell incidences
(97)	Y_B, (10.14)	the reference shading and good-ball factor
(98)–(99)	$\bar{Y}$, (10.22)–(10.23)	buffered degree and separate scale losses
(100)–(101)	$Y^*, Z^{\angle}, \bar{Y}$, (10.26)–(10.27)	synchronized fine mass and c_*^{-1}
(102)–(103)	$\bar{Y}, Z^{\angle}$, (10.28)–(10.29)	fixed cap input and resolved angle

TABLE 2. Correspondence with the Section 9 equations of [GWZ26].

quantity	size or definition	role
h	$c_0 a$	cellular grid scale
q	$2-\sigma$	rectangular density power
$L_m, L_{a,b}, L_N$	dyadic level counts	fine multiplicity, dimensions, child count
L_{reg}	$L_{J1}L_{R1}L_{R2}$	positive edge and two degree levels
J_v	(10.11)	physical child volume per selected cell
P_{tag}	$L_\mu L_\omega L_d$	first whole-tag synchronization
P_{angtag}	$L_\theta^{\mathrm{tag}} L_{d_\angle}^{\mathrm{tag}}$	angle/degree whole-tag synchronization
P_{sync}	(10.16b)	full represented-mass bridge in the thin branches
c_{ang}	$A_{\mathrm{ang}}^{-K}/(J_\theta J_d)$	lower bound for typical-angle parent mass
H_{ang}	$4A_{\mathrm{ang}}^{K}$	converts input degree to output threshold
c_*	(10.25)	final fine-tube refinement fraction
κ_{inc}	fine-incidence representation	bounded across tags at one point
κ_{loc}	fixed-tag label representation	concentration and cap comparison inside one B
κ_{amb}	bounded overlap	tagged-to-physical recombination
H_{col}	(10.30)	repeated-label conflict coloring
$\omega_{\mathrm{src}}(\delta)$	(10.16d), tending to zero	uniform envelope for the subpower factors

TABLE 3. Quantitative factors used in the cellular construction.

source	use and local verification	status
DOV Theorem 1.12/4.1 [DOV22]	unweighted superlevels in (A.5); $t = 1 + \sigma > 1$, parameter separation, the (δ, t)-condition, and the cutoff are checked in (A.4)–(A.6)	published input; weighted form proved here
Córdoba L^2 method [Cór77]	plain-Frostman endpoint and section B; labelled atom mass, coincident parameters, angular layers, and measurable shadings are handled in Theorems A.5 and B.3	historical source; both labelled estimates are proved here
streamlined reduction [GWZ26]	Sections 6 and 9, cap uniformity, and the non-sticky count; propositions 9.5 and 9.7, (10.8f′), and (I1)–(I4) record the local interfaces	conditional source interface; all subsequent closure steps are proved locally
Löwner–John inclusion [Hen12, Joh48]	containing ellipsoid and factor-three inner copy; ordered axes and the repeated-eigenvalue rule are checked in lemma 8.3	Henk equation (2); frame transfer proved locally
convex factoring framework [WZ25b]	parent geometry; lemma 8.1 proves the labelled form	contextual; no theorem imported
Wang–Zahl background [WZ26, WZ25a]	multiscale and non-sticky structure; background for the surrounding scales	contextual
expository reconstructions [Coh26, Gut26]	surrounding reduction; used for comparison	contextual
independent correction note [Zer26]	comparison and shaded-slab calculation; Theorem B.3 is proved locally	contextual

TABLE 4. Proof-bearing and contextual source map.

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